Lecture Notes for STAT 433/833
Stochastic Processes 2

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Chapter 1

Probability, Random Variables, and Expectation

1.1 Probability Space

This is *not* meant to be a fully rigorous set of notes on measure theoretical probability. However, there are significant benefits to using the language of measure theory even if the underlying details are not fully justified. This set of course notes intends to explore the approach to stochastic processes within this fine line between fully rigorous measure theory and fully non-rigorous probability. I hope the reader will appreciate the conceptual gains we can make with this approach to probability without all the technical burdens.

We will motivate our concepts here with a stylized model of Lehman Brothers' stock during 2008, right before their infamous bankruptcy. Place yourself just before a future trading day, and let $S_t \in \mathbb{R}$ denote its stock price at time $t \geq 0$. The question we would like ask is:

What is the best way to model the randomness of the Lehman stock price as a sequence $\{S_t\}_{t\geq 0}$, with either a discrete or continuous time index t?

The standard approach to model random objects such as this one is to use a consider an abstract **sample space** Ω of all possible outcomes, and S_t is a function of the outcome $\omega \in \Omega$. Think of this abstract sample space as a "God's space" of all possible worlds, where each $\omega \in \Omega$ is "God's plan" for one such world: trades, news, solvency, and the resulting price path. The nice thing about this approach is that we can model multiple objects from the same sample space, e.g. the stock price at different times are just different functions $S_1, S_2, \ldots$ on the same sample space.

Now suppose our God makes His decisions randomly. How would we model this? A naive approach is to assign a probability (i.e. a number in $[0, 1]$) to each possible world $\omega \in \Omega$. This is problematic because there are infinitely many possible worlds, i.e. the stock price can take infinitely many possible values in $\mathbb{R}$. So an adapted approach is to instead assign probabilities to **events**, which are subsets of Ω. More precisely, we need a collection of events $\mathcal{F}$, where all of its elements are just subsets of Ω, and then a **probability** $\mathbb{P}$ is a function

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1] . \tag{1.1.1}$$

So now instead of thinking about "What's the probability of $S_t = 10$", we can think about "What's the probability of $S_t \in [10, 11]$" or "What's the probability of bankruptcy, i.e. $S_t = 0$?"

We will denote this event as $A \in \mathcal{F}$, e.g. $A = \{\omega \in \Omega : S_t(\omega) = 0\}$ for the bankruptcy event, and we can say the probability of bankruptcy is $\mathbb{P}(A)$. Note the event A notation means that "all possible worlds ω in which the price $S_t(\omega) = 0$".

So we have now reached our first important definition.

Definition 1.1 (Probability Space)**.** A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- Ω is the sample space,
- $\mathcal{F}$ is a collection of events (subsets of Ω),
- $\mathbb{P}$ is a probability (function from $\mathcal{F}$ to $[0, 1]$).

This description tells us the roles played by Ω, $\mathcal{F}$, and $\mathbb{P}$, but it leaves two questions unanswered:

1. What properties should the collection $\mathcal{F}$ have?
2. What rules should the function $\mathbb{P}$ satisfy?

The collection $\mathcal{F}$ should be closed under the usual logical operations on events. If A is an event, then "A does not occur" should also be an event. If $A_1, A_2, \ldots$ are events, then "at least one of the A_i occurs" should be an event. These requirements motivate the following definition.

Definition 1.2 (σ-Algebra)**.** A collection $\mathcal{F}$ of subsets of Ω is a **σ-algebra** if:

1. $\Omega \in \mathcal{F}$,
2. if $A \in \mathcal{F}$, then $A^c = \Omega \setminus A \in \mathcal{F}$,
3. if $A_1, A_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Given any collection $\mathcal{C}$ of subsets of Ω, we write $\sigma(\mathcal{C})$ for the **σ-algebra generated by $\mathcal{C}$**: the smallest σ-algebra that contains every set in $\mathcal{C}$. Equivalently, it is the intersection of all σ-algebras on Ω that contain $\mathcal{C}$. For events $A_1, \ldots, A_m$, we use the shorthand

$$\sigma(A_1, \ldots, A_m) := \sigma(\{A_1, \ldots, A_m\}). \tag{1.1.2}$$

If $A_1, \ldots, A_m$ form a partition of Ω, then $\sigma(A_1, \ldots, A_m)$ consists exactly of all unions of the partition cells. In particular, $\sigma(A) = \{\varnothing, A, A^c, \Omega\}$.

The countable-union condition here is a technicality that we will briefly explain. In particular, one might ask: Why not always let $\mathcal{F}$ contain every subset of Ω? This works just fine for finite and countable spaces.

On the real line however, even the construction of a uniform distribution leads to a **contradiction**, and therefore "breaks" the theory. The interested reader can find more details in the Vitali non-measurability theorem, which essentially shows that there are subsets of $\mathbb{R}$ that cannot be assigned a probability in a consistent way. We will not cover this result, but it is important to keep in mind why the technicality leads us to restrict our attention to σ-algebras instead of all subsets of Ω.

The second question can be answered by stating the probability axioms.

Definition 1.3 (Probability Measure)**.** Let $\mathcal{F}$ be a σ-algebra on Ω. A function $\mathbb{P} : \mathcal{F} \longrightarrow [0, 1]$ is a **probability measure** if:

1. $\mathbb{P}(\Omega) = 1$,
2. whenever $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \tag{1.1.3}$$

Before moving on, let us ask why these particular probability axioms are natural. The range $[0,1]$ says that probabilities are nonnegative and that no event is more likely than certainty. Moreover, one of the possible worlds in Ω must occur, so the event Ω is certain and should have probability one.

The second property is also known as **countable additivity**, which describes what should happen when an event can occur in several mutually exclusive ways. The finite version of this property is intuitive: if an event A can occur in one of n mutually exclusive ways, then the probability of A should be the sum of the probabilities of each way.

The reason we require *countable* and not *uncountable* additivity is a bit subtle. Let us suppose we allow uncountable additivity, and consider the probability space $[0,1]$ with the uniform probability distribution. Then we can write

$$1 = \mathbb{P}([0,1]) = \mathbb{P}\left(\bigcup_{x\in[0,1]} \{x\}\right) = \sum_{x\in[0,1]} \mathbb{P}(\{x\}), \tag{1.1.4}$$

but $\mathbb{P}(\{x\}) = 0$ for all $x \in [0,1]$, so the right-hand side is zero, which is a contradiction. Therefore countable additivity is the strongest property we can require without running into contradictions.

The axioms immediately give us the familiar rules of probability that we will use throughout the course.

Proposition 1.4 (Useful Consequences of the Probability Axioms)**.** *If $A, B \in \mathcal{F}$, then:*

1. $\mathbb{P}(\varnothing) = 0$,
2. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$,
3. *if* $A \subseteq B$, *then* $\mathbb{P}(A) \leq \mathbb{P}(B)$,
4. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

More generally, if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i = \Omega$, then for every $B \in \mathcal{F}$,

$$\mathbb{P}(B) = \sum_{i=1}^{\infty} \mathbb{P}(B \cap A_i). \tag{1.1.5}$$

Proof. A σ-algebra is closed under intersections and set differences, so every set used below belongs to $\mathcal{F}$. Apply countable additivity to the disjoint sequence $\Omega, \varnothing, \varnothing, \ldots$. Since all terms are nonnegative and $\mathbb{P}(\Omega) = 1$, this gives $\mathbb{P}(\varnothing) = 0$. Consequently, countable additivity applied to $E, F, \varnothing, \varnothing, \ldots$ shows that $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F)$ whenever E and F are disjoint.

Since $\Omega = A \dot\cup A^c$, finite additivity gives $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. If $A \subseteq B$, then $B = A \dot\cup (B \setminus A)$, and nonnegativity gives $\mathbb{P}(B) \geq \mathbb{P}(A)$. Also, $A \cup B = A \dot\cup (B \setminus A)$ and $B = (A \cap B) \dot\cup (B \setminus A)$. Eliminating $\mathbb{P}(B \setminus A)$ from these two identities yields the inclusion–exclusion formula in part (4).

Finally, if $A_1, A_2, \ldots$ form the stated partition, then $B = \bigcup_{i=1}^{\infty} (B \cap A_i)$, and this union is disjoint. Countable additivity therefore gives (1.1.5). □

So far, probability has answered questions about events in God's space. In practice, however, we do not observe the world ω directly; we observe quantities such as the stock price $S_t(\omega)$. To connect these two viewpoints, we need to ensure that questions about the observed price correspond to events in $\mathcal{F}$. This requirement will lead us to random variables as measurable functions.

Exercise 1.1 (What Does the Report Tell Us?)**.** Let $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ and $A = \{\omega_1, \omega_2\}$. A news report tells us only whether A occurred. We use the event collection $\mathcal{F} = \{\varnothing, A, A^c, \Omega\}$, not every subset of Ω.

(a) Explain why these are precisely the events whose occurrence can be determined from the report, whatever the outcome. Verify that $\mathcal{F}$ is a σ-algebra.
(b) Let $\mathbb{P}$ be a probability measure on $\mathcal{F}$ with $\mathbb{P}(A) = 0.3$. Determine its value on each of the four events in $\mathcal{F}$.
(c) Does this probability space assign a probability to $\{\omega_1\}$? Explain.

Exercise 1.2 (When All Conditions Must Hold)**.** Let $\mathcal{F}$ be a σ-algebra and let $A_1, A_2, \ldots \in \mathcal{F}$. Show directly from the definition that $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$. *Hint: Think about the event that at least one condition fails.*

Exercise 1.3 (Avoid Counting Twice)**.** In a toy model, let A be the event of bad earnings news and B the event of a credit downgrade. Suppose $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.2$. Find the probability that (a) at least one occurs, (b) neither occurs, and (c) exactly one occurs.

1.2 Random Variables

We have already written the stock price as a function $S_t : \Omega \longrightarrow \mathbb{R}$, but so far this is only a function from possible worlds to numbers. What additional condition makes this function compatible with the probability space?

Suppose we ask whether an observed quantity X belongs to a set $B \subseteq \mathbb{R}$, where we might think of the event as $\{X \in B\}$ informally. To write this precisely in the language of probability spaces, we will define the event as

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}. \tag{1.2.1}$$

This is called the **preimage** of B, and note this does not require X to have an inverse function.

Recall the probability $\mathbb{P}$ can only assign values to sets in the σ-algebra $\mathcal{F}$, therefore it would be problematic if $X^{-1}(B)$ is not an element of $\mathcal{F}$. Thus we need to restrict the possible events B to a collection that is compatible with $\mathcal{F}$. Also recall that taking every subset of $\mathbb{R}$ gives us a contradiction with the Vitali theorem, so we cannot take every subset of $\mathbb{R}$ as a possible event B. Then we might ask:

What is the smallest σ-algebra that still lets us ask the useful questions?

The standard approach here is to require us to be able to ask whether the observed quantity X lies in any interval, for example is the Lehman stock today S_t trading between \$10 and \$11. We therefore want every interval to be included. That being said, the intervals alone do not form a σ-algebra: a complement or a countable union of intervals need not be an interval. So we also include exactly the additional sets required to make it a σ-algebra.

Definition 1.5 (Borel σ-Algebra)**.** The **Borel σ-algebra** $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra on $\mathbb{R}$ that contains every interval.

We will not construct $\mathcal{B}(\mathbb{R})$ explicitly. For our purposes, it contains the intervals and all the sets obtained from them using complements and countable set operations. And most importantly, the Vitali type contradictions won't bother us here.

Now we are ready to define a random variable with the exact condition that makes it compatible with the probability space.

Definition 1.6 (Random Variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : \Omega \longrightarrow \mathbb{R}$ is a **random variable** if $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$.

The information revealed by observing X is represented by the smallest sub-σ-algebra of $\mathcal{F}$ with respect to which X is measurable. It is

$$\sigma(X) := \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}, \tag{1.2.2}$$

and is called the **σ-algebra generated by** X. Thus $\sigma(X)$ contains exactly the events whose occurrence can be determined from the observed value of X.

We also sometimes say X is **$\mathcal{F}$-measurable**, which just means exactly the same thing here. Once again, it does not mean that we can observe the underlying world ω, but we can ask questions about the observed quantity $X(\omega)$ within the Borel sets $\mathcal{B}(\mathbb{R})$. This is how a random variable translates information from God's space into a quantity that we can understand in our world.

Returning to our example, the price S_t is a real valued random variable for each fixed time t, in other words $S_t : \Omega \longrightarrow \mathbb{R}$ is measurable. We can now assign values of probability to events such as "the stock price is between \$10 and \$11" by using the preimage $S_t^{-1}([10, 11])$. More precisely

$$\mathbb{P}(S_t(\omega) \in [10, 11]) = \mathbb{P}(\omega \in \Omega : S_t(\omega) \in [10, 11]) = \mathbb{P}(S_t^{-1}([10, 11])). \tag{1.2.3}$$

This approach also lets us describe probabilities directly on the real line. For a real-valued random variable X, its **distribution**, or **law**, is the probability measure μ_X defined by

$$\mu_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)), \qquad B \in \mathcal{B}(\mathbb{R}). \tag{1.2.4}$$

This measure is called the **pushforward** of $\mathbb{P}$ through X.

We are now ready to bring the probability space approach back to the setting you are familiar with from your introduction course on probability. Here we observe the probability axioms carry over to $\mathbb{R}$ because $X^{-1}(\mathbb{R}) = \Omega$ and preimages preserve disjoint unions. The pushforward therefore gives us a *new probability space* $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X)$. Its sample points are real numbers, its events are Borel sets, and its probability measure is the law of X.

For example, the stock price S_t has the distribution μ_{S_t} on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, which satisfies

$$\mu_{S_t}([10, 11]) = \mathbb{P}(S_t \in [10, 11])\,. \tag{1.2.5}$$

This approach also allows us to model the joint behaviour of multiple random variables, such as the stock price at different times. If every S_t is a random variable on the same probability space, then the collection $\{S_t\}_{t \geq 0}$ is called a **stochastic process**. Using one common space also lets us ask joint questions about prices at different times.

Here if $t \in \{0, 1, 2, \ldots, T\}$ is a discrete time index, then we can think about the stochastic process as a single random variable

$$S : \Omega \longrightarrow \mathbb{R}^{T+1}, \quad S(\omega) = (S_0(\omega), S_1(\omega), \ldots, S_T(\omega)) \in \mathbb{R}^{T+1}. \tag{1.2.6}$$

For the vector $(S_0, \ldots, S_T)$, we might want to ask whether each price S_t lies in a specified interval I_t, and what the probability of that event is. These questions correspond to rectangles $I_0 \times \cdots \times I_T$ in $\mathbb{R}^{T+1}$, or more precisely,

$$\{S_0 \in I_0, S_1 \in I_1, \ldots, S_T \in I_T\} = \{S \in I_0 \times I_1 \times \cdots \times I_T\} \tag{1.2.7}$$

We call the smallest σ-algebra containing these rectangles the **Borel σ-algebra** $\mathcal{B}(\mathbb{R}^{T+1}) = \mathcal{B}(\mathbb{R})^{\otimes(T+1)}$. Sometimes this is also called the **product σ-algebra** on $\mathbb{R}^{T+1}$, since it's generated by products of sets in $\mathcal{B}(\mathbb{R})$. The same pushforward operation gives the law μ_S of the vector on $(\mathbb{R}^{T+1}, \mathcal{B}(\mathbb{R}^{T+1}))$.

Similarly, if $t \in [0, T]$ is a continuous time index and the sample paths are continuous, then we can think about the stochastic process as a random variable taking values in continuous functions (or paths) on $[0, T]$, i.e.

$$S : \Omega \longrightarrow C([0,T]), \quad S(\omega) = (S_t(\omega))_{t \in [0,T]} \in C([0,T]). \tag{1.2.8}$$

Similarly, to construct the appropriate σ-algebra on $C([0, T])$, we start with questions about finitely many times, such as "is the price at time t_1 in interval I_1, at time t_2 in interval I_2, and so on?" The analogues of intervals and rectangles are **cylinder sets** of the form

$$\{f \in C([0,T]) : f(t_1) \in I_1, \ldots, f(t_m) \in I_m\}, \tag{1.2.9}$$

where $t_1, \ldots, t_m \in [0, T]$ and the I_i are intervals. Such a set specifies conditions at finitely many times and leaves the rest of the path unspecified.

The smallest σ-algebra containing these sets is called the **cylindrical σ-algebra** on $C([0, T])$. On this space, it is also the usual Borel σ-algebra $\mathcal{B}(C([0, T]))$. We won't worry too much about the technical construction of this σ-algebra in this course, but it's good to know this is the smallest σ-algebra that lets us ask all these finite-time questions.

The pushforward then gives the law μ_S on $(C([0, T]), \mathcal{B}(C([0, T])))$, just as it gave a law on the real line. The difference is that the observed object is now an entire path rather than a single number.

Exercise 1.4 (Measurability and an Interval Event)**.** Let $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$, $A = \{\omega_1, \omega_2\}$, and $\mathcal{F} = \{\varnothing, A, A^c, \Omega\}$. Let $\mathbb{P}$ be the probability measure on $\mathcal{F}$ determined by $\mathbb{P}(A) = 0.3$, and define $X, Y : \Omega \to \mathbb{R}$ by

ω	ω_1	ω_2	ω_3	ω_4
$X(\omega)$	0	0	10	10
$Y(\omega)$	0	1	10	10

(a) Determine which of X and Y is a random variable from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Justify each answer using preimages of Borel sets.

(b) For whichever of X and Y is a random variable, find the preimage of $[9, 11]$, and hence calculate the probability that its value lies in $[9, 11]$.

Exercise 1.5 (Same One-Day Probabilities, Different Joint Behaviour)**.** For this exercise, let $\Omega = \{\omega_1, \omega_2\}$, and let $\mathcal{F}$ consist of $\varnothing$, $\{\omega_1\}$, $\{\omega_2\}$, and Ω. Let $\mathbb{P}(\{\omega_1\}) = \mathbb{P}(\{\omega_2\}) = 1/2$. Define two random vectors $S^{(A)}, S^{(B)} : \Omega \to \mathbb{R}^2$ by

ω	ω_1	ω_2
$S^{(A)}(\omega)$	$(10, 10)$	$(12, 12)$
$S^{(B)}(\omega)$	$(10, 12)$	$(12, 10)$

Write $S^{(A)} = (S_0^{(A)}, S_1^{(A)})$ and $S^{(B)} = (S_0^{(B)}, S_1^{(B)})$, where the coordinates are the prices on Day 0 and Day 1.

(a) For each $t \in \{0, 1\}$, find the preimages $(S_t^{(A)})^{-1}([9, 11])$ and $(S_t^{(B)})^{-1}([9, 11])$, and hence calculate the corresponding four probabilities.

(b) Let $R = [9, 11] \times [9, 11]$. Find the two preimages $(S^{(A)})^{-1}(R)$ and $(S^{(B)})^{-1}(R)$, and hence calculate the probability that both prices lie in $[9, 11]$ for each vector.

(c) Do the four one-day probabilities from part (a) determine the probability that both prices lie in $[9, 11]$? Explain using your answer to part (b).

Exercise 1.6 (What Do Endpoint Observations Tell Us?)**.** Let $\Omega = \{\omega_1, \omega_2\}$, and let $\mathcal{F}$ consist of $\varnothing$, $\{\omega_1\}$, $\{\omega_2\}$, and Ω, with $\mathbb{P}(\{\omega_1\}) = 0.7$ and $\mathbb{P}(\{\omega_2\}) = 0.3$. Define $S : \Omega \to C([0, 1])$ by $S(\omega_1) = f_1$ and $S(\omega_2) = f_2$, where $f_1(t) = 10$ and $f_2(t) = 10 + 8t(1 - t)$. Thus $S(\omega)$ is an entire continuous price path.

(a) Write the cylinder set $B_{\text{end}} \subseteq C([0, 1])$ saying that the prices at times zero and one both lie in $[9, 11]$. Find $S^{-1}(B_{\text{end}})$ and $\mathbb{P}(S \in B_{\text{end}})$.

(b) Does observing this endpoint event guarantee that the price remains in $[9, 11]$ throughout the time interval? Find the set $\{\omega \in \Omega : S(\omega)(t) \in [9, 11] \text{ for every } t \in [0, 1]\}$ and its probability.

1.3 Laws and Probability Measures

We now return to the law of a real-valued random variable. Recall from (1.2.4) that it is the pushforward of $\mathbb{P}$ through $X : \Omega \to \mathbb{R}$

$$\mu_X(B) = \mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)), \qquad B \in \mathcal{B}(\mathbb{R}). \tag{1.3.1}$$

Once again, X pushes the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to a space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_X)$ that we can observe.

In your introductory probability course, you probably learned that a law can be described by a probability mass function (pmf) or a probability density function (pdf). Indeed, this remains consistent with our probability space approach. While in general probability measures can take other forms, we will only consider laws that can be represented by a mixture of point masses and a density, which we will define below.

Definition 1.7 (Dirac Point Mass)**.** For $a \in \mathbb{R}$, the **Dirac measure** at a, denoted by δ_a, is the probability measure (i.e. a function on $\mathcal{B}(\mathbb{R})$) satisfying

$$\delta_a(B) = \begin{cases} 1, & a \in B, \\ 0, & a \notin B, \end{cases} \qquad B \in \mathcal{B}(\mathbb{R}). \tag{1.3.2}$$

Thus, for $p \geq 0$, $p\delta_a$ represents a point mass of size p: it assigns mass p to every set containing a, and zero to every set not containing a.

Example 1.8 (A Bernoulli Law)**.** Let $0 \leq p \leq 1$, and let X be a Bernoulli random variable with $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. Its law is

$$\mu_X = p\,\delta_1 + (1-p)\,\delta_0. \tag{1.3.3}$$

To verify this, take any $B \in \mathcal{B}(\mathbb{R})$. The first term contributes p precisely when $1 \in B$, and the second contributes $1 - p$ precisely when $0 \in B$. These are the only two values X can take, and the events $\{X = 1\}$ and $\{X = 0\}$ are disjoint. Consequently,

$$\mu_X(B) = p\delta_1(B) + (1-p)\delta_0(B) = \mathbb{P}(X \in B). \tag{1.3.4}$$

For example, $\mu_X(\{1\}) = p$ and $\mu_X(\{0\}) = 1 - p$, while $\mu_X(\{0, 1\}) = 1$ and $\mu_X((0, 1)) = 0$.

We will also define the integral as the action of a measure on a set

$$\int_B \delta_a(dx) = \delta_a(B), \tag{1.3.5}$$

where we have that $\int_B p\,\delta_a(dx) = p$ if $a \in B$.

Likewise, integration against δ_a evaluates a function at the point a: for a Borel-measurable function $\varphi : \mathbb{R} \to \mathbb{R}$ (i.e. for all $B \in \mathcal{B}(\mathbb{R})$, we have $\varphi^{-1}(B) \in \mathcal{B}(\mathbb{R})$), we have

$$\int_{\mathbb{R}} \varphi(x)\,\delta_a(dx) = \varphi(a). \tag{1.3.6}$$

All the mass is at a, so only the value $\varphi(a)$ contributes to the integral. We will use this property without developing the technical construction of the integral.

In this course we won't go into the technical details of measures, but we will emphasize that δ_a is **not a function**. We can only make sense of δ_a as either an integral, or equivalently applied as a function on a Borel set B.

This allows us to recover the definition of a probability mass function for a discrete law, which all it is saying is X only takes finite (or countably many) values x_j with probabilities p_j

$$\mu_X = \sum_j p_j\,\delta_{x_j}, \qquad \sum_{j \in J} p_j = 1. \tag{1.3.7}$$

On the other hand, continuous laws are based on a building block similar to the Dirac, but instead of assigning all mass to a single point, it assigns mass to an interval.

Definition 1.9 (Lebesgue Measure)**.** The **Lebesgue measure** λ is a function on $\mathcal{B}(\mathbb{R})$ satisfying countable additivity and

$$\lambda([a, b]) = b - a, \qquad a < b. \tag{1.3.8}$$

For the rest of this course, we will drop the notation of λ often when writing an integral with respect to the Lebesgue measure, and simply write dx instead of $\lambda(dx)$, i.e.

$$\int_{[a,b]} \lambda(dx) = \int_{[a,b]} dx = b - a. \tag{1.3.9}$$

We note that the integral against the Lebesgue measure matches the Riemann integral for piecewise-continuous functions, but this approach allows us to also integrate against point masses, which is not possible with the Riemann integral.

This also allows us to recover the definition of a probability density function for a continuous law, which is saying that X has a nonnegative density f satisfying $\int_{\mathbb{R}} f(x)\,dx = 1$, and the law is given by

$$\mu_X(B) = \int_B f(x)\,\lambda(dx) = \int_B f(x)\,dx, \qquad B \in \mathcal{B}(\mathbb{R}). \tag{1.3.10}$$

For this course, we only consider laws consisting of a finite or countable collection of point masses together with a density. For distinct points $x_j \in \mathbb{R}$, weights $p_j \geq 0$, and a nonnegative piecewise-continuous function f, we write

$$\begin{aligned}\mu_X(B) &= \sum_{j\in J} p_j \int_B \delta_{x_j}(dx) + \int_B f(x)\,\lambda(dx) \\ &= \sum_{\substack{j\in J \\ x_j\in B}} p_j + \int_B f(x)\,dx, \qquad B \in \mathcal{B}(\mathbb{R}),\end{aligned} \tag{1.3.11}$$

and we require that $\sum_{j\in J} p_j + \int_{\mathbb{R}} f(x)\,dx = 1$.

Here J is finite or countable. If $f = 0$, the law is discrete; if $J = \varnothing$, the law has a density; and if both terms are present, the law is mixed. This allows us to model a variety of situations, including the possibility of bankruptcy, which is a point mass at zero.

Example 1.10 (A Bankruptcy Point Mass)**.** Suppose that, at a fixed time t, the stock is worth zero with probability $0 \leq q < 1$, its positive price has a nonnegative density g satisfying $\int_0^\infty g(x)\,dx = 1$. Its law can be written as

$$\mu_{S_t}(B) = q\int_B \delta_0(dx) + (1-q)\int_{B\cap(0,\infty)} g(x)\,dx. \tag{1.3.12}$$

The two terms record bankruptcy and the ordinary range of positive trading prices, respectively.

Up to this point, we have only considered the law of a single real valued random variable. Knowing the law of S_t at every single time t still does not tell us how prices at different times move together. In Exercise 1.5, the one-day probabilities considered there agree in two models, while the probabilities of a two-day event are different. The missing information is contained in a joint law, which we can consider as the law of a sequence of random variables.

More precisely, if $X_1, \ldots, X_m$ are random variables on the same probability space, then we can consider the random vector $X = [X_1, \ldots, X_m]^\top \in \mathbb{R}^m$ as a single random variable $X : \Omega \to \mathbb{R}^m$. Then the pushforward distribution μ_X of $\mathbb{P}$ through X is called the **joint distribution** or **joint law** of the random variables $X_1, \ldots, X_m$. We can write the joint probability as

$$\mu_X(B) = \mathbb{P}\big((X_1, \ldots, X_m) \in B\big), \qquad B \in \mathcal{B}(\mathbb{R}^m). \tag{1.3.13}$$

In particular, if $B_1, \ldots, B_m \in \mathcal{B}(\mathbb{R})$, then the joint law assigns the rectangle $B_1 \times \cdots \times B_m$ the probability of the simultaneous event

$$\mu_{X_1,\ldots,X_m}(B_1 \times \cdots \times B_m) = \mathbb{P}(X_1 \in B_1, \ldots, X_m \in B_m). \tag{1.3.14}$$

For a vector $x \in \mathbb{R}^m$, we use δ_x for the Dirac point mass at x, defined in exactly the same way as the real-valued point mass in (1.3.2).

The law of one coordinate is recovered by leaving every other coordinate unrestricted. For $i \in \{1, \dots, m\}$ and $B \in \mathcal{B}(\mathbb{R})$,

$$\mu_{X_i}(B) = \mu_{X_1,\dots,X_m}(\mathbb{R} \times \cdots \times \underbrace{B}_{i\text{th coordinate}} \times \cdots \times \mathbb{R}). \tag{1.3.15}$$

The coordinate laws $\mu_{X_1}, \dots, \mu_{X_m}$ are called the **marginal laws** of the joint distribution. The joint law determines all of its marginals, but the marginals generally do not determine the joint law: they do not record how the coordinates move together.

While we won't focus on the details, we note that it is possible to have laws that are not a combination of point masses and a density. For example, it's possible to have a law concentrated on the diagonal in $\mathbb{R}^2$, which has no area, and therefore is not a point mass but also cannot have a density.

When probability is spread across a region in the plane, a joint law can sometimes be described by a **joint density** $f_{X,Y}$. In this case, $f_{X,Y}$ is nonnegative, has total integral one, and assigns probability to a Borel set $B \subseteq \mathbb{R}^2$ according to

$$\mu_{X,Y}(B) = \iint_B f_{X,Y}(x, y)\, dx\, dy. \tag{1.3.16}$$

The marginal densities are obtained by leaving the other coordinate unrestricted:

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)\, dy, \qquad f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y)\, dx. \tag{1.3.17}$$

For the piecewise-continuous densities used in this course, these are ordinary iterated integrals. A joint density is only one useful kind of joint law; the diagonal example above shows why not every joint law has one.

For example, a random vector $X = (X_1, \dots, X_d)^\top$ is **jointly Gaussian** with mean $m \in \mathbb{R}^d$ and (symmetric positive semidefinite) covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ if its joint density is

$$f_X(x) = \frac{1}{(2\pi)^{d/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x - m)^\top \Sigma^{-1}(x - m)\right), \qquad x \in \mathbb{R}^d. \tag{1.3.18}$$

We write this law as $\mathcal{N}(m, \Sigma)$.

In Section 1.2, we represented a process with continuous sample paths as a random element of $C([0, T])$ (although continuous index processes need not to be continuous)

$$S : \Omega \longrightarrow C([0, T]), \qquad S(\omega)(t) = S_t(\omega). \tag{1.3.19}$$

Recall the relevant event space is generated by the **cylinder sets**

$$C(t_1, \dots, t_k; B_1, \dots, B_k) = \{x \in C([0, T]) : x(t_i) \in B_i \text{ for } i = 1, \dots, k\}, \tag{1.3.20}$$

where $0 \le t_1 < \cdots < t_k \le T$ and $B_1, \dots, B_k \in \mathcal{B}(\mathbb{R})$.

The joint law of $(S_{t_1}, \dots, S_{t_k})$ is called the **finite-dimensional distribution** of the process at $t_1, \dots, t_k$. If this joint law has a density $f_{t_1,\dots,t_k}$, then the process law μ_S assigns the cylinder the probability

$$\begin{aligned} \mu_S(C(t_1, \dots, t_k; B_1, \dots, B_k)) &= \mathbb{P}(S_{t_1} \in B_1, \dots, S_{t_k} \in B_k) \\ &= \int_{B_1 \times \cdots \times B_k} f_{t_1,\dots,t_k}(x_1, \dots, x_k)\, dx_1 \cdots dx_k. \end{aligned} \tag{1.3.21}$$

A good example of such a process is a **Gaussian process**, where each of its finite-dimensional distributions is jointly Gaussian. There are a mean function $m : [0,T] \to \mathbb{R}$ and a covariance function $\Sigma : [0,T] \times [0,T] \to \mathbb{R}$ that determine all of these laws. For every $k \geq 1$ and every choice of finitely many times $\boldsymbol{t} = [t_1, \ldots, t_k] \in [0,T]^k$, we have

$$\begin{aligned} &[S_{t_1}, \ldots, S_{t_k}]^\top \sim \mathcal{N}(m(\boldsymbol{t}), \Sigma_{\boldsymbol{t}}), \\ &m(\boldsymbol{t}) = [m(t_1), \ldots, m(t_k)]^\top, \qquad (\Sigma_{\boldsymbol{t}})_{ij} = \Sigma(t_i, t_j). \end{aligned} \tag{1.3.22}$$

Here the covariance entry formula holds for $1 \leq i, j \leq k$.

For instance, we will eventually study a process called Brownian motion, which is a centered (zero mean) Gaussian process with $m(t) = 0$ and covariance function $\Sigma(s,t) = \min\{s,t\}$. For $0 < t_1 < \cdots < t_k \leq T$, we then have $(S_{t_1}, \ldots, S_{t_k})^\top \sim \mathcal{N}(0, \Sigma_{\boldsymbol{t}})$, and the finite-dimensional distribution has density

$$f_{t_1,\ldots,t_k}(x) = \frac{1}{(2\pi)^{k/2} \det(\Sigma_{\boldsymbol{t}})^{1/2}} \exp\left(-\frac{1}{2} x^\top \Sigma_{\boldsymbol{t}}^{-1} x\right), \qquad x \in \mathbb{R}^k. \tag{1.3.23}$$

Exercise 1.7 (A Joint Law and Its Marginals)**.** Let $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$, $\mathcal{F} = 2^\Omega$, and define the probability measure and the random variables $X, Y : \Omega \to \mathbb{R}$ by

ω	ω_1	ω_2	ω_3	ω_4
$\mathbb{P}(\{\omega\})$	0.1	0.2	0.3	0.4
$(X(\omega), Y(\omega))$	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$

(a) Write the joint law $\mu_{X,Y}$ as a weighted sum of Dirac point masses on $\mathbb{R}^2$.
(b) Find the marginal laws μ_X and μ_Y as weighted sums of Dirac point masses on $\mathbb{R}$.
(c) Use the joint law to calculate $\mathbb{P}(X \leq Y)$.

Exercise 1.8 (A Continuous Joint Law and Its Marginals)**.** Consider the function

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 \leq x \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{1.3.24}$$

(a) Verify that $f_{X,Y}$ is a joint probability density.
(b) Suppose that (X,Y) has the joint law described by $f_{X,Y}$. Find the marginal laws of X and Y by giving their densities on all of $\mathbb{R}$.
(c) Use the joint density to calculate $\mathbb{P}(X + Y \leq 1)$.

1.4 Expectation as an Integral

Another important object you might recall from your introductory probability course is the **expectation** of a random variable. As before, the intuition is that the expectation is a probability-weighted average of the possible values of the random variable. Eventually, we will see that the law of large numbers gives us a precise sense in which the expectation is the long-run average of repeated independent observations of the random variable.

In this course, we shall also think about the expectation as an integral with respect to the law of the random variable, especially when the law is a mixture of point masses and a density.

Definition 1.11 (Expectation and Integrability)**.** Suppose that $X : \Omega \to \mathbb{R}$ has a law of the form

$$\mu_X(B) = \sum_{\substack{j \in J \\ x_j \in B}} p_j + \int_B f(x)\,dx, \qquad B \in \mathcal{B}(\mathbb{R}), \tag{1.4.1}$$

where J is finite or countable, the points x_j are distinct, the weights p_j and the piecewise-continuous function f are nonnegative, and $\sum_{j\in J} p_j + \int_{\mathbb{R}} f(x)\,dx = 1$. We call X **integrable** if

$$\int_{\mathbb{R}} |x|\,\mu_X(dx) = \sum_{j\in J} p_j|x_j| + \int_{\mathbb{R}} |x| f(x)\,dx < \infty. \tag{1.4.2}$$

In this case, the **expectation** or **mean** of X is

$$\mathbb{E}X = \int_{\mathbb{R}} x\,\mu_X(dx) = \sum_{j\in J} p_j x_j + \int_{\mathbb{R}} x f(x)\,dx. \tag{1.4.3}$$

One might ask why we require integrability in order to define expectation. The reason is that some probability laws do not have a well-defined expectation. For example, the Cauchy density is

$$f(x) = \frac{1}{\pi(1+x^2)}\,, \tag{1.4.4}$$

which is a well-defined probability density function, but the integral $\int_{\mathbb{R}} |x| f(x)\,dx$ diverges. A Cauchy random variable is therefore not integrable, and its expectation is *undefined*.

For the rest of this course, we will often write $\mu(dx)$ or $d\mu$ interchangeably when integrating with respect to a law μ. Both notations have their relative conveniences, and we will use whichever is better suited to the context.

We will also sometimes write an integral with respect to the probability measure $\mathbb{P}$ on the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. More generally, if $\varphi : \mathbb{R} \to \mathbb{R}$ is integrable with respect to μ_X, then the pushforward relationship gives

$$\int_{\mathbb{R}} \varphi(x)\,\mu_X(dx) = \int_{\Omega} \varphi\big(X(\omega)\big)\,\mathbb{P}(d\omega). \tag{1.4.5}$$

We will use these integrals through their standard properties, but we will not construct the general Lebesgue integral or develop its measure-theoretic foundations.

Writing the integral this way has a convenient application when we talk about the notion of almost sure events. We say that an event $A \in \mathcal{F}$ occurs *surely* if $A = \Omega$, and A occurs **almost surely** if $\mathbb{P}(A) = 1$. One might ask what is the difference between these two notions? Essentially, in probability theory we can "throw out" events of zero probability. This includes for integrals on zero probability events like A^c

$$\int_{A^c} X(\omega)\,\mathbb{P}(d\omega) = 0. \tag{1.4.6}$$

For example, if U is the uniform random variable on $[0,1]$, then the event that U takes any finite number of values has probability zero, and we can ignore these events. In fact, we can even ignore countably many events of probability zero, which is a consequence of the countable subadditivity of probability measures. For a lack of a better word, probabilists do not care about events of probability zero!

Example 1.12 (Expected Price with Bankruptcy)**.** Return to the stock-price law in (1.3.12), and assume that $\int_0^\infty xg(x)\,dx < \infty$. Its expected price is

$$\mathbb{E}S_t = q \cdot 0 + (1-q)\int_0^\infty xg(x)\,dx = (1-q)\int_0^\infty xg(x)\,dx. \tag{1.4.7}$$

The bankruptcy branch contributes $q \cdot 0 = 0$, while the positive-price branch contributes its average under g, weighted by its probability $1-q$.

As we hinted at earlier, we can also represent the probability of an event in terms of an integral, or equivalently an expectation. To define this precisely, we will introduce the indicator function or random variable. For an event $A \in \mathcal{F}$, its **indicator random variable** is defined as

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases} \tag{1.4.8}$$

Its law places mass $\mathbb{P}(A)$ at one and the remaining mass at zero. Consequently,

$$\begin{aligned} \mu_{\mathbf{1}_A} &= (1-\mathbb{P}(A))\delta_0 + \mathbb{P}(A)\delta_1, \\ \mathbb{E}\mathbf{1}_A &= \int_{\mathbb{R}} x\,\mu_{\mathbf{1}_A}(dx) = (1-\mathbb{P}(A))\cdot 0 + \mathbb{P}(A)\cdot 1 = \mathbb{P}(A). \end{aligned} \tag{1.4.9}$$

Thus, ordinary event probabilities are already a special case of expectation. This connection will become particularly useful when we define conditional expectation using the events whose information is available to us.

Proposition 1.13 (Basic Properties of Expectation)**.** *Let X and Y be integrable random variables, and let $a, b \in \mathbb{R}$. Then*

(a) The random variable $aX + bY$ is integrable, and expectation is linear:

$$\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y. \tag{1.4.10}$$

(b) If $X \le Y$ almost surely, then $\mathbb{E}X \le \mathbb{E}Y$.

(c) The expectation is bounded by the average absolute size:

$$|\mathbb{E}X| \le \mathbb{E}|X|. \tag{1.4.11}$$

(d) If $X \ge 0$ almost surely and $\mathbb{E}X = 0$, then $X = 0$ almost surely.

Proof. We prove the four claims in order.

(a) For every possible world $\omega \in \Omega$, the ordinary triangle inequality gives

$$|aX(\omega) + bY(\omega)| \le |a||X(\omega)| + |b||Y(\omega)|. \tag{1.4.12}$$

Averaging both sides preserves the inequality. Therefore,

$$\begin{aligned} &\int_\Omega |aX(\omega) + bY(\omega)|\,\mathbb{P}(d\omega) \\ &\qquad \le |a|\int_\Omega |X(\omega)|\,\mathbb{P}(d\omega) + |b|\int_\Omega |Y(\omega)|\,\mathbb{P}(d\omega) < \infty. \end{aligned} \tag{1.4.13}$$

The final quantity is finite because X and Y are integrable, so (1.4.13) proves that $aX+bY$ is integrable. We may now split the integral and pull out the constants:

$$\begin{aligned}\mathbb{E}[aX+bY] &= \int_\Omega \big(aX(\omega)+bY(\omega)\big)\,\mathbb{P}(d\omega)\\ &= a\int_\Omega X(\omega)\,\mathbb{P}(d\omega) + b\int_\Omega Y(\omega)\,\mathbb{P}(d\omega)\\ &= a\mathbb{E}X + b\mathbb{E}Y. \end{aligned} \tag{1.4.14}$$

(b) The assumption $X \le Y$ almost surely means that $Y - X \ge 0$, except possibly on an event of probability zero. That event has no effect on the integral, and averaging nonnegative values cannot give a negative result. Part (a) therefore gives

$$0 \le \mathbb{E}[Y-X] = \mathbb{E}Y - \mathbb{E}X. \tag{1.4.15}$$

Rearranging proves that $\mathbb{E}X \le \mathbb{E}Y$.

(c) Because X is integrable, $|X|$ is also integrable. For every $\omega \in \Omega$,

$$-|X(\omega)| \le X(\omega) \le |X(\omega)|. \tag{1.4.16}$$

Applying part (b) to both inequalities and part (a) to the leftmost expectation gives

$$-\mathbb{E}|X| = \mathbb{E}[-|X|] \le \mathbb{E}X \le \mathbb{E}|X|. \tag{1.4.17}$$

Thus $\mathbb{E}X$ lies between the negative and positive values of $\mathbb{E}|X|$, which is exactly (1.4.11).

(d) Suppose that $X \ge 0$ almost surely and $\mathbb{E}X = 0$. For each integer $n \ge 1$, let

$$A_n = \{\omega \in \Omega : X(\omega) \ge 1/n\}. \tag{1.4.18}$$

The interval $[1/n, \infty)$ is a Borel set, so A_n is an event by the measurability of X. On A_n, the random variable X is at least $1/n$. Outside A_n, the indicator is zero while X is nonnegative, except possibly on an event of probability zero. Hence

$$X \ge \frac{1}{n}\mathbf{1}_{A_n} \qquad \text{almost surely.} \tag{1.4.19}$$

Parts (a) and (b), together with (1.4.9), now give

$$0 = \mathbb{E}X \ge \frac{1}{n}\mathbb{E}\mathbf{1}_{A_n} = \frac{1}{n}\mathbb{P}(A_n) \ge 0. \tag{1.4.20}$$

It follows that $\mathbb{P}(A_n) = 0$ for every n. Every positive value is at least $1/n$ for some sufficiently large n, so

$$\{X > 0\} = \bigcup_{n=1}^{\infty} A_n. \tag{1.4.21}$$

Countable subadditivity then gives

$$\mathbb{P}(X > 0) \le \sum_{n=1}^{\infty} \mathbb{P}(A_n) = 0. \tag{1.4.22}$$

The assumption $X \ge 0$ almost surely also gives $\mathbb{P}(X < 0) = 0$. Therefore,

$$\mathbb{P}(X \ne 0) \le \mathbb{P}(X < 0) + \mathbb{P}(X > 0) = 0, \tag{1.4.23}$$

which says exactly that $X = 0$ almost surely.

□

These properties are the familiar rules for sums and integrals applied to general random variables. In particular, linearity of expectation does *not* require X and Y to be independent, a concept that you have probably seen in an introductory course and we will review soon.

One of the most useful applications of linearity and indicator functions is splitting up a random variable or expectation into branches of mutually exclusive (or disjoint) events. More precisely, if $A_1, \dots, A_m$ are disjoint events whose union is Ω, and

$$X = \sum_{i=1}^{m} X \mathbf{1}_{A_i}, \tag{1.4.24}$$

and linearity gives

$$\mathbb{E}X = \sum_{i=1}^{m} \mathbb{E}[X \mathbf{1}_{A_i}]. \tag{1.4.25}$$

In a special case, if $X = x_i$ is constant on A_i, then each expectation in the sum simplifies further to $x_i \mathbb{P}(A_i)$. This type of calculation is particularly useful later when we separate a complex expectation into simpler parts to control.

The bankruptcy calculation in Example 1.12 follows the same logic: the bankruptcy branch contributes zero, while the surviving branch contributes its probability times its conditional average.

Exercise 1.9 (Expectation Under a Mixed Law)**.** Let $q \in [0,1]$ and $r > 0$. Suppose that X has law

$$\mu_X(B) = q\delta_0(B) + (1-q) \int_{B \cap (0,\infty)} r e^{-rx}\, dx, \qquad B \in \mathcal{B}(\mathbb{R}). \tag{1.4.26}$$

(a) Verify that the displayed law has total mass one.
(b) Show that X is integrable and compute $\mathbb{E}X$.

Exercise 1.10 (Linearity and Fixed Points)**.** A deck of $n \geq 2$ cards labelled $1, \dots, n$ is shuffled uniformly among all $n!$ possible orders. Let N be the number of cards that finish in their original positions, and let I_i be the indicator that card i finishes in position i.

(a) Explain why $N = \sum_{i=1}^{n} I_i$.
(b) Compute $\mathbb{P}(I_i = 1)$ and $\mathbb{E}I_i$ for each i.
(c) Use linearity of expectation to compute $\mathbb{E}N$.

Exercise 1.11 (A Finite Tail-Sum Identity)**.** Let m be a positive integer, and let N be a random variable taking values in $\{0, 1, \dots, m\}$.

(a) Verify the pointwise identity

$$N = \sum_{k=1}^{m} \mathbf{1}_{\{N \geq k\}}. \tag{1.4.27}$$

(b) Use linearity and the expectation of an indicator to show that

$$\mathbb{E}N = \sum_{k=1}^{m} \mathbb{P}(N \geq k). \tag{1.4.28}$$

(c) Let $m = 3$ and suppose that

$$\mathbb{P}(N = 0) = 0.1, \quad \mathbb{P}(N = 1) = 0.2, \quad \mathbb{P}(N = 2) = 0.3, \quad \mathbb{P}(N = 3) = 0.4. \tag{1.4.29}$$

Compute $\mathbb{E}N$ using the identity from part (b).

1.5 Independence

Recall that the joint law of two random variables X and Y is a probability measure $\mu_{X,Y}$ on $\mathbb{R}^2$. As we have previously mentioned, knowing the marginal laws μ_X and μ_Y does not usually determine this joint law. Independence is most naturally a statement that two sources of information reveal nothing about one another, so we begin with σ-algebras.

Definition 1.14 (Independent σ-Algebras and Random Variables)**.** Let $\mathcal{G}_1, \ldots, \mathcal{G}_n$ be sub-σ-algebras of $\mathcal{F}$. They are **independent** if, for every choice $A_i \in \mathcal{G}_i$,

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \prod_{i=1}^{n} \mathbb{P}(A_i). \tag{1.5.1}$$

An arbitrary family of sub-σ-algebras is independent if every finite subfamily is independent.

In particular, two sub-σ-algebras $\mathcal{G}$ and $\mathcal{H}$ are independent precisely when $\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C)$ for every $A \in \mathcal{G}$ and $C \in \mathcal{H}$.

The random variables $X_1, \ldots, X_n$ are **independent** when $\sigma(X_1), \ldots, \sigma(X_n)$ are independent in the preceding sense. Equivalently, for every choice of Borel sets $B_1, \ldots, B_n \in \mathcal{B}(\mathbb{R})$,

$$\mu_{X_1,\ldots,X_n}(B_1 \times \cdots \times B_n) = \prod_{i=1}^{n} \mu_{X_i}(B_i). \tag{1.5.2}$$

A sequence $(X_i)_{i \geq 1}$ is independent if every finite subcollection is independent.

For two random variables, (1.5.2) can be written more familiarly as

$$\mathbb{P}(X \in B_1,\, Y \in B_2) = \mathbb{P}(X \in B_1)\mathbb{P}(Y \in B_2), \qquad B_1, B_2 \in \mathcal{B}(\mathbb{R}). \tag{1.5.3}$$

A loose intuitive way to think about independence is in terms of the area of rectangles. Here the probability $\mathbb{P}(X \in B_1,\, Y \in B_2)$ is like the area of a rectangle, and the length is $\mathbb{P}(X \in B_1)$ and the width is $\mathbb{P}(Y \in B_2)$. Obviously, the weight of the probability is not uniformly distributed across the rectangle, but the analogy is still useful to keep in mind, as probabilities are really a type of "area" in a more general sense.

Similarly, two events $A, C \in \mathcal{F}$ are **independent** when the σ-algebras $\sigma(A)$ and $\sigma(C)$ are independent. Equivalently, $\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C)$, which is also equivalent to the independence of their indicators $\mathbf{1}_A$ and $\mathbf{1}_C$.

Example 1.15 (Detecting Dependence from a Joint Law)**.** In Exercise 1.7,

$$\mathbb{P}(X = 0, Y = 0) = 0.1, \qquad \mathbb{P}(X = 0)\mathbb{P}(Y = 0) = 0.3 \cdot 0.4 = 0.12. \tag{1.5.4}$$

The factorization fails for this one pair of sets, so we say X and Y are **dependent**.

Independence is especially useful for expectation because it allows a product involving two random variables to separate into two ordinary averages.

Proposition 1.16 (Expectation of a Product)**.** *Let X and Y be independent. If X, Y, and XY are integrable, then*

$$\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y. \tag{1.5.5}$$

Proof. Independence says that the joint law separates into the two marginal laws. The integrability assumptions allow us to evaluate the integral one coordinate at a time. For the point masses and densities used in this course, this is the familiar separation of a double sum or an iterated integral. Therefore,

$$\begin{aligned} \mathbb{E}[XY] &= \int_{\mathbb{R}} \int_{\mathbb{R}} xy \, \mu_Y(dy) \, \mu_X(dx) \\ &= \left(\int_{\mathbb{R}} x \, \mu_X(dx) \right) \left(\int_{\mathbb{R}} y \, \mu_Y(dy) \right) = \mathbb{E}X\mathbb{E}Y. \end{aligned} \tag{1.5.6}$$

□

The converse is not true: the single numerical identity in (1.5.5) does not imply independence. The following finite example shows how different these two properties are.

Example 1.17 (Factoring Expectation Does Not Imply Independence)**.** Let X be uniformly distributed on $\{-1, 0, 1\}$, and let $Y = X^2$. Then

$$\mathbb{E}X = 0, \qquad \mathbb{E}Y = \frac{2}{3}, \qquad \mathbb{E}[XY] = \mathbb{E}X^3 = 0. \tag{1.5.7}$$

Thus $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$. However, Y is completely determined by X, and

$$\mathbb{P}(X = 0, Y = 0) = \frac{1}{3} \neq \frac{1}{9} = \mathbb{P}(X = 0)\mathbb{P}(Y = 0). \tag{1.5.8}$$

Therefore X and Y are dependent even though their expectation of a product factors.

Furthermore, transforming X using only X, and Y using only Y, cannot create dependence between them. The following proposition makes this precise.

Proposition 1.18 (Functions of Independent Random Variables)**.** *Suppose that X and Y are independent, and let $g, h : \mathbb{R} \to \mathbb{R}$ be measurable functions. Then $g(X)$ and $h(Y)$ are independent.*

Proof. For any Borel sets $B_1, B_2 \in \mathcal{B}(\mathbb{R})$, the preimages $g^{-1}(B_1)$ and $h^{-1}(B_2)$ are Borel because g and h are measurable. Independence of X and Y therefore gives

$$\begin{aligned} &\mathbb{P}\big(g(X) \in B_1,\, h(Y) \in B_2\big) \\ &\quad = \mathbb{P}\big(X \in g^{-1}(B_1),\, Y \in h^{-1}(B_2)\big) \\ &\quad = \mathbb{P}\big(X \in g^{-1}(B_1)\big)\mathbb{P}\big(Y \in h^{-1}(B_2)\big) \\ &\quad = \mathbb{P}\big(g(X) \in B_1\big)\mathbb{P}\big(h(Y) \in B_2\big). \end{aligned} \tag{1.5.9}$$

This is exactly the factorization required for $g(X)$ and $h(Y)$ to be independent. □

Repeated observations are often modeled as random variables that are both independent and governed by the same law.

Definition 1.19 (Independent and Identically Distributed Random Variables)**.** A sequence $(X_i)_{i\geq 1}$ is **independent and identically distributed**, abbreviated **i.i.d.**, if the random variables are independent and X_i has the same law as X_1 for every $i \geq 1$.

The two parts of the definition have different roles: independence describes how the variables relate to one another, while identical distribution says that each individual observation follows the same probability law. We will be using this setting for the well known laws of large numbers and central limit theorem.

We will sometimes also say X and Y are equal in distribution, denoted with $X \stackrel{d}{=} Y$ to indicate that X and Y have the same law.

Exercise 1.12 (Checking Independence from a Joint Law)**.** Suppose that X and Y take values in $\{0, 1\}$ and have the joint law shown in the following table.

	$Y = 0$	$Y = 1$
$X = 0$	0.12	0.18
$X = 1$	0.28	0.42

(a) Find the marginal laws of X and Y.
(b) Determine whether X and Y are independent. Justify your answer by checking the four possible pairs of values.

Exercise 1.13 (Independent Repetitions)**.** Let $n \geq 2$, and let $I_1, \ldots, I_n$ be independent random variables taking values in $\{0, 1\}$, with $\mathbb{P}(I_i = 1) = p$ and $\mathbb{P}(I_i = 0) = 1 - p$ for every i. Calculate the probability that

(a) all of the random variables equal zero;
(b) at least one of the random variables equals one;
(c) exactly one of the random variables equals one.

1.6 Variance and Probability Bounds

Recall the definition of variance and covariance from your introductory probability course.

Definition 1.20 (Variance and Covariance)**.** Suppose that X and Y have finite second moments, meaning that $\mathbb{E}X^2 < \infty$ and $\mathbb{E}Y^2 < \infty$. The **variance** of X and the **covariance** of X and Y are defined by

$$\operatorname{Var}(X) = \mathbb{E}\left(X - \mathbb{E}X\right)^2, \tag{1.6.1}$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}\left[\left(X - \mathbb{E}X\right)\left(Y - \mathbb{E}Y\right)\right]. \tag{1.6.2}$$

The variance is always nonnegative. Moreover, Proposition 1.13 shows that $\operatorname{Var}(X) = 0$ exactly when $X = \mathbb{E}X$ (is a constant) almost surely.

The following familiar identities are the ones we will use most often.

Proposition 1.21 (Basic Variance and Covariance Identities)**.** *Suppose that $n \geq 2$, that $X_1, \ldots, X_n$ have finite second moments, and let $a, b \in \mathbb{R}$. Then*

$$\operatorname{Var}(X_1) = \mathbb{E}X_1^2 - (\mathbb{E}X_1)^2, \tag{1.6.3}$$

$$\operatorname{Cov}(X_1, X_2) = \mathbb{E}[X_1 X_2] - \mathbb{E}X_1 \mathbb{E}X_2, \tag{1.6.4}$$

$$\operatorname{Var}(aX_1 + b) = a^2 \operatorname{Var}(X_1), \tag{1.6.5}$$

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \operatorname{Cov}(X_i, X_j). \tag{1.6.6}$$

Proof. Expanding the products in (1.6.1) and (1.6.2), and then using linearity of expectation, gives the first two identities. For the affine identity, linearity gives

$$aX_1 + b - \mathbb{E}[aX_1 + b] = a\left(X_1 - \mathbb{E}X_1\right). \tag{1.6.7}$$

Squaring both sides and taking expectations proves (1.6.5). Finally,

$$\sum_{i=1}^{n} X_i - \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \left(X_i - \mathbb{E}X_i\right). \tag{1.6.8}$$

Expanding the square of the right-hand side and taking expectations gives (1.6.6). □

Independence significantly simplifies the expression for the variance of a sum.

Corollary 1.22 (Variance of an Independent Sum)**.** *If $n \geq 2$ and $X_1, \ldots, X_n$ are independent and have finite second moments, then*

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i). \tag{1.6.9}$$

Proof. The bound $2|X_i X_j| \leq X_i^2 + X_j^2$ shows that $X_i X_j$ is integrable. For $i \neq j$, independence and Proposition 1.16 give $\mathbb{E}[X_i X_j] = \mathbb{E}X_i \mathbb{E}X_j$. Hence (1.6.4) gives $\operatorname{Cov}(X_i, X_j) = 0$, and the result follows from (1.6.6). □

Random variables with covariance zero are called **uncorrelated**, but uncorrelated random variables need not be independent. For random variables X, Y with finite second moments, we define the **correlation coefficient** $\rho(X, Y)$ as the normalized covariance

$$\rho(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}}. \tag{1.6.10}$$

Variance becomes particularly useful when it is turned into a bound on the probability of a large deviation. We begin with an even simpler bound. A nonnegative random variable cannot be large very often unless its expectation is also large.

Proposition 1.23 (Markov's Inequality)**.** *If X is a nonnegative integrable random variable and $a > 0$, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}. \tag{1.6.11}$$

Proof. In every possible world, i.e. for all $\omega \in \Omega$,

$$X(\omega) \geq a\mathbf{1}_{\{X(\omega)\geq a\}}. \tag{1.6.12}$$

Therefore taking expectations we have

$$\mathbb{E}X \geq a\,\mathbb{E}\mathbf{1}_{\{X\geq a\}} = a\,\mathbb{P}(X \geq a), \tag{1.6.13}$$

where we used monotonicity of expectations. □

Applying Markov's inequality to the squared distance from the mean gives the most common elementary variance bound.

Corollary 1.24 (Chebyshev's Inequality)**.** *If X has a finite second moment and $\varepsilon > 0$, then*

$$\mathbb{P}\left(|X - \mathbb{E}X| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}. \tag{1.6.14}$$

Proof. Apply Markov's inequality to $(X - \mathbb{E}X)^2$ with threshold ε^2. □

Remark 1.25. Markov's inequality was actually first discovered by Chebyshev (Markov's teacher), and Chebyshev's inequality was first discovered by Bienaymé. So perhaps don't take naming conventions too seriously.

The importance of these identities becomes visible when we average repeated observations, in particular in establishing the well known law of large numbers.

Theorem 1.26 (Weak Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$, and define their sample average by*

$$S_n = \frac{1}{n}\sum_{i=1}^{n} X_i. \tag{1.6.15}$$

Then, for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left(|S_n - \mu| \geq \varepsilon\right) = 0. \tag{1.6.16}$$

Proof. Linearity of expectation gives

$$\mathbb{E}S_n = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}X_i = \mu. \tag{1.6.17}$$

For $n \geq 2$, (1.6.5) and Corollary 1.22 give

$$\mathrm{Var}(S_n) = \frac{1}{n^2}\,\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\sigma^2}{n}. \tag{1.6.18}$$

Chebyshev's inequality therefore gives

$$0 \leq \mathbb{P}\left(|S_n - \mu| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0. \tag{1.6.19}$$

Since probabilities are nonnegative, this proves (1.6.16). □

We recall it is common to interpret the expectation as an average, and the law of large numbers formalizes this intuition. We will eventually discuss why this result is considered "weak", and we will introduce a "strong" version later.

Exercise 1.14 (Constructing an Uncorrelated Residual)**.** Let X and Y have finite second moments, with $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = 3$.

(a) Compute the correlation coefficient $\rho(X, Y)$.
(b) For $a \in \mathbb{R}$, define $Z_a = Y - aX$. Compute $\mathrm{Cov}(X, Z_a)$ as a function of a, and choose a so that X and Z_a are uncorrelated.
(c) Let $Z = Z_{3/4}$. Compute $\mathrm{Var}(Z)$, and use $Y = \frac{3}{4}X + Z$ and $\mathrm{Cov}(X, Z) = 0$ to verify the variance decomposition

$$\mathrm{Var}(Y) = \mathrm{Var}\left(\frac{3}{4}X\right) + \mathrm{Var}(Z). \tag{1.6.20}$$

Exercise 1.15 (A Finite-Variance Accuracy Guarantee)**.** Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean 10 and variance 9, and let $S_n = n^{-1}\sum_{i=1}^{n} X_i$.

(a) Compute $\mathbb{E}S_n$ and $\mathrm{Var}(S_n)$.
(b) Use Chebyshev's inequality to show that $\mathbb{P}(|S_n - 10| \geq 1) \leq 9/n$.
(c) Find the smallest integer n for which this bound is at most 0.05.
(d) For an arbitrary $\varepsilon > 0$, derive a bound for $\mathbb{P}(|S_n - 10| \geq \varepsilon)$ and use it to verify the conclusion of the weak law of large numbers in this example.

1.7 Modes of Convergence

In calculus, we say that a sequence of numbers $x_n \in \mathbb{R}$ converges to a limit x^* if for every $\varepsilon > 0$, there exists an N such that $|x_n - x^*| < \varepsilon$ for all $n \geq N$. We will often use the notations $x_n \to x^*$ or $x_n \xrightarrow{n\to\infty} x^*$ to denote this convergence. Here we will consider several different notions of convergence for random variables.

To start, we will recall that random variables are functions $X : \Omega \to \mathbb{R}$. The first notion of convergence for functions is pointwise convergence, which is the same as ordinary convergence of a sequence of numbers in every possible world $\omega \in \Omega$. In particular, we say a sequence of random variables X_n converges **surely** or pointwise to a random variable X if for every $\omega \in \Omega$, we have

$$\lim_{n\to\infty} X_n(\omega) = X(\omega). \tag{1.7.1}$$

This is the strongest notion of convergence, but it is often too strong to be useful in practice. For the rest of this section, we will introduce three weaker notions of convergence that are more useful in probability theory.

Definition 1.27 (Almost Sure Convergence)**.** A sequence of random variables X_n converges **almost surely** to a random variable X if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1. \tag{1.7.2}$$

We write this as $X_n \xrightarrow{\text{a.s.}} X$ or $X_n \to X$ a.s. (or with probability one).

This is essentially the same idea as almost sure events, where there may be worlds ω where $X_n(\omega)$ does not converge to $X(\omega)$, but the probability of those worlds is zero. To be precise, $\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \nrightarrow X(\omega)\}) = 0$. Again, the intuition is that probabilists don't care zero probability events, so essentially the strongest useful notion of convergence is almost sure convergence.

Next, we will weaken the notion of convergence even further such that it's possible that no single world ω has $X_n(\omega)$ converging to $X(\omega)$, but the probability of the worlds where X_n is far from X goes to zero.

Definition 1.28 (Convergence in Probability)**.** A sequence of random variables X_n converges **in probability** to a random variable X if for every $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| \geq \varepsilon) \to 0 \quad \text{as } n \to \infty. \tag{1.7.3}$$

We write this as $X_n \xrightarrow{\mathbb{P}} X$ or $X_n \to X$ in probability.

This is a strange notion of convergence. Let us consider the following example.

Example 1.29 (Convergence in Probability Without Almost-Sure Convergence)**.** Let $\Omega = (0, 1]$ with the uniform probability measure, and let $X = 0$. For each integer $m \geq 0$ and each $k \in \{0, 1, \ldots, 2^m - 1\}$, define

$$X_{2^m+k} = \mathbf{1}_{(k/2^m,(k+1)/2^m]}. \tag{1.7.4}$$

For each fixed m, the intervals in (1.7.4) partition $(0, 1]$ and each has probability 2^{-m}. Hence, if $2^m \leq n < 2^{m+1}$, then for every $0 < \varepsilon \leq 1$,

$$\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}(X_n = 1) = 2^{-m} \longrightarrow 0. \tag{1.7.5}$$

On the other hand, in every possible world $\omega \in (0, 1]$, exactly one random variable in each block $X_{2^m}, X_{2^m+1}, \ldots, X_{2^{m+1}-1}$ equals one. Once $m \geq 1$, at least one other random variable in the same block equals zero. Thus $X_n(\omega)$ has infinitely many ones and infinitely many zeros, so it does not converge in any possible world. In particular,

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 0, \tag{1.7.6}$$

and therefore X_n does not converge almost surely to X.

One might ask then, why is convergence in probability useful if it doesn't guarantee convergence in any particular world? This is indeed a strange phenomenon, but if all we care about is the probability of certain events in the limit, then convergence in probability is exactly what we need. Note, the key here is we *only care about probabilities*, and not the actual values of the random variables.

For example, suppose we want to know the probability of $X \in A$ but have access only to the sequence X_n, where A is a Borel set satisfying $\mathbb{P}(X \in \partial A) = 0$. Then convergence in probability guarantees that

$$\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A) \quad \text{as } n \to \infty. \tag{1.7.7}$$

The other thing to keep in mind is that *weaker* notions of convergence are often *easier* to establish. For practitioners, the strongest notions of convergence are typically not more useful, but they are often harder to prove.

In fact, we will continue to weak the notion of convergence even further, as it is often not ony the easiest to prove, but also the most useful in practice.

Definition 1.30 (Convergence in Distribution)**.** The random variables $X_1, X_2, \ldots$ and X need not be defined on the same probability space. We say that X_n **converges in distribution**, or **weakly**, to X if for all continuous and bounded test functions $\varphi : \mathbb{R} \to \mathbb{R}$, we have

$$\mathbb{E}\,\varphi(X_n) \longrightarrow \mathbb{E}\,\varphi(X). \tag{1.7.8}$$

We write this as $X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} \mu_X$ since only the law matters.

Immediately, we will note that not needing to be defined on the same probability space is a very significant weakening of the notion of convergence. Secondly, you may be familiar with the definition of convergence in distribution in terms of the cumulative distribution functions (CDFs), where $F_Y(x) = \mathbb{P}(Y \leq x)$ for a real-valued random variable Y. This definition says that

$$F_{X_n}(x) \longrightarrow F_X(x) \quad \text{at every point } x \text{ where } F_X \text{ is continuous.} \tag{1.7.9}$$

It turns out that these two definitions are equivalent, but our definition generalizes to random variables in higher dimensions.

Proposition 1.31. *Let $X_1, X_2, \ldots$ and X be real-valued random variables, and write $F_n = F_{X_n}$ and $F = F_X$. The test-function condition* (1.7.8)*, required for every bounded continuous function $\varphi : \mathbb{R} \to \mathbb{R}$, is equivalent to the CDF condition* (1.7.9)*.*

Proof Sketch. We approximate an indicator by continuous ramps in one direction, and a continuous test function by a step function in the other.

First assume the test-function condition, and fix a continuity point x of F. Choose continuous ramps g_δ^- and g_δ^+: the first equals one to the left of $x - \delta$, decreases linearly to zero at x, and stays zero afterward; the second makes the same transition between x and $x + \delta$. Then

$$\mathbb{E}g_\delta^-(X_n) \leq F_n(x) \leq \mathbb{E}g_\delta^+(X_n). \tag{1.7.10}$$

The two expectations converge by assumption, and their limits lie between $F(x-\delta)$ and $F(x+\delta)$. Given any $\varepsilon > 0$, first choose δ so that these CDF values are within ε of $F(x)$, then choose n large enough that both expectations are within ε of their limits. Thus $|F_n(x) - F(x)| < 2\varepsilon$, proving CDF convergence.

Conversely, assume the CDF condition and fix a bounded continuous test function φ. Choose a large interval $(a, b]$, with endpoints where F is continuous, so that the contribution to the expectation from outside it is arbitrarily small. Boundedness of φ makes this possible, and CDF convergence at a and b makes the tail contribution small for X_n as well, once n is sufficiently large.

On $[a, b]$, continuity lets us approximate φ uniformly by a step function with finitely many bins $(t_{j-1}, t_j]$, where $a = t_0 < t_1 < \cdots < t_m = b$ and the endpoints are continuity points of F. The probability of each bin converges because

$$\mathbb{P}(t_{j-1} < X_n \leq t_j) = F_n(t_j) - F_n(t_{j-1}) \longrightarrow F(t_j) - F(t_{j-1}). \tag{1.7.11}$$

Therefore the expectation of the step function, taken to be zero outside $(a, b]$, converges: it is just a finite weighted sum of these probabilities. The step-function approximation error and the tail contributions can be made arbitrarily small, giving $\mathbb{E}\varphi(X_n) \to \mathbb{E}\varphi(X)$. □

These three modes of convergence satisfy the following useful hierarchy.

Proposition 1.32 (Comparison of Modes of Convergence)**.** *Let $X_1, X_2, \dots$ and X be real-valued random variables on the same probability space. Then*

$$X_n \xrightarrow{\text{a.s.}} X \quad \Longrightarrow \quad X_n \xrightarrow{\mathbb{P}} X \quad \Longrightarrow \quad X_n \xrightarrow{d} X. \tag{1.7.12}$$

Neither implication can be reversed in general.

Proof. Assume first that $X_n \xrightarrow{\text{a.s.}} X$. Fix $\varepsilon > 0$ and, for $N \geq 1$, let $A_N = \bigcup_{n \geq N} \{|X_n - X| \geq \varepsilon\}$. The events A_N decrease, and their intersection is the event that $|X_n - X| \geq \varepsilon$ occurs infinitely often. Almost-sure convergence therefore gives $\mathbb{P}(\bigcap_{N \geq 1} A_N) = 0$. By continuity from above of probability, $\mathbb{P}(A_N) \to 0$. Since $\{|X_N - X| \geq \varepsilon\} \subseteq A_N$, it follows that

$$0 \leq \mathbb{P}(|X_N - X| \geq \varepsilon) \leq \mathbb{P}(A_N) \longrightarrow 0. \tag{1.7.13}$$

Hence $X_n \xrightarrow{\mathbb{P}} X$.

Now assume that $X_n \xrightarrow{\mathbb{P}} X$, and write $F_n = F_{X_n}$ and $F = F_X$. Fix a continuity point x of F and $\delta > 0$. The event inclusions

$$\{X \leq x - \delta\} \subseteq \{X_n \leq x\} \cup \{|X_n - X| \geq \delta\}, \qquad \{X_n \leq x\} \subseteq \{X \leq x + \delta\} \cup \{|X_n - X| \geq \delta\} \tag{1.7.14}$$

imply

$$F(x - \delta) - \mathbb{P}(|X_n - X| \geq \delta) \leq F_n(x) \leq F(x + \delta) + \mathbb{P}(|X_n - X| \geq \delta). \tag{1.7.15}$$

For fixed δ, the error probabilities tend to zero. Taking lower and upper limits in n, then letting $\delta \downarrow 0$, gives $F_n(x) \to F(x)$ by continuity of F at x. Thus Proposition 1.31 gives $X_n \xrightarrow{d} X$.

Finally, Example 1.29 shows that convergence in probability need not imply almost-sure convergence, while Exercise 1.16 shows that convergence in distribution need not imply convergence in probability. □

The weak law of large numbers in Theorem 1.26 can now be written simply as

$$S_n \xrightarrow{\mathbb{P}} \mu. \tag{1.7.16}$$

Thus the word "weak" refers to convergence in probability. The strong law of large numbers replaces this conclusion by almost-sure convergence. The central limit theorem answers a different question: it describes the size and shape of the fluctuations that remain around μ.

The central limit theorem can instead be understood by replacing one summand at a time with a Gaussian random variable having the same mean and variance. This is Lindeberg's replacement argument. We give the main part of the argument for smooth test functions, where the Taylor expansion is transparent, and discuss the remaining approximation step afterward.

Theorem 1.33 (Central Limit Theorem)**.** *Let $X_1, X_2, \dots$ be i.i.d. random variables with mean μ and variance $0 < \sigma^2 < \infty$. Then*

$$\frac{1}{\sqrt{n}} \sum_{j=1}^{n} \frac{X_j - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1). \tag{1.7.17}$$

Here $\mathcal{N}(0,1)$ denotes the standard normal distribution with density $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.

Proof Sketch. For simplicity of this sketch, we will assume $\mathbb{E}|X_1|^3 < \infty$ and consider only smooth test functions $\varphi : \mathbb{R} \to \mathbb{R}$ with compact support. By centering and scaling, we may also assume without loss of generality that $\mu = 0$ and $\sigma^2 = 1$. Thus $\mathbb{E}X_j = 0$ and $\mathbb{E}X_j^2 = 1$.

Let $Z_1, Z_2, \ldots$ be i.i.d. standard normal random variables, independent of the X_j. Thus X_j and Z_j have the same mean and variance. For $0 \le k \le n$, define the sum in which the first k variables are the original variables and the remaining ones are Gaussian:

$$W_{n,k} = \frac{1}{\sqrt{n}} \left(\sum_{j=1}^{k} X_j + \sum_{j=k+1}^{n} Z_j \right). \tag{1.7.18}$$

In particular, $W_{n,n}$ is the standardized sum in the theorem, while $W_{n,0} \sim \mathcal{N}(0,1)$. Recall the goal is to show $\mathbb{E}\,\varphi(W_{n,n}) \to \mathbb{E}\,\varphi(Z) = \mathbb{E}\,\varphi(W_{n,0})$ as $n \to \infty$, where $Z \sim \mathcal{N}(0,1)$. This means the difference of the expectations is what we want to control, and we start by writing it as a telescoping sum:

$$\mathbb{E}\,\varphi(W_{n,n}) - \mathbb{E}\,\varphi(W_{n,0}) = \sum_{k=1}^{n} \left(\mathbb{E}\,\varphi(W_{n,k}) - \mathbb{E}\,\varphi(W_{n,k-1}) \right). \tag{1.7.19}$$

Here note that we can write a Taylor expansion for $\varphi(W_{n,k})$ and $\varphi(W_{n,k-1})$ around the common partial sum $W_{n,k-1} - Z_k/\sqrt{n}$, which is independent of X_k and Z_k. Since φ is smooth with compact support, φ''' is bounded. Taylor's theorem therefore gives a constant $C < \infty$ such that, for all $w, x \in \mathbb{R}$ and $n \ge 1$,

$$\left| \varphi\left(w + \frac{x}{\sqrt{n}} \right) - \varphi(w) - \frac{x}{\sqrt{n}}\varphi'(w) - \frac{x^2}{2n}\varphi''(w) \right| \le \frac{C|x|^3}{n^{3/2}}. \tag{1.7.20}$$

In particular, we can apply this bound to $x = X_k$ and $x = Z_k$, and then take expectations, which gives us

$$|\mathbb{E}\varphi(W_{n,k}) - \mathbb{E}\varphi(W_{n,k-1})| \le \frac{C}{n^{3/2}} \left(\mathbb{E}|X_1|^3 + \mathbb{E}|Z_1|^3 \right). \tag{1.7.21}$$

In the above calculation, we are left only with the cubic remainders because the constant, linear, and quadratic terms cancel in expectation. The constant terms are identical, and the linear and quadratic terms cancel because X_k and Z_k have the same mean and variance and are independent of the common partial sum.

Replacing the variables one at a time and summing the n errors gives

$$|\mathbb{E}\varphi(W_{n,n}) - \mathbb{E}\varphi(W_{n,0})| \le \frac{C}{\sqrt{n}} \left(\mathbb{E}|X_1|^3 + \mathbb{E}|Z_1|^3 \right) \to 0. \tag{1.7.22}$$

Thus the expectations of every smooth compactly supported test function agree asymptotically for the standardized sum and the standard normal law. □

Remark 1.34 (The Approximation Step). The definition of convergence in distribution uses every bounded continuous test function, not only smooth ones with compact support. To extend the sketch to these test functions, we can first restrict our attention to a large bounded interval, which is okay because we can use Chebyshev's inequality to control the tails. On the bounded interval, a bounded continuous test function can be approximated uniformly by smooth functions. Combining these two approximations gives the convergence in (1.7.17).

To remove the bounded third moment condition, we can combine the cubic remainder bound for bounded increments with a quadratic bound for large increments. The expected contribution from the large increments can be made small using only the finite second moment, also giving us control over the Lindeberg error term.

Exercise 1.16 (Same Laws Need Not Mean Close Values)**.** Let X take the values zero and one with equal probability, and define $X_n = 1 - X$ for every n.

(a) Show that X_n and X have the same law, and conclude that $X_n \xrightarrow{d} X$.
(b) Show that X_n does not converge in probability to X.

Exercise 1.17 (Bernoulli Averages)**.** Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = 0) = 1 - p$, where $0 < p < 1$, and let $S_n = n^{-1} \sum_{i=1}^{n} X_i$.

(a) Compute $\mathbb{E}S_n$ and $\mathrm{Var}(S_n)$, and state the weak-law conclusion.
(b) State the central-limit conclusion for S_n.
(c) For a fixed $c \in \mathbb{R}$, find

$$\lim_{n\to\infty} \mathbb{P}\left(S_n \le p + \frac{c}{\sqrt{n}}\right). \tag{1.7.23}$$

Chapter 2

Conditional Expectation and Martingales

2.1 Conditional Expectation and Its Properties

In an introductory probability course, conditional probability is defined, when $\mathbb{P}(B) > 0$, as a ratio of probabilities

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}\,, \tag{2.1.1}$$

and the conditional expectation is defined with respect to this probability.

In this course, we will take a slightly more precise approach to conditional expectation, which is defined as a random variable rather than a single number. As we will see later, this approach has significant advantages when it comes to studying stochastic processes.

To define conditional expectations, we consider a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$.

Definition 2.1 (Conditional Expectation)**.** Let X be an integrable random variable. The **conditional expectation of X given $\mathcal{G}$** is an integrable, $\mathcal{G}$-measurable random variable Z such that

$$\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A] \qquad \text{for every } A \in \mathcal{G}. \tag{2.1.2}$$

We denote such a random variable by $\mathbb{E}[X \,|\, \mathcal{G}]$.

There are two technical points about this definition. First, it determines the conditional expectation only up to almost-sure equality. Indeed, if Z and Z' both satisfy the definition, then testing $Z - Z'$ on the $\mathcal{G}$-measurable events $\{Z > Z'\}$ and $\{Z < Z'\}$ shows that $Z = Z'$ almost surely. We therefore understand all identities involving conditional expectations as almost-sure identities. Second, we will assume throughout this course that the conditional expectation exists.

Lemma 2.2 (Almost-Sure Uniqueness)**.** *Suppose that Z and Z' both satisfy Definition 2.1 for the same random variable X and sub-σ-algebra $\mathcal{G}$. Then $Z = Z'$ almost surely.*

Proof. Since Z and Z' are $\mathcal{G}$-measurable, the event $A = \{Z > Z'\}$ belongs to $\mathcal{G}$. The testing identity gives

$$\mathbb{E}[(Z - Z')\mathbf{1}_A] = \mathbb{E}[Z\mathbf{1}_A] - \mathbb{E}[Z'\mathbf{1}_A] = 0. \tag{2.1.3}$$

The random variable $(Z - Z')\mathbf{1}_A$ is nonnegative, so it must vanish almost surely. Hence $\mathbb{P}(Z > Z') = 0$. Interchanging Z and Z' also gives $\mathbb{P}(Z' > Z) = 0$, and therefore $Z = Z'$ almost surely. □

We will interpret $\mathcal{G}$ as the information available to an observer: its events are exactly the questions that the observer can answer. A $\mathcal{G}$-measurable random variable uses only this information. The following are two extreme cases of $\mathcal{G}$ that can be insightful to consider.

Example 2.3 (Complete Information and No Information)**.** If X is $\mathcal{G}$-measurable, take $Z = X$. Then Z is $\mathcal{G}$-measurable and, for every $A \in \mathcal{G}$, $\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A]$. Thus Z satisfies the definition, and uniqueness gives

$$\mathbb{E}[\,X\,|\,\mathcal{G}] = X \qquad \text{almost surely.} \tag{2.1.4}$$

At the other extreme, let $\mathcal{G} = \{\varnothing, \Omega\}$ and take $Z = \mathbb{E}X$. Since Z is constant, it is $\mathcal{G}$-measurable. If $A = \varnothing$, then both sides of the testing identity are zero, while if $A = \Omega$, then $\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}Z = \mathbb{E}X = \mathbb{E}[X\mathbf{1}_A]$. Thus Z satisfies the definition, and uniqueness gives

$$\mathbb{E}[\,X\,|\,\mathcal{G}] = \mathbb{E}X \qquad \text{almost surely.} \tag{2.1.5}$$

Thus complete information leaves X unchanged, while no information reduces it to its ordinary average. In particular, this conditional expectation is a *constant*.

Taking $A = \Omega$ in the testing identity gives the **law of total expectation**

$$\mathbb{E}\left[\mathbb{E}[\,X\,|\,\mathcal{G}]\mathbf{1}_\Omega\right] = \mathbb{E}[X\mathbf{1}_\Omega] = \mathbb{E}X. \tag{2.1.6}$$

Conditioning changes the prediction according to the available information, but averaging the updated prediction recovers the original average.

The next example turns the abstract definition into a direct calculation.

Example 2.4 (The Finite Partition Case)**.** Suppose that $A_1, \ldots, A_m$ are disjoint events of positive probability whose union is Ω, and let $\mathcal{G} = \sigma(A_1, \ldots, A_m)$. Define

$$Z = \sum_{i=1}^{m} \frac{\mathbb{E}[X\mathbf{1}_{A_i}]}{\mathbb{P}(A_i)}\mathbf{1}_{A_i}. \tag{2.1.7}$$

This finite sum is integrable and constant on each cell, so it is $\mathcal{G}$-measurable. Every $B \in \mathcal{G}$ is a union $B = \bigcup_{i\in I} A_i$ for some $I \subseteq \{1, \ldots, m\}$, and

$$\begin{aligned}\mathbb{E}[Z\mathbf{1}_B] &= \sum_{i\in I} \frac{\mathbb{E}[X\mathbf{1}_{A_i}]}{\mathbb{P}(A_i)}\mathbb{P}(A_i) \\ &= \sum_{i\in I} \mathbb{E}[X\mathbf{1}_{A_i}] = \mathbb{E}[X\mathbf{1}_B].\end{aligned} \tag{2.1.8}$$

Therefore, Z satisfies both defining conditions and is almost surely equal to $\mathbb{E}[\,X\,|\,\mathcal{G}]$.

For an event A with $\mathbb{P}(A) > 0$, we use the scalar notation

$$\mathbb{E}[\,X\,|\,A] := \frac{\mathbb{E}[X\mathbf{1}_A]}{\mathbb{P}(A)}. \tag{2.1.9}$$

It is the average of X over the possible worlds in which A occurs. With this notation, the random variable in (2.1.7) is $\sum_{i=1}^{m} \mathbb{E}[X|\, A_i]\mathbf{1}_{A_i}$. The scalar conditional average on one event is therefore a value taken by a conditional-expectation random variable, rather than the primary definition.

Conditional probability is the special case obtained by taking an indicator. For $B \in \mathcal{F}$, define the $\mathcal{G}$-measurable random variable

$$\mathbb{P}(B \mid \mathcal{G}) := \mathbb{E}[\mathbf{1}_B|\, \mathcal{G}]. \tag{2.1.10}$$

When the available information is the occurrence of an event A with positive probability, the corresponding scalar value is the familiar ratio $\mathbb{P}(B \mid A) = \mathbb{P}(A \cap B)/\mathbb{P}(A)$.

Often the information comes from observing another random variable Y. Recall from (1.2.2) that the information revealed by Y is the sub-σ-algebra $\sigma(Y)$. The introductory conditional expectation often uses

$$\mathbb{E}[X|\, Y] = \mathbb{E}[X|\, \sigma(Y)], \tag{2.1.11}$$

We will try to avoid using this notation and stick to σ-algebra conventions.

If Y takes exactly the distinct values $y_1, \ldots, y_m$, each with positive probability, the finite-partition formula gives

$$\mathbb{E}[X|\, \sigma(Y)] = \sum_{i=1}^{m} \frac{\mathbb{E}[X\mathbf{1}_{\{Y=y_i\}}]}{\mathbb{P}(Y = y_i)}\mathbf{1}_{\{Y=y_i\}}. \tag{2.1.12}$$

The definition through $\sigma(Y)$ remains meaningful even when every event $\{Y = y\}$ has probability zero; a pointwise conditional expectation at $Y = y$ then requires additional interpretation and is not our definition.

Lemma 2.5 (Conditioning on an Independent Random Variable)**.** *If X is integrable and X and Y are independent, then*

$$\mathbb{E}[X|\, \sigma(Y)] = \mathbb{E}X \qquad \textit{almost surely.} \tag{2.1.13}$$

Proof. Since X and Y are independent, X and $\mathbf{1}_A$ are independent for every $A \in \sigma(Y)$. By Proposition 1.16,

$$\mathbb{E}[X\mathbf{1}_A] = \mathbb{E}[X]\mathbb{P}(A) = \mathbb{E}\left[\mathbb{E}[X]\mathbf{1}_A\right] \qquad \text{for every } A \in \sigma(Y). \tag{2.1.14}$$

The constant random variable $\mathbb{E}X$ is $\sigma(Y)$-measurable, so it satisfies the definition. The conclusion follows from almost-sure uniqueness. □

Example 2.6 (A Gaussian Signal and Independent Noise)**.** Let Y and Z be independent standard Gaussian random variables. For $\mu \in \mathbb{R}$, $\sigma > 0$, and $-1 < \rho < 1$, define

$$X = \mu + \sigma\left(\rho Y + \sqrt{1-\rho^2}\, Z\right). \tag{2.1.15}$$

The pair (X, Y) is jointly Gaussian, and ρ is their correlation. Observing Y reveals the first random term in (2.1.15), while the independent noise Z is replaced by its mean. Therefore,

$$\begin{aligned} \mathbb{E}[X|\, \sigma(Y)] &= \mu + \sigma\rho Y + \sigma\sqrt{1-\rho^2}\, \mathbb{E}[Z|\, \sigma(Y)] \\ &= \mu + \sigma\rho Y. \end{aligned} \tag{2.1.16}$$

Notice that Y survives the conditional expectation with respect to $\sigma(Y)$.

Remark 2.7. It's worth considering the intuition of conditional expectations here using the example above. Here when computing the conditional expectation of X given $\sigma(Y)$, we are essentially treating Y as a constant *inside* the conditional expectation. However, when we are *outside* the conditional expectation, Y remains random and $\mathbb{E}[X|\,\sigma(Y)]$ is a random variable with exactly the sources of randomness from Y preserved. In other words, we can summarize the above as

$$Y \text{ is } \begin{cases} \text{constant} & \text{inside of } \mathbb{E}[\cdot|\,\sigma(Y)], \\ \text{random} & \text{outside of } \mathbb{E}[\cdot|\,\sigma(Y)]. \end{cases} \tag{2.1.17}$$

Example 2.8 (An Unchanged Conditional Mean Without Independence)**.** Let X take the values $-1, 0, 1$ with probabilities $1/4, 1/2, 1/4$, respectively, and define $Y = X^2$. Observing Y tells us whether $X = 0$ or $|X| = 1$. When $Y = 1$, the values -1 and 1 are equally likely, so the conditional average is zero on both events in the partition generated by Y. Therefore,

$$\mathbb{E}[X|\,\sigma(Y)] = 0\mathbf{1}_{\{Y=0\}} + 0\mathbf{1}_{\{Y=1\}} = 0 = \mathbb{E}X. \tag{2.1.18}$$

Nevertheless,

$$\mathbb{P}(X = 0, Y = 0) = \frac{1}{2} \neq \frac{1}{4} = \mathbb{P}(X = 0)\mathbb{P}(Y = 0), \tag{2.1.19}$$

so X and Y are not independent. Thus the converse of Lemma 2.5 does not hold: preserving the mean after conditioning does not mean that Y carries no information about X.

Conditional expectation indeed is also an expectation, beyond being a random variable. Therefore it behaves like an ordinary expectation with respect to the first input.

Proposition 2.9. *Let X and Y be integrable, let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra, and let $a, b \in \mathbb{R}$.*

(a) Conditional expectation is linear:

$$\mathbb{E}[aX + bY|\,\mathcal{G}] = a\mathbb{E}[X|\,\mathcal{G}] + b\mathbb{E}[Y|\,\mathcal{G}] \qquad \textit{almost surely.} \tag{2.1.20}$$

(b) If $X \geq 0$ almost surely, then $\mathbb{E}[X|\,\mathcal{G}] \geq 0$ almost surely.

(c) If $X \leq Y$ almost surely, then

$$\mathbb{E}[X|\,\mathcal{G}] \leq \mathbb{E}[Y|\,\mathcal{G}] \qquad \textit{almost surely.} \tag{2.1.21}$$

Proof. For linearity, the right-hand side of (2.1.20) is integrable and $\mathcal{G}$-measurable. For every $A \in \mathcal{G}$, its tested average is

$$\begin{aligned} &\mathbb{E}\left[\left(a\mathbb{E}[X|\,\mathcal{G}] + b\mathbb{E}[Y|\,\mathcal{G}]\right)\mathbf{1}_A\right] \\ &\quad = a\mathbb{E}[X\mathbf{1}_A] + b\mathbb{E}[Y\mathbf{1}_A] = \mathbb{E}[(aX + bY)\mathbf{1}_A]. \end{aligned} \tag{2.1.22}$$

Thus it satisfies the definition of $\mathbb{E}[aX + bY|\,\mathcal{G}]$.

For positivity, write $Z = \mathbb{E}[X|\,\mathcal{G}]$ and suppose that $X \geq 0$ almost surely. For $k \geq 1$, the event $A_k = \{Z \leq -1/k\}$ belongs to $\mathcal{G}$, and the testing identity gives

$$0 \leq \mathbb{E}[X\mathbf{1}_{A_k}] = \mathbb{E}[Z\mathbf{1}_{A_k}] \leq -\frac{1}{k}\mathbb{P}(A_k). \tag{2.1.23}$$

Hence $\mathbb{P}(A_k) = 0$ for every k. Since $\{Z < 0\} = \bigcup_{k\geq 1} A_k$, we obtain $Z \geq 0$ almost surely.

Finally, if $X \leq Y$ almost surely, then $Y - X \geq 0$ almost surely. Positivity and linearity give $\mathbb{E}[Y|\,\mathcal{G}] - \mathbb{E}[X|\,\mathcal{G}] \geq 0$ almost surely, which is (2.1.21). □

For square-integrable random variables, we can also interpret conditional expectation as an *orthogonal projection*.

Let $L^2 = L^2(\Omega, \mathcal{F}, \mathbb{P})$ be the space of random variables X with finite second moment $\|X\|_{L^2}^2 := \mathbb{E}X^2 < \infty$. We can equip L^2 with the inner product $\langle U, V \rangle = \mathbb{E}[UV]$, and let $L^2(\mathcal{G})$ be the closed linear subspace consisting of the $\mathcal{G}$-measurable random variables in L^2.

In general, we can let L^p be the space of random variables X with finite p-th moment $\|X\|_{L^p}^p := \mathbb{E}|X|^p < \infty$. In particular, we can say that X is *integrable* if $X \in L^1$, and X is *square-integrable* if $X \in L^2$.

Lemma 2.10 (Conditional Expectation as an Orthogonal Projection)**.** *If $X \in L^2$, then $\mathbb{E}[X \,|\, \mathcal{G}]$ is the orthogonal projection of X onto $L^2(\mathcal{G})$. That is,*

$$\mathbb{E}\left(X - \mathbb{E}[X \,|\, \mathcal{G}]\right)^2 = \min_{Y \in L^2(\mathcal{G})} \mathbb{E}(X - Y)^2. \tag{2.1.24}$$

Proof. Let M be a minimizer of the L^2 norm, i.e.

$$M \in \arg\min_{Y \in L^2(\mathcal{G})} \mathbb{E}(X - Y)^2. \tag{2.1.25}$$

We will use without proof the fact that the minimizer of the L^2 norm is unique and it's characterized by the orthogonality condition (see the projection theorem in Hilbert spaces)

$$\langle X - M, W \rangle = 0\,, \quad \text{for every } W \in L^2(\mathcal{G}). \tag{2.1.26}$$

For every $A \in \mathcal{G}$, we may take $W = \mathbf{1}_A$ to obtain $\mathbb{E}[M\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A]$. Thus M satisfies Definition 2.1, and uniqueness gives $M = \mathbb{E}[X \,|\, \mathcal{G}]$ almost surely. Therefore the conditional expectation is the unique minimizer in (2.1.24). □

Another property of conditional expectation is that it is based on a sense of "information" that is available. As we will see in the next couple of results, we can really see the conditioning σ-algebra $\mathcal{G}$ as information.

Suppose that Y is $\mathcal{G}$-measurable. Once the information in $\mathcal{G}$ is available, the value of Y is known, so it should behave like a constant inside a conditional expectation.

Proposition 2.11 (Taking Out What Is Known)**.** *Let $X \in L^1$ and let Y be a $\mathcal{G}$-measurable random variable such that $XY \in L^1$. Then*

$$\mathbb{E}[XY \,|\, \mathcal{G}] = Y\mathbb{E}[X \,|\, \mathcal{G}] \qquad \textit{almost surely.} \tag{2.1.27}$$

Proof sketch. Assume for simplicity that $X \in L^2$ and Y is bounded. Then both XY and $Y\mathbb{E}[X \,|\, \mathcal{G}]$ are square-integrable, and the latter belongs to $L^2(\mathcal{G})$, which we will identify as the projection of XY onto $L^2(\mathcal{G})$.

Let $W \in L^2(\mathcal{G})$. The product YW also belongs to $L^2(\mathcal{G})$, so the orthogonality condition in (2.1.26) gives

$$0 = \langle X - \mathbb{E}[X \,|\, \mathcal{G}], YW \rangle = \langle XY - Y\mathbb{E}[X \,|\, \mathcal{G}], W \rangle \tag{2.1.28}$$

Therefore $Y\mathbb{E}[X \,|\, \mathcal{G}]$ is the orthogonal projection of XY onto $L^2(\mathcal{G})$. Finally, Lemma 2.10 implies $Y\mathbb{E}[X \,|\, \mathcal{G}]$ is also the conditional expectation of XY given $\mathcal{G}$ (under these additional assumptions). □

Remark 2.12. To remove both strengthenin of the assumptions in the proof sketch, we can use a standard truncation argument. Firstly, truncating both X and Y to be bounded, we can apply the above argument to obtain the result for bounded X and Y. Then, we can first take the limit as the truncation on X goes to infinity, and then take the limit as the truncation on Y goes to infinity, which would recover the same results.

The next rule compares predictions based on two levels of information. If $\mathcal{H} \subseteq \mathcal{G}$, then $\mathcal{G}$ contains at least as much information as $\mathcal{H}$. Conditioning first on $\mathcal{G}$ and then on $\mathcal{H}$ amounts to discarding the extra information. Essentially, the smallest σ-algebra $\mathcal{H}$ is the one that survives in the end.

Proposition 2.13 (Tower Property)**.** *Let $X \in L^1$ and let $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ be σ-algebras. Then*

$$\mathbb{E}[\mathbb{E}[X|\,\mathcal{G}]|\,\mathcal{H}] = \mathbb{E}[\mathbb{E}[X|\,\mathcal{H}]|\,\mathcal{G}] = \mathbb{E}[X|\,\mathcal{H}], \tag{2.1.29}$$

Proof sketch. Assume first that $X \in L^2$. Since $\mathcal{H} \subseteq \mathcal{G}$, we have $L^2(\mathcal{H}) \subseteq L^2(\mathcal{G})$. Projecting X first onto $L^2(\mathcal{G})$ and then onto the smaller subspace $L^2(\mathcal{H})$ is therefore the same as projecting X directly onto $L^2(\mathcal{H})$. This gives $\mathbb{E}[\mathbb{E}[X|\,\mathcal{G}]|\,\mathcal{H}] = \mathbb{E}[X|\,\mathcal{H}]$.

On the other hand, $\mathbb{E}[X|\,\mathcal{H}]$ already belongs to $L^2(\mathcal{H}) \subseteq L^2(\mathcal{G})$, so projecting it onto $L^2(\mathcal{G})$ leaves it unchanged. This proves the other equality. The result for integrable X follows by truncating X and passing to the L^1 limit. □

Taking $\mathcal{H} = \{\varnothing, \Omega\}$ in (2.1.29) recovers the law of total expectation. Thus that earlier identity is the no-information case of the tower property.

The pull-out property lets quantities determined by the current information behave like constants inside conditional expectations. The tower property keeps predictions consistent when we pass between nested levels of information. When these levels are indexed by time, they form a filtration, which is the setting for martingales.

Exercise 2.1 (Conditioning on a Two-Cell Partition)**.** Let $\Omega = \{\omega_1, \ldots, \omega_6\}$, with each outcome having probability $1/6$. Let $A = \{\omega_1, \omega_2\}$ and $\mathcal{G} = \sigma(A)$, and suppose that

$$\big(X(\omega_1), \ldots, X(\omega_6)\big) = (0, 2, 4, 6, 8, 10). \tag{2.1.30}$$

(a) List the events in $\mathcal{G}$.
(b) Use the finite-partition formula to find $\mathbb{E}[X|\,\mathcal{G}]$.
(c) Verify the testing identity for every event in $\mathcal{G}$ and check the law of total expectation.

Exercise 2.2 (Conditional Probability Is Random)**.** Let D be the outcome of a fair six-sided die. Set $A = \{D \le 3\}$, $B = \{D \in \{2, 4, 6\}\}$, and $\mathcal{G} = \sigma(A)$.

(a) Find the random variable $\mathbb{P}(B \mid \mathcal{G})$ by giving its value on A and on A^c.
(b) Verify its testing identity on A and A^c.
(c) Check that its ordinary expectation equals $\mathbb{P}(B)$.

Exercise 2.3 (Independent Noise and an Observed Signal)**.** Let U and V be independent random variables, each taking the values zero and one with equal probability, and let $\mathcal{G} = \sigma(V)$.

(a) Find $\mathbb{E}[U|\,\mathcal{G}]$ and explain why it is constant.

(b) Use the finite-partition formula to find $\mathbb{E}[U+V\,|\,\mathcal{G}]$.
(c) Explain which part of the answer remains random after conditioning and which part has been replaced by an average.

Exercise 2.4 (Conditional Absolute-Value Bound)**.** Let X be integrable and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra.

(a) Use monotonicity to prove that

$$|\mathbb{E}[X\,|\,\mathcal{G}]| \leq \mathbb{E}[|X|\,|\,\mathcal{G}] \qquad \text{almost surely.} \tag{2.1.31}$$

(b) Deduce that

$$\mathbb{E}|\mathbb{E}[X\,|\,\mathcal{G}]| \leq \mathbb{E}|X|. \tag{2.1.32}$$

Exercise 2.5 (The Conditional Residual)**.** Let X be an integrable random variable, let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-algebra, and set

$$R = X - \mathbb{E}[X\,|\,\mathcal{G}]. \tag{2.1.33}$$

(a) Show that $\mathbb{E}[R|\,\mathcal{G}] = 0$ almost surely.
(b) Show that $\mathbb{E}[RY] = 0$ for every bounded, $\mathcal{G}$-measurable random variable Y. Thus no bounded quantity determined by the available information has a nonzero expected product with the residual.

2.2 Filtrations and Martingales

Recall that we have referred to a σ-algebra $\mathcal{F}$ as the information available to an observer. In this section, we will consider a sequence of σ-algebras that represent the information available at different times. This will be directly useful when describing a sequence of random variables that evolves over time, in particular also when describing useful structure surrounding conditional expectations.

Definition 2.14 (Filtration and Natural Filtration)**.** A **filtration** on $(\Omega, \mathcal{F}, \mathbb{P})$ is a sequence $(\mathcal{F}_n)_{n\geq 0}$ of sub-σ-algebras of $\mathcal{F}$ such that

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}. \tag{2.2.1}$$

The quadruple $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\geq 0}, \mathbb{P})$ is called a **filtered probability space**.

For a process $(X_n)_{n\geq 0}$, its **natural filtration** is

$$\mathcal{F}_n^X = \sigma(X_0, \ldots, X_n), \qquad n \geq 0. \tag{2.2.2}$$

It records exactly the information contained in the observations of the process up to time n.

We won't always be working with a natural filtration, but it is a useful special case to study carefully. In general, a filtration may contain additional information that is not directly observed in the process itself.

Definition 2.15 (Adapted and Predictable Processes)**.** A process $(X_n)_{n\geq 0}$ is **adapted** to $(\mathcal{F}_n)_{n\geq 0}$ if X_n is $\mathcal{F}_n$-measurable for every $n \geq 0$. A process $(H_n)_{n\geq 1}$ is **predictable** if H_n is $\mathcal{F}_{n-1}$-measurable for every $n \geq 1$.

We should think about these two processes as follows. Imagine you are trading stocks on a daily basis, then

- An adapted process is like the *stock price* at the end of day n, which can only be determined after you observe the trading day n.
- A predictable process is like the *number of shares* you own before day n starts trading, which is only determined by the information available up to day $n-1$.

Note that all stochastic processes $(X_n)_{n\geq 0}$ are adapted to their natural filtration $(\mathcal{F}_n^X)_{n\geq 0}$, but generally speaking the space's filtration $(\mathcal{F}_n)_{n\geq 0}$ may contain more information than just the observations of the process. Remember, if a random variable X_n is $\mathcal{F}_n$-measurable, then it means the conditional expectation satisfies

$$\mathbb{E}[\,X_n|\,\mathcal{F}_n] = X_n\,, \qquad \text{almost surely,} \tag{2.2.3}$$

but note this does not apply for $\mathbb{E}[\,X_{n+1}|\,\mathcal{F}_n]$.

Example 2.16 (Information in a Random Walk)**.** Let $\xi_1, \xi_2, \ldots$ be independent random variables satisfying $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = 1/2$, and define

$$S_0 = 0, \qquad S_n = \sum_{k=1}^{n} \xi_k, \qquad \mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n) = \sigma(S_0, \ldots, S_n). \tag{2.2.4}$$

The equality of the two σ-algebras follows because each S_k is determined by $\xi_1, \ldots, \xi_k$, while $\xi_k = S_k - S_{k-1}$. The process $(S_n)_{n\geq 0}$ is adapted, and $H_n = \mathbf{1}_{\{S_{n-1}\geq 0\}}$ is predictable. In contrast, ξ_{n+1} is not known from $\mathcal{F}_n$.

One particularly useful class of adapted processes is the martingales, which are defined by a conditional expectation property.

Definition 2.17 (Martingales, Submartingales, and Supermartingales)**.** Let $(M_n)_{n\geq 0}$ be adapted to $(\mathcal{F}_n)_{n\geq 0}$, and suppose that every M_n is integrable. It is a **martingale** if

$$\mathbb{E}[\,M_{n+1}|\,\mathcal{F}_n] = M_n \qquad \text{almost surely for every } n \geq 0. \tag{2.2.5}$$

It is a **submartingale** if the left-hand side is at least ($\geq$) M_n almost surely, and a **supermartingale** if the left-hand side is at most ($\leq$) M_n almost surely.

In Example 2.16, independence and $\mathbb{E}\xi_{n+1} = 0$ give

$$\mathbb{E}[\,S_{n+1}|\,\mathcal{F}_n] = S_n + \mathbb{E}[\,\xi_{n+1}|\,\mathcal{F}_n] = S_n, \tag{2.2.6}$$

so the centered random walk is a martingale.

Historically, the name "martingale" was used for the gambling strategy that doubles the amount you bet each round after every loss, so that the next win recovers all previous losses plus a profit equal to the original stake. The next example explains how this betting strategy is related to martingales.

Example 2.18 (Predictable Betting on a Fair Game)**.** Fix a horizon N. Let $(M_n)_{0\leq n\leq N}$ be a martingale and let $(H_n)_{1\leq n\leq N}$ be a bounded predictable process. Interpret H_n as the stake chosen before the change $M_n - M_{n-1}$ is observed, and define the cumulative gain by

$$X_0 = 0, \qquad X_n = \sum_{k=1}^{n} H_k(M_k - M_{k-1}), \qquad 1 \leq n \leq N. \tag{2.2.7}$$

For $0 \le n < N$, the random variables H_{n+1} and X_n are $\mathcal{F}_n$-measurable. Taking out what is known gives

$$\begin{aligned}\mathbb{E}[X_{n+1}|\,\mathcal{F}_n] &= X_n + H_{n+1}\mathbb{E}[M_{n+1} - M_n|\,\mathcal{F}_n] \\ &= X_n.\end{aligned} \tag{2.2.8}$$

Thus $(X_n)_{0\le n\le N}$ is again a martingale.

To see the classical doubling strategy inside this construction, let $\xi_1, \dots, \xi_N$ be independent fair-coin payoffs, where $\xi_n = 1$ for a win and $\xi_n = -1$ for a loss. Set $M_0 = 0$, $M_n = \sum_{k=1}^{n} \xi_k$, $\mathcal{F}_0 = \{\varnothing, \Omega\}$, and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ for $n \ge 1$. The strategy stakes one unit on the first round, doubles after each loss, and stops betting after the first win. In symbols,

$$H_1 = 1, \qquad H_n = 2^{n-1}\mathbf{1}_{\{\xi_1=\cdots=\xi_{n-1}=-1\}}, \qquad 2 \le n \le N. \tag{2.2.9}$$

The event in (2.2.9) depends only on the previous outcomes, so the stakes are predictable. For example, suppose the first four rounds are losses and the fifth is a win. The evolution of the strategy is

Turn n	Bet size H_n	Outcome $\xi_n = M_n - M_{n-1}$	Net wealth X_n
1	1	-1	-1
2	2	-1	-3
3	4	-1	-7
4	8	-1	-15
5	16	1	1

The fifth-round gain of 16 offsets the accumulated loss of 15, leaving a net gain of one unit. More generally, if the first win occurs on round $n \le N$, then

$$X_n = -\sum_{k=1}^{n-1} 2^{k-1} + 2^{n-1} = 1, \tag{2.2.10}$$

which recovers all previous losses and earns one unit. If all N rounds are losses, however, then $X_N = -(2^N - 1)$. Since this happens with probability 2^{-N}, the expected terminal gain is

$$\left(1 - 2^{-N}\right) \cdot 1 + 2^{-N}\left[-\left(2^N - 1\right)\right] = 0. \tag{2.2.11}$$

Thus the small chance of a large loss exactly offsets the one-unit gain from every sequence in which a win occurs.

However, since $(X_n)_{0\le n\le N}$ is a martingale, this means the martingale betting strategy still yields *zero expected return* at every round!

Conditional expectation gives another basic construction. For any integrable random variable X, define $M_n = \mathbb{E}[X|\,\mathcal{F}_n]$. By definition, each M_n is integrable and $\mathcal{F}_n$-measurable. The tower property gives

$$\mathbb{E}[M_{n+1}|\,\mathcal{F}_n] = \mathbb{E}[\mathbb{E}[X|\,\mathcal{F}_{n+1}]|\,\mathcal{F}_n] = \mathbb{E}[X|\,\mathcal{F}_n] = M_n, \tag{2.2.12}$$

so $(M_n)_{n\ge0}$ is a martingale. As information accumulates, this process records the successive predictions of the same random variable.

This class of martingales is often called **Doob martingales**.

For a martingale with a finite horizon N, the repeated martingale property below gives

$$M_n = \mathbb{E}[\,M_N|\,\mathcal{F}_n], \qquad 0 \le n \le N. \tag{2.2.13}$$

Thus every finite-horizon martingale is a Doob martingale with terminal variable M_N. For an infinite-horizon martingale, representing every M_n by conditioning one terminal random variable requires additional hypotheses; we will return to this point later.

We will next explore several related properties of martingales.

Proposition 2.19 (Repeated Martingale Property)**.** *Let $0 \le m \le n$.*

(a) If $(M_n)_{n\ge0}$ is a martingale, then

$$\mathbb{E}[\,M_n|\,\mathcal{F}_m] = M_m \qquad \textit{almost surely.} \tag{2.2.14}$$

In particular, $\mathbb{E}M_n = \mathbb{E}M_0$ for every n.

(b) If $(X_n)_{n\ge0}$ is a submartingale, then $\mathbb{E}[X_n|\,\mathcal{F}_m] \ge X_m$ almost surely. If it is a supermartingale, the inequality is reversed. Consequently, the expectations of a submartingale are nondecreasing, while those of a supermartingale are nonincreasing.

Proof. The result is immediate when $m = n$. If $m < n$, the tower property and the one-step martingale condition give

$$\begin{aligned}\mathbb{E}[\,M_n|\,\mathcal{F}_m] &= \mathbb{E}[\mathbb{E}[\,M_n|\,\mathcal{F}_{n-1}]|\,\mathcal{F}_m] \\ &= \mathbb{E}[\,M_{n-1}|\,\mathcal{F}_m].\end{aligned} \tag{2.2.15}$$

Repeating this step gives (2.2.14). For a submartingale, the tower property followed by conditional monotonicity gives

$$\mathbb{E}[\,X_n|\,\mathcal{F}_m] \ge \mathbb{E}[\,X_{n-1}|\,\mathcal{F}_m] \ge \cdots \ge X_m, \tag{2.2.16}$$

and the supermartingale case is analogous. Taking ordinary expectations proves the remaining statements. □

Using the martingale property, the differences $D_n = M_n - M_{n-1}$ satisfy

$$\mathbb{E}[\,D_n|\,\mathcal{F}_{n-1}] = 0. \tag{2.2.17}$$

It turns out this property characterizes martingales.

Definition 2.20 (Martingale Difference Sequence)**.** A sequence $(D_n)_{n\ge1}$ adapted to $(\mathcal{F}_n)_{n\ge0}$ is a martingale difference sequence if D_n is integrable and

$$\mathbb{E}[\,D_n|\,\mathcal{F}_{n-1}] = 0 \qquad \text{almost surely for every } n \ge 1. \tag{2.2.18}$$

Proposition 2.21 (Martingales and Their Differences)**.** *An adapted process $(M_n)_{n\ge0}$ is a martingale if and only if its differences $D_n = M_n - M_{n-1}$ form a martingale difference sequence and M_0 is integrable and $\mathcal{F}_0$ measurable.*

Proof. A martingale has integrable differences. Conversely, if M_0 is integrable and $(D_n)_{n\ge1}$ is a martingale difference sequence, then $M_n = M_0 + \sum_{k=1}^{n} D_k$ is integrable for every n. Thus the conditional expectations below are well defined in either direction.

Since M_{n-1} is $\mathcal{F}_{n-1}$-measurable, linearity gives

$$\mathbb{E}[\,D_n|\,\mathcal{F}_{n-1}] = \mathbb{E}[\,M_n|\,\mathcal{F}_{n-1}] - M_{n-1}. \tag{2.2.19}$$

Thus the conditional mean in (2.2.18) is zero exactly when the martingale identity holds. □

Note that this is a representation theorem: every martingale can be reconstructed from its martingale differences. Here's an example where the increments are not iid. but they form martingale differences.

Example 2.22 (A Geometric Random Walk)**.** Let $\xi_1, \xi_2, \ldots$ be independent random variables satisfying $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = 1/2$, and let $\mathcal{F}_0 = \{\varnothing, \Omega\}$ and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ for $n \geq 1$. Define

$$M_0 = 1, \qquad M_n = \prod_{k=1}^{n} \left(1 + \frac{\xi_k}{2}\right), \qquad n \geq 1. \tag{2.2.20}$$

At each step, the process is multiplied by $3/2$ or $1/2$, each with probability $1/2$. Its differences are

$$D_n = M_n - M_{n-1} = \frac{1}{2} M_{n-1}\xi_n. \tag{2.2.21}$$

Note the differences D_n are not independent, since they share the common factor M_{n-1}. However, since M_{n-1} is $\mathcal{F}_{n-1}$-measurable while ξ_n is independent of $\mathcal{F}_{n-1}$ and has mean zero, we have

$$\mathbb{E}[\,D_n|\,\mathcal{F}_{n-1}] = \frac{1}{2} M_{n-1}\mathbb{E}[\,\xi_n|\,\mathcal{F}_{n-1}] = 0. \tag{2.2.22}$$

Thus $(D_n)_{n\geq 1}$ is a martingale difference sequence, and Proposition 2.21 shows that $(M_n)_{n\geq 0}$ is a martingale. This process is a geometric, or multiplicative, random walk: its successive changes are proportional to its current value.

We will also use the familiar principle that applying a convex function after averaging gives no larger a value than applying the function first and then averaging.

Proposition 2.23 (Jensen's Inequality)**.** *Let X be integrable and let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex. Then*

$$\varphi\left(\mathbb{E}X\right) \leq \mathbb{E}\varphi(X). \tag{2.2.23}$$

Proof. Set $m = \mathbb{E}X$. Since φ is convex, there is a supporting line at m: for some slope a, $\varphi(x) \geq \varphi(m) + a(x - m)$ for every x. Substituting X for x and taking expectations gives

$$\mathbb{E}\varphi(X) \geq \varphi(m) + a\left(\mathbb{E}X - m\right) = \varphi(m), \tag{2.2.24}$$

which is (2.2.23). □

Remark 2.24 (Conditional Jensen's Inequality). If X and $\varphi(X)$ are integrable, Jensen's inequality also holds with ordinary expectations replaced by conditional expectations:

$$\varphi\left(\mathbb{E}[\,X|\,\mathcal{G}]\right) \leq \mathbb{E}[\,\varphi(X)|\,\mathcal{G}] \qquad \text{almost surely.} \tag{2.2.25}$$

We will use this conditional form in exactly the same way as the usual inequality.

Example 2.25 (Convex Functions of Martingales)**.** Let $(M_n)_{n\geq 0}$ be a martingale and let φ be convex, with $\varphi(M_n)$ integrable for every n. Conditional Jensen's inequality gives

$$\mathbb{E}[\,\varphi(M_{n+1})|\,\mathcal{F}_n] \geq \varphi\left(\mathbb{E}[\,M_{n+1}|\,\mathcal{F}_n]\right) = \varphi(M_n). \tag{2.2.26}$$

Thus $(\varphi(M_n))_{n\geq 0}$ is a submartingale. In particular, $(|M_n|)_{n\geq 0}$ is a submartingale, and $(M_n^2)_{n\geq 0}$ is a submartingale when the M_n are square-integrable. For example, the centered random walk $(S_n)_{n\geq 0}$ in Example 2.16 is a martingale, while $(S_n^2)_{n\geq 0}$ is a submartingale. Although the conditional mean of a martingale stays fixed, its conditional size or squared size can increase.

Exercise 2.6 (Adaptedness, Predictability, and a Martingale)**.** Let $\xi_1, \xi_2, \ldots$ be independent random variables taking the values -1 and 1 with equal probability. Set $S_0 = 0$, $S_n = \sum_{k=1}^n \xi_k$, and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$.

(a) Prove that $\mathcal{F}_n = \sigma(S_0, \ldots, S_n)$.
(b) Show that $(S_n)_{n\geq 0}$ is adapted and that $H_n = \mathbf{1}_{\{S_{n-1}=0\}}$, $n \geq 1$, is predictable. Explain why ξ_{n+1} is not $\mathcal{F}_n$-measurable.
(c) Show that $M_n = S_n^2 - n$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$.

Exercise 2.7 (Product and Exponential Martingales)**.** (a) Let $(X_n)_{n\geq 1}$ be independent random variables. Assume that they are nonnegative and $\mathbb{E}X_n = 1$. Set $\mathcal{F}_0 = \{\varnothing, \Omega\}$, $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, and

$$M_0 = 1, \qquad M_n = \prod_{k=1}^n X_k, \quad n \geq 1. \tag{2.2.27}$$

Show that $(M_n)_{n\geq 0}$ is a martingale.

(b) Let $\xi_1, \xi_2, \ldots$ be independent and identically distributed, and fix $\theta \in \mathbb{R}$ such that $\phi(\theta) = \mathbb{E}e^{\theta\xi_1} < \infty$. Use part (a) to show that

$$\frac{\exp(\theta \sum_{k=1}^n \xi_k)}{\phi(\theta)^n}, \qquad n \geq 0, \tag{2.2.28}$$

is a martingale with respect to $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$.

Exercise 2.8 (A Random Walk with Drift)**.** Let $\xi_1, \xi_2, \ldots$ be independent and identically distributed with $\mathbb{E}\xi_1 = \mu$. Set $S_0 = 0$, $S_n = \sum_{k=1}^n \xi_k$, and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$.

(a) Show that $(S_n - n\mu)_{n\geq 0}$ is a martingale.
(b) Determine when $(S_n)_{n\geq 0}$ is a submartingale, a supermartingale, or a martingale.
(c) Suppose that $\mathbb{P}(\xi_1 = 1) = p$ and $\mathbb{P}(\xi_1 = -1) = 1 - p$. Express the centered martingale and the classification in part (b) in terms of p.

Exercise 2.9 (The Information Matters)**.** Let $\xi_1, \xi_2, \ldots$ be independent fair payoffs taking the values -1 and 1. Set $S_0 = 0$, $S_n = \sum_{k=1}^n \xi_k$, $\mathcal{F}_0 = \{\varnothing, \Omega\}$, and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ for $n \geq 1$. Now let $\mathcal{G}_n = \mathcal{F}_{n+1}$ for $n \geq 0$, so that the observer knows the next payoff one turn early.

(a) Show that $(\mathcal{G}_n)$ is a filtration and that (S_n) is adapted to both $(\mathcal{F}_n)$ and $(\mathcal{G}_n)$.
(b) Compute $\mathbb{E}[\,S_{n+1}|\,\mathcal{F}_n]$ and $\mathbb{E}[\,S_{n+1}|\,\mathcal{G}_n]$. Is (S_n) a martingale with respect to both filtrations?

(c) Set $H_n = \xi_n$ and $X_0 = 0$, $X_n = \sum_{k=1}^{n} H_k \xi_k$. Show that (H_n) is predictable with respect to $(\mathcal{G}_n)$, but not $(\mathcal{F}_n)$. Find X_n and explain why this does not contradict the fair-game betting example.

Exercise 2.10 (Orthogonal Martingale Differences)**.** Let $(M_n)_{n\geq 0}$ be a martingale with $M_0 = 0$ and $\mathbb{E}M_n^2 < \infty$ for every n. Write $D_n = M_n - M_{n-1}$.

(a) For $j < k$, use $\mathbb{E}[D_k | \mathcal{F}_{k-1}] = 0$ to show that $\mathbb{E}[D_j D_k] = 0$. Explain why the taking-out property applies even though D_j need not be bounded.

(b) Expand M_n^2 to prove

$$\mathbb{E}M_n^2 = \sum_{k=1}^{n} \mathbb{E}D_k^2. \tag{2.2.29}$$

(c) Let ξ_1, ξ_2 be independent fair payoffs taking the values -1 and 1, and set $D_1 = \xi_1$, $D_2 = (2 + \xi_1)\xi_2$. With $\mathcal{F}_0$ trivial and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ for $n = 1, 2$, show that D_1, D_2 are martingale differences. Show that they are uncorrelated but not independent.

2.3 Stopping Times

Recall the example about the martingale betting strategy in Example 2.18. If you can see the future, you would want to stop betting when your wealth is at its highest. However, given that you cannot, you can only rely on the past information to decide when to stop. In other words, there is a sense that the time you stop betting is a random time that can only depend on the past information you have so far. We can formalize this idea with the next definition.

Definition 2.26 (Stopping Time)**.** Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration. A random variable $\tau : \Omega \to \{0, 1, 2, \ldots\} \cup \{\infty\}$ is a **stopping time** with respect to this filtration if

$$\{\tau \leq n\} \in \mathcal{F}_n \qquad \text{for every } n \geq 0. \tag{2.3.1}$$

The event $\{\tau \leq n\}$ says that the decision to stop has been made by time n. The definition requires this question to be answerable using the information available at that time. In discrete time, it is equivalent to require $\{\tau = n\} \in \mathcal{F}_n$ for every n, since

$$\{\tau = n\} = \{\tau \leq n\} \setminus \{\tau \leq n - 1\}, \qquad n \geq 1. \tag{2.3.2}$$

We will also use the consequence

$$\{\tau \geq n\} = \{\tau \leq n - 1\}^c \in \mathcal{F}_{n-1}, \qquad n \geq 1. \tag{2.3.3}$$

Thus, one step before time n, we know whether the process has not yet been stopped.

Example 2.27 (First Hitting Times)**.** Let $(X_n)_{n\geq 0}$ be an adapted real-valued process and let $B \subseteq \mathbb{R}$ be Borel. Define the first time that the process enters B by

$$\tau_B = \inf\{n \geq 0 : X_n \in B\}, \qquad \inf \varnothing = \infty. \tag{2.3.4}$$

Then

$$\{\tau_B \leq n\} = \bigcup_{k=0}^{n} \{X_k \in B\} \in \mathcal{F}_n, \tag{2.3.5}$$

so τ_B is a stopping time. Deterministic times are also stopping times, and $\tau_B \wedge N$ is the rule that stops upon entering B or at the fixed horizon N, whichever comes first.

By contrast, a time defined by looking one day ahead need not be a stopping time. Knowing whether we should stop today may require information from tomorrow.

Example 2.28 (The Day Before Bankruptcy)**.** Consider a toy model for Lehman's stock price. On each day before bankruptcy the price moves up or down by one unit with equal probability. Once the price reaches zero, it stays there.

Let $\xi_1, \xi_2, \dots$ be independent fair payoffs taking the values -1 and 1. Set $S_0 = 1$ and, for $n \geq 1$, define

$$S_n = \begin{cases} S_{n-1} + \xi_n, & S_{n-1} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2.3.6}$$

Use the natural filtration $\mathcal{F}_n = \sigma(S_0, \dots, S_n)$. The bankruptcy time $\tau = \inf\{n \geq 1 : S_n = 0\}$ is a stopping time, since we can observe when the price reaches zero.

However, the day before bankruptcy, $\sigma = \tau - 1$, is not a stopping time. To see this, observe that $\{\sigma \leq 0\} = \{\tau = 1\} = \{\xi_1 = -1\}$. This event has probability one half, so it does not belong to the trivial sigma-algebra $\mathcal{F}_0 = \sigma(S_0) = \{\varnothing, \Omega\}$. Thus, we cannot determine whether we are on the day before bankruptcy using only the information available up to day n and sell the stock before it goes bankrupt.

From here onwards, we will define $t \wedge s = \min(t, s)$ and $t \vee s = \max(t, s)$ for any real numbers t and s.

Proposition 2.29 (Minimum and Maximum of Stopping Times)**.** *If σ and τ are stopping times, then $\sigma \wedge \tau$ and $\sigma \vee \tau$ are stopping times. In particular, $\tau \wedge N$ is a stopping time for every deterministic N.*

Proof. For every $n \geq 0$,

$$\begin{aligned} \{\sigma \wedge \tau \leq n\} &= \{\sigma \leq n\} \cup \{\tau \leq n\}, \\ \{\sigma \vee \tau \leq n\} &= \{\sigma \leq n\} \cap \{\tau \leq n\}. \end{aligned} \tag{2.3.7}$$

Both events belong to $\mathcal{F}_n$. A deterministic N is a stopping time, which gives the final claim. □

Once a stopping rule has been chosen, we can freeze a process at the time when the rule is triggered.

Definition 2.30 (Stopped Process)**.** Let $(X_n)_{n\geq 0}$ be adapted and let τ be a stopping time. The **process stopped at** τ is defined by $(X_{n\wedge\tau})_{n\geq 0}$.

Stopped processes are very useful. For example, if we only want to study a process until it first hits a certain boundary, we can stop it at that time. Very often, this is used to control a process that might otherwise become unbounded.

For example, we can stop the process with the stopping time $\tau = \inf\{n \geq 0 : |X_n| \geq a\}$ for some level a. If $|X_0| < a$ and X_n has increments of at most size b, then $|X_{n\wedge\tau}|$ will be bounded by $a + b$ for all n. As a result, we can focus on studying a bounded process, which is often easier to analyze.

Proposition 2.31 (Stopping Preserves Martingales)**.** *If $(M_n)_{n\geq 0}$ is a martingale and τ is a stopping time, then $(M_{n\wedge\tau})_{n\geq 0}$ is a martingale. The same statement holds for submartingales and supermartingales.*

Proof. The stopped process is adapted because

$$M_{n\wedge\tau} = \sum_{k=0}^{n} M_k \mathbf{1}_{\{\tau=k\}} + M_n \mathbf{1}_{\{\tau>n\}}, \tag{2.3.8}$$

and every term on the right is $\mathcal{F}_n$-measurable. It is also integrable because the sum is finite. Its next increment is

$$M_{(n+1)\wedge\tau} - M_{n\wedge\tau} = \mathbf{1}_{\{\tau\geq n+1\}}(M_{n+1} - M_n). \tag{2.3.9}$$

By (2.3.3), the indicator is $\mathcal{F}_n$-measurable. Taking out what is known gives

$$\mathbb{E}[\, M_{(n+1)\wedge\tau} - M_{n\wedge\tau} \Big|\, \mathcal{F}_n] = \mathbf{1}_{\{\tau\geq n+1\}}\mathbb{E}[\, M_{n+1} - M_n |\, \mathcal{F}_n] = 0. \tag{2.3.10}$$

This proves the martingale claim. For a submartingale, the last conditional expectation is nonnegative because the indicator is nonnegative; for a supermartingale it is nonpositive. □

In particular, we can apply the repeated martingale property at a fixed time N to get a nice identity

$$\mathbb{E}M_{\tau\wedge N} = \mathbb{E}M_0. \tag{2.3.11}$$

Example 2.32 (A Random Walk Stopped at Two Barriers)**.** Let $\xi_1, \xi_2, \ldots$ be independent random variables taking the values -1 and 1 with equal probability. Set $S_0 = 0$, $S_n = \sum_{k=1}^{n} \xi_k$, and let $\mathcal{F}_0 = \{\varnothing, \Omega\}$ and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ for $n \geq 1$. For positive integers a and b, set

$$\tau = \inf\{n \geq 0 : S_n \in \{-a, b\}\}. \tag{2.3.12}$$

This is a stopping time by Example 2.27, so $(S_{n\wedge\tau})_{n\geq 0}$ is a martingale. Also, $S_n^2 - n$ is a martingale: this follows by expanding $S_{n+1} = S_n + \xi_{n+1}$ and using the conditional mean zero and conditional second moment one of ξ_{n+1}. Stopping this martingale gives $(S_{n\wedge\tau}^2 - (n \wedge \tau))_{n\geq 0}$. Therefore, for every fixed n,

$$\mathbb{E}S_{n\wedge\tau} = 0, \qquad \mathbb{E}S_{n\wedge\tau}^2 = \mathbb{E}[n \wedge \tau]. \tag{2.3.13}$$

The intuition of this example is quite nice: stopping a martingale does not change its expected value. In particular, if τ is representing a strategic decision to stop a gambling game, then as long as τ is using only the information available up to that time, there is no advantage to be gained.

The only minor caveat here is the stopping time is required to be bounded in this example, but this is not required in general. We will next present a theorem that provides possible conditions for a general unbounded stopping time to also preserve the expected value of a martingale.

Theorem 2.33 (Optional Stopping)**.** *Let $(M_n)_{n\geq 0}$ be a martingale and let τ be a stopping time that is finite almost surely. Suppose that either*

(a) $\tau \leq N$ almost surely for some deterministic integer N, or
(b) there is a constant $C < \infty$ such that $|M_{n\wedge\tau}| \leq C$ almost surely for every $n \geq 0$.

Then

$$\mathbb{E}M_\tau = \mathbb{E}M_0. \tag{2.3.14}$$

Under the same conditions, a submartingale satisfies $\mathbb{E}M_\tau \geq \mathbb{E}M_0$, and a supermartingale satisfies the reversed inequality.

Proof. In case (a), Proposition 2.31 shows that $(M_{n\wedge\tau})_{n\geq 0}$ is a martingale. Applying the martingale property repeatedly up to the deterministic time N gives $\mathbb{E}M_{N\wedge\tau} = \mathbb{E}M_0$. Since $\tau \leq N$ almost surely, $N \wedge \tau = \tau$ almost surely, and hence $\mathbb{E}M_\tau = \mathbb{E}M_0$.

In case (b), $|M_\tau| \leq C$ almost surely, since τ is finite. On $\{\tau \leq n\}$, $M_\tau = M_{\tau\wedge n}$, while on $\{\tau > n\}$ their difference has absolute value at most $2C$. Consequently, as $n \to \infty$,

$$\begin{aligned} |\mathbb{E}M_\tau - \mathbb{E}M_{\tau\wedge n}| &= \left|\mathbb{E}\left[(M_\tau - M_{\tau\wedge n})\mathbf{1}_{\{\tau>n\}}\right]\right| \\ &\leq \mathbb{E}\left[|M_\tau - M_{\tau\wedge n}|\,\mathbf{1}_{\{\tau>n\}}\right] \\ &\leq 2C\,\mathbb{P}(\tau > n). \end{aligned} \tag{2.3.15}$$

Here $\mathbb{P}(\tau > n) \to 0$ because τ is finite almost surely. Combining this bound with $\mathbb{E}M_{\tau\wedge n} = \mathbb{E}M_0$ proves the claim.

For submartingales and supermartingales, use the corresponding fixed-horizon inequality and the same limit argument. □

The two conditions control different things: (a) limits how long we wait, whereas (b) limits the values of the process while we wait. In particular, (b) allows an unbounded stopping time, provided it is finite almost surely.

We will see next that optional stopping can be used to compute some seemingly complicated probabilities.

Example 2.34 (Gambler's Ruin)**.** Let $\xi_1, \xi_2, \ldots$ be independent fair payoffs taking the values -1 and 1. Define $S_0 = 0$, $S_n = \sum_{k=1}^n \xi_k$, and $\mathcal{F}_0 = \{\varnothing, \Omega\}$, $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$ for $n \geq 1$. For positive integers a, b, let $\tau = \inf\{n \geq 0 : S_n \in \{-a, b\}\}$. Our goal here is to compute the probability of reaching the target fortune b before ruin $-a$ (sometimes also called a stop loss).

To apply optional stopping, we must verify that τ is finite. From any position between the barriers, $a + b$ consecutive positive increments force an exit. Therefore we have that

$$\mathbb{P}(\tau > a + b) \leq \mathbb{P}\left(\{\xi_1 = \cdots = \xi_{a+b} = +1\}^c\right) = 1 - 2^{-(a+b)}. \tag{2.3.16}$$

Next for every integer $r \geq 1$, we will attempt to control the probability $\mathbb{P}(\tau > r(a + b))$ by splitting up the event. In particular, we can let $A_r = \{\xi_{(r-1)(a+b)+1} = \cdots = \xi_{r(a+b)} = +1\}$, and observe that

$$\{\tau > (r-1)(a+b)\} \cap A_r \subset \{(r-1)(a+b) < \tau \leq r(a+b)\}, \tag{2.3.17}$$

which implies that

$$\{\tau > r(a+b)\} \subset \{\tau > (r-1)(a+b)\} \cap A_r^c. \tag{2.3.18}$$

The event A_r depends only on the increments in the new block, so it is independent of $\{\tau > (r-1)(a+b)\}$, as it depends only on the preceding increments. Return to the probability $\mathbb{P}(\tau > r(a+b))$, we get

$$\begin{aligned} \mathbb{P}(\tau > r(a+b)) &\leq \left(1 - 2^{-(a+b)}\right)\mathbb{P}(\tau > (r-1)(a+b)) \\ &\leq \left(1 - 2^{-(a+b)}\right)^r \longrightarrow 0. \end{aligned} \tag{2.3.19}$$

Therefore $\tau < \infty$ almost surely, and since $-a \leq S_{n\wedge\tau} \leq b$, optional stopping gives $\mathbb{E}S_\tau = \mathbb{E}S_0 = 0$.

Writing $p = \mathbb{P}(S_\tau = b)$, the two possible terminal values give $0 = -a(1-p) + bp$. Solving for p, we obtain

$$\mathbb{P}(S_\tau = b) = \frac{a}{a+b}, \qquad \mathbb{P}(S_\tau = -a) = \frac{b}{a+b}. \tag{2.3.20}$$

The chance of reaching the target is the initial fortune divided by the target fortune.

Example 2.35 (A Finite Stopping Time Is Not Enough)**.** Let $S_0 = 0$ and $S_n = \sum_{k=1}^n \xi_k$, where the ξ_k are independent fair payoffs taking the values -1 and 1, and use the natural filtration. Set $\tau = \inf\{n \geq 0 : S_n = 1\}$.

Note this is the case of Gambler's ruin with $b = 1$ and $a \to \infty$, but this fails both conditions (in particular (b)) of Theorem 2.33).

For every positive integer a, reaching 1 before $-a$ implies $\tau < \infty$. The preceding gambler's-ruin calculation with $b = 1$ therefore gives

$$\mathbb{P}(\tau < \infty) \geq \frac{a}{a+1} \longrightarrow 1. \tag{2.3.21}$$

Thus τ is finite almost surely. Nevertheless, $S_\tau = 1$, so

$$\mathbb{E}S_\tau = 1 \neq 0 = \mathbb{E}S_0. \tag{2.3.22}$$

Indeed, in the event that we have k consecutive negative increments, we would have $S_{k\wedge\tau} = -k$ and the event has probability 2^{-k}. So the stopped values have no common deterministic bound, which is the second requirement of the optional stopping theorem.

This example also shows that bounding the terminal value alone is not enough.

Example 2.36 (Almost-Sure Convergence Is Not Enough)**.** Consider the martingale betting strategy again in a game with independent fair payoffs $\xi_n \in \{-1, 1\}$, and let $\mathcal{F}_0 = \{\varnothing, \Omega\}$ and $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$.

Start with $X_0 = 0$, bet one unit on the first round, double the bet after each loss, and stop betting after the first win. Thus $X_n = \sum_{k=1}^n H_k \xi_k$, where $H_1 = 1$ and $H_k = 2^{k-1}\mathbf{1}_{\{\xi_1 = \cdots = \xi_{k-1} = -1\}}$ for $k \geq 2$. The stakes depend only on previous outcomes, so (X_n) is a martingale. Let $\tau = \inf\{n \geq 1 : X_n = 1\}$, the time of the first win.

Since $\mathbb{P}(\tau > n) = 2^{-n} \to 0$, τ is finite almost surely. A first win recovers all previous losses and leaves one unit of profit, whereas n consecutive losses leave wealth $-(2^n - 1)$. Hence

$$X_{n\wedge\tau} = \begin{cases} 1, & \tau \leq n, \\ -(2^n - 1), & \tau > n. \end{cases} \tag{2.3.23}$$

On almost every path, a win eventually occurs and the wealth stays at 1 thereafter. Thus $X_{n\wedge\tau} \to X_\tau = 1$ almost surely. However,

$$\mathbb{E}X_{n\wedge\tau} = (1 - 2^{-n}) - (2^n - 1)2^{-n} = 0, \qquad \mathbb{E}X_\tau = 1. \tag{2.3.24}$$

The positive contribution tends to 1, but the negative contribution tends to -1, not zero.

The losses become rarer but larger, so their contribution to the expectation does not disappear. In this sense, a negative contribution escapes to negative infinity, even though the probability of an outstanding loss tends to zero.

In fact, on $\{\tau > n\}$ the distance from 1 is 2^n, while on $\{\tau \le n\}$ it is zero. Therefore

$$\mathbb{E}|X_{n\wedge\tau} - 1| = 2^n \mathbb{P}(\tau > n) = 1 \tag{2.3.25}$$

for every n.

This motivates the following definition for a mode of convergence.

Definition 2.37 (Convergence in L^p)**.** For $p \ge 1$, we say that X_n converges to X in L^p, if $X_n, X \in L^p$ and $\|X_n - X\|_{L^p} = (\mathbb{E}|X_n - X|^p)^{1/p} \to 0$ as $n \to \infty$.

As a result, we can say that the wealth process of the martingale strategy $X_{n\wedge\tau}$ converges almost surely to $X_\tau = 1$ but not in L^1.

Exercise 2.11 (Stopping Times and a Future-Looking Time)**.** Let $S_0 = 0$, $S_n = \sum_{k=1}^n \xi_k$, where the ξ_k are independent and take the values -1 and 1 with equal probability, and let $\mathcal{F}_0 = \{\varnothing, \Omega\}$ and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ for $n \ge 1$.

(a) Show that $\inf\{n \ge 0 : S_n = 2\}$ is a stopping time.
(b) Show that $\inf\{n \ge 0 : S_n \notin \{-1, 0, 1\}\}$ is a stopping time.
(c) Let $L = \max\{k \in \{0, 1, 2\} : S_k = 0\}$. Show that L is not a stopping time by examining the event $\{L \le 0\}$.

Exercise 2.12 (Delaying and Anticipating a Stopping Time)**.** (a) Let τ be a stopping time. Prove that $\tau + 1$ is also a stopping time, with the convention $\infty + 1 = \infty$.
(b) Let $\xi_1, \xi_2, \dots$ be independent fair coin tosses taking values -1 and 1, with their natural filtration and trivial $\mathcal{F}_0$. Set $\tau = \inf\{n \ge 1 : \xi_n = 1\}$. Show that τ is a stopping time but $\tau - 1$ is not, with the convention $\infty - 1 = \infty$.
(c) Explain why delaying a decision uses only past information, whereas moving it one step earlier may require future information.

Exercise 2.13 (Doob's Maximal Inequality)**.** Let $(M_n)_{n\ge0}$ be a martingale. Fix an integer $N \ge 0$ and $\lambda > 0$. We will prove that

$$\mathbb{P}\left(\max_{0\le n\le N} |M_n| \ge \lambda\right) \le \frac{\mathbb{E}|M_N|}{\lambda}. \tag{2.3.26}$$

(a) Use conditional Jensen's inequality to show that $(|M_n|)_{n\ge0}$ is a nonnegative submartingale.
(b) Set $\tau = \inf\{n \ge 0 : |M_n| \ge \lambda\} \wedge N$. Show that τ is a stopping time and that

$$\left\{\max_{0\le n\le N} |M_n| \ge \lambda\right\} = \{|M_\tau| \ge \lambda\}. \tag{2.3.27}$$

(c) Check the pathwise identity

$$|M_N| - |M_\tau| = \sum_{k=1}^{N} \mathbf{1}_{\{\tau<k\}}(|M_k| - |M_{k-1}|). \tag{2.3.28}$$

Explain why $\mathbf{1}_{\{\tau<k\}}$ is $\mathcal{F}_{k-1}$-measurable, and use (a) to conclude that $\mathbb{E}|M_\tau| \le \mathbb{E}|M_N|$.
(d) On the event in (b), $|M_\tau| \ge \lambda$. Use this and (c) to prove (2.3.26).

Exercise 2.14 (Expected Time to Reach a Barrier)**.** Let $\xi_1, \xi_2, \dots$ be independent fair payoffs taking the values -1 and 1, and set $S_0 = 0$, $S_n = \sum_{k=1}^{n} \xi_k$. For positive integers a, b, let $\tau = \inf\{n \geq 0 : S_n \in \{-a, b\}\}$.

(a) Show that $\mathbb{E}[n \wedge \tau] = \sum_{k=1}^{n} \mathbb{P}(\tau \geq k)$. You may use the analogous identity $\mathbb{E}\tau = \sum_{k=1}^{\infty} \mathbb{P}(\tau \geq k)$, with an infinite value allowed.

(b) Using $\mathbb{E}S_{n\wedge\tau}^2 = \mathbb{E}[n \wedge \tau]$ and $-a \leq S_{n\wedge\tau} \leq b$, show that $\mathbb{E}\tau < \infty$.

(c) Bound the difference between $\mathbb{E}S_\tau^2$ and $\mathbb{E}S_{n\wedge\tau}^2$ on the event $\{\tau > n\}$. Use the two terminal values to conclude that $\mathbb{E}\tau = ab$.

Exercise 2.15 (Biased Gambler's Ruin)**.** Let $\xi_1, \xi_2, \dots$ be independent with $\mathbb{P}(\xi_n = 1) = p$ and $\mathbb{P}(\xi_n = -1) = 1 - p$, where $0 < p < 1$ and $p \neq 1/2$. Set $S_0 = 0$, $S_n = \sum_{k=1}^{n} \xi_k$, $\mathcal{F}_0 = \{\varnothing, \Omega\}$, and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$ for $n \geq 1$. For positive integers a, b, let $\tau = \inf\{n \geq 0 : S_n \in \{-a, b\}\}$ and $r = (1-p)/p$.

(a) Check that $pr + (1-p)r^{-1} = 1$, and use this to show that $(r^{S_n})_{n\geq 0}$ is a martingale.

(b) Show that $\mathbb{P}(\tau > m(a+b)) \leq (1 - p^{a+b})^m$ for every integer $m \geq 1$. Use blocks of $a + b$ consecutive positive increments. Conclude that $\tau < \infty$ almost surely.

(c) Check that $(r^{S_{n\wedge\tau}})$ has a common deterministic bound. Apply optional stopping and use the two possible values of S_τ to show that

$$\mathbb{P}(S_\tau = b) = \frac{1 - r^a}{1 - r^{a+b}}. \tag{2.3.29}$$

2.4 Martingale Convergence

One of the most important results in martingale theory is about convergence as $n \to \infty$. In this section, we will study the convergence of martingales and submartingales. To start, we will first show the equivalence of convergence with a notion of upcrossings.

Definition 2.38 (Upcrossings)**.** Let $(x_n)_{n\geq 0}$ be a real sequence and let $a < b$. Set

$$\begin{aligned} \sigma_1 &= \inf\{n \geq 0 : x_n \leq a\}, \\ \tau_1 &= \inf\{n > \sigma_1 : x_n \geq b\}. \end{aligned} \tag{2.4.1}$$

Recursively, let

$$\begin{aligned} \sigma_{j+1} &= \inf\{n > \tau_j : x_n \leq a\}, \\ \tau_{j+1} &= \inf\{n > \sigma_{j+1} : x_n \geq b\}. \end{aligned} \tag{2.4.2}$$

We use $\inf \varnothing = \infty$.

The number of completed **upcrossings** by time N is

$$U_N(a, b) = \#\{j : \tau_j \leq N\}. \tag{2.4.3}$$

For a stochastic process, these times and counts are defined on each sample path $\omega \in \Omega$.

As we will see in the next lemma, upcrossings can directly characterize convergence.

Lemma 2.39 (Upcrossings and Convergence)**.** *If a real sequence has only finitely many upcrossings of $[a, b]$ for every rational $a < b$, then it converges to a limit in $\mathbb{R} \cup \{-\infty, \infty\}$.*

Proof. Suppose for contradiction the sequence does not converge to any limit in $\mathbb{R} \cup \{-\infty, \infty\}$.

Every real sequence has a subsequence that converges to a finite or infinite limit, i.e. $x_{n_k} \to y_1$ as $k \to \infty$. Since the full sequence does not converge to y_1, we can find another subsequence with a different limit. If y_1 is finite, infinitely many terms satisfy $|x_n - y_1| > \epsilon$ for some $\epsilon > 0$. If $y_1 = +\infty$, infinitely many terms lie below some fixed finite level; if $y_1 = -\infty$, infinitely many terms lie above some fixed finite level.

In each case, select a subsequence of these terms that converges to a finite or infinite limit y_2. The way these terms were selected ensures that $y_2 \neq y_1$. Without loss of generality we can assume $y_1 < y_2$ (we can always relabel the limits). Next, we can choose rational numbers a and b such that $y_1 < a < b < y_2$.

Due to the convergence of the two subsequences, the original sequence must therefore visit below a infinitely many times and above b infinitely many times and arbitrarily late. As a result, we can choose a subsequence of x_n such that it visits below a then above b and then below a again and so on, infinitely many times.

This constructs infinitely many upcrossings of $[a, b]$, contradicting the assumption. □

The reason we choose rational a, b will be clear later, for now we will just note the fact that there are only countably many rational numbers, and they are dense in the real line.

Next we turn our attention to studying the upcrossings of a martingale. Suppose we consider the "buy low sell high" trading strategy for the upcrossings of a martingale, that is buy at a and sell at $b > a$. If the martingale crosses a and b infinitely many times, then we can make infinitely many of these trades, which would yield infinite profit.

It turns out our expected profit from using this strategy can be bounded by an expected value of the positive part of the martingale.

Lemma 2.40 (Upcrossing Inequality)**.** *Let $(X_n)_{n\geq 0}$ be a submartingale, and let $a < b$. Write $x^+ = \max\{x, 0\}$. Then, for every $N \geq 0$,*

$$(b-a)\,\mathbb{E}U_N(a,b) \leq \mathbb{E}(X_N - a)^+ - \mathbb{E}(X_0 - a)^+. \tag{2.4.4}$$

Proof. Set $Y_n = (X_n - a)^+$. The function $x \mapsto (x-a)^+$ is convex and increasing. Thus conditional Jensen's inequality and the submartingale property give

$$\mathbb{E}[Y_k | \mathcal{F}_{k-1}] \geq (\mathbb{E}[X_k | \mathcal{F}_{k-1}] - a)^+ \geq (X_{k-1} - a)^+ = Y_{k-1}. \tag{2.4.5}$$

Hence (Y_n) is a nonnegative submartingale.

Set $\tau_0 = 0$. For $j \geq 1$, recursively define the buying and selling times by

$$\sigma_j = \inf\{n \geq \tau_{j-1} : X_n \leq a\}, \qquad \tau_j = \inf\{n > \sigma_j : X_n \geq b\}. \tag{2.4.6}$$

We use $\inf \varnothing = \infty$. The first purchase can occur at time 0; for $j \geq 2$, $X_{\tau_{j-1}} \geq b > a$ whenever $\tau_{j-1} < \infty$, so the next purchase occurs strictly after the previous sale. Set

$$H_k = \sum_{j=1}^{N} \mathbf{1}_{\{\sigma_j < k \leq \tau_j\}}, \qquad 1 \leq k \leq N. \tag{2.4.7}$$

At most one crossing is in progress, so H_k is either zero or one.

The times σ_j and τ_j are stopping times: at time $k-1$, we can tell from the observed path whether an upcrossing has begun and whether it has ended. Thus H_k is $\mathcal{F}_{k-1}$-measurable, so the strategy is predictable.

Each completed trade starts with $Y_{\sigma_j} = 0$ and ends with $Y_{\tau_j} \geq b - a$. If a trade is still open at time N, its gain is $Y_N - Y_{\sigma_j} = Y_N \geq 0$. Adding the gains of the trades gives the pathwise bound

$$(b-a)U_N(a,b) \leq \sum_{k=1}^{N} H_k(Y_k - Y_{k-1}). \tag{2.4.8}$$

Also, $1 - H_k$ is nonnegative and $\mathcal{F}_{k-1}$-measurable. Since (Y_n) is a submartingale, taking out what is known gives $\mathbb{E}[(1-H_k)(Y_k - Y_{k-1})] \geq 0$. Therefore

$$\mathbb{E}\left[\sum_{k=1}^{N} H_k(Y_k - Y_{k-1})\right] \leq \mathbb{E}[Y_N - Y_0]. \tag{2.4.9}$$

Combining the last two bounds proves (2.4.4). □

Once again, the upcrossing inequality shows that a submartingale upcrossing can be bounded by the expected positive part of the process. In combination with the upcrossing characterization of convergence, we can now prove that a submartingale with uniformly bounded positive parts converges almost surely to a finite limit.

Theorem 2.41 (Martingale Convergence Theorem)**.** *Let $(X_n)_{n\geq 0}$ be a submartingale such that*

$$\sup_{n\geq 0} \mathbb{E}X_n^+ < \infty. \tag{2.4.10}$$

Then X_n converges almost surely to a finite random variable.

Proof. Let $K = \sup_n \mathbb{E}X_n^+$, and fix rational numbers $a < b$.

Since $(X_n - a)^+ \leq X_n^+ + |a|$, Lemma 2.40 gives

$$\mathbb{E}U_N(a,b) \leq \frac{K + |a|}{b-a} \qquad \text{for every } N. \tag{2.4.11}$$

For any positive integer r, Markov's inequality and continuity of probability for increasing events give

$$\mathbb{P}\left(\sup_N U_N(a,b) \geq r\right) = \lim_{N\to\infty} \mathbb{P}(U_N(a,b) \geq r) \leq \frac{K+|a|}{r(b-a)}. \tag{2.4.12}$$

Letting r increase shows that the number of upcrossings of $[a,b]$ is finite almost surely.

There are only countably many pairs of rational levels, so we can use countable additivity to show that

$$\mathbb{P}\left(\bigcap_{a<b\in\mathbb{Q}} \sup_N U_N(a,b) < \infty\right) = 1. \tag{2.4.13}$$

Then the upcrossing inequality Lemma 2.39 shows that X_n has a limit, possibly infinite, almost surely.

It remains to rule out an infinite limit. Since the expectations of a submartingale are nondecreasing, $\mathbb{E}X_n \geq \mathbb{E}X_0$. Thus

$$\mathbb{E}|X_n| = 2\mathbb{E}X_n^+ - \mathbb{E}X_n \leq 2K - \mathbb{E}X_0 =: C. \tag{2.4.14}$$

By Markov's inequality

$$\mathbb{P}(|X_n| \geq R) \leq \frac{C}{R} \quad \text{for every } n, R > 0. \tag{2.4.15}$$

For each fixed R, we define the event

$$A_R = \{|X_n| \geq R \text{ eventually}\} = \bigcup_{m=0}^{\infty} \{|X_n| \geq R, \forall n \geq m\}. \tag{2.4.16}$$

Each of these events is contained in the event $\{|X_n| \geq R\}$, which has probability at most C/R, so their union does too.

A path tending to ∞ or $-\infty$ belongs to this union for every R. Letting R increase shows that the probability of an infinite limit is zero. □

Remark 2.42 (Doob Martingale Representation). If $(X_n)_{n\geq 0}$ is a martingale and $X_n \to X_\infty$ in L^1, then

$$X_n = \mathbb{E}[\,X_\infty |\, \mathcal{F}_n], \qquad n \geq 0. \tag{2.4.17}$$

Thus the process is a Doob martingale: its value at each time is the conditional expectation of its limiting value given the information available then.

To see this, fix n and use the martingale property $X_n = \mathbb{E}[\,X_m |\, \mathcal{F}_n]$ for every $m \geq n$. The conditional absolute-value bound gives

$$\begin{aligned} \mathbb{E}\,|X_n - \mathbb{E}[\,X_\infty |\, \mathcal{F}_n]| &= \mathbb{E}\,|\mathbb{E}[\,X_m - X_\infty |\, \mathcal{F}_n]| \\ &\leq \mathbb{E}|X_m - X_\infty| \longrightarrow 0. \end{aligned} \tag{2.4.18}$$

The left-hand side does not depend on m, so it must be zero, proving (2.4.17) almost surely. Almost-sure convergence alone does not guarantee this representation.

Two useful consequences follow immediately.

Corollary 2.43 (Convergence of Martingales and Nonnegative Supermartingales)**.** *The following processes converge almost surely to finite limits:*

(a) a martingale such that $\sup_n \mathbb{E}|M_n| < \infty$*;*
(b) a nonnegative supermartingale.

Proof. For (a), a martingale is a submartingale and $M_n^+ \leq |M_n|$, so Theorem 2.41 applies.

For (b), if (X_n) is a nonnegative supermartingale, then $(-X_n)$ is a submartingale and $(-X_n)^+ = 0$. Apply the theorem to $(-X_n)$ and change the sign of its limit. □

Example 2.44 (Pólya's Urn)**.** Start with one red ball and one blue ball in an urn. At each turn, draw a ball uniformly, return it to the urn, and add one more ball of the same colour. Let R_n be the number of red balls after n draws, so $R_0 = 1$ and the urn contains $n + 2$ balls in total. Write

$M_n = R_n/(n+2)$ for the proportion of red balls, and let $\mathcal{F}_n$ record the colours drawn in the first n turns.

Given $\mathcal{F}_n$, the next draw is red with probability M_n. A red draw increases R_n by one, while a blue draw leaves it unchanged. In either case, the total number of balls becomes $n+3$. Therefore

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \frac{R_n + 1}{n+3} + (1 - M_n)\frac{R_n}{n+3} = M_n. \tag{2.4.19}$$

Thus (M_n) is a martingale. Since $0 \le M_n \le 1$, Corollary 2.43 shows that M_n converges almost surely to a limit in $[0,1]$.

Each draw reinforces the colour that was drawn, but the proportion has no conditional drift. The limiting proportion is random, not necessarily $1/2$: early draws can influence the eventual balance. We will not compute its distribution here.

Example 2.45 (A Random Walk Absorbed at Zero)**.** Let $\xi_1, \xi_2, \dots$ be independent random variables taking the values -1 and 1 with equal probability. Start at $S_0 = 1$ and set

$$S_n = S_{n-1} + \mathbf{1}_{\{S_{n-1}>0\}}\xi_n, \qquad n \ge 1. \tag{2.4.20}$$

The walk takes fair steps until it reaches zero and then stays there. It is a nonnegative martingale with respect to $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$, since its next increment has conditional mean zero.

By Corollary 2.43, S_n converges almost surely. On any path that never reaches zero, however, every increment has absolute value one, so S_n cannot converge. Therefore the walk reaches zero almost surely.

Exercise 2.16 (A Product Martingale That Vanishes)**.** Let $\xi_1, \xi_2, \dots$ be independent random variables taking the values -1 and 1 with equal probability. Set $M_0 = 1$, $M_n = \prod_{k=1}^{n}(1 + \xi_k/2)$, and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$, with $\mathcal{F}_0$ trivial.

(a) Show that (M_n) is a nonnegative martingale and $\mathbb{E}M_n = 1$ for every n.
(b) Explain why M_n has an almost-sure limit.
(c) Set $\rho = (\sqrt{3/2} + \sqrt{1/2})/2$. Show that $\rho < 1$ and $\mathbb{E}\sqrt{M_n} = \rho^n$. Use Markov's inequality to show that M_n converges in probability to zero, and identify its almost-sure limit.

This exercise shows why almost-sure convergence alone does not allow us to pass a limit through expectation.

Exercise 2.17 (Why a Random Walk Needs a Convergence Hypothesis)**.** Let $S_0 = 0$ and $S_n = \sum_{k=1}^{n} \xi_k$, where the ξ_k are independent random variables taking the values -1 and 1 with equal probability. Show that (S_n) is a martingale but cannot converge on any sample path. Use Corollary 2.43 to conclude that $\sup_n \mathbb{E}|S_n| = \infty$.

Exercise 2.18 (A Bounded Martingale Converges in L^1)**.** Let $(M_n)_{n\ge 0}$ be a martingale such that $|M_n| \le K$ almost surely for every n, where K is a fixed positive constant.

(a) Use the martingale convergence theorem to show that $M_n \to M_\infty$ almost surely for some random variable satisfying $|M_\infty| \le K$ almost surely.

(b) For $\varepsilon > 0$, split according to whether $|M_n - M_\infty| \le \varepsilon$ and prove

$$\mathbb{E}|M_n - M_\infty| \le \varepsilon + 2K\mathbb{P}(|M_n - M_\infty| > \varepsilon). \tag{2.4.21}$$

Use almost-sure convergence implying convergence in probability to conclude that $M_n \to M_\infty$ in L^1.

(c) Show that $\mathbb{E}M_\infty = \mathbb{E}M_0$. Why can this conclusion fail for a nonnegative martingale that is not bounded by a fixed constant?

2.5 Backward Martingales and the Strong Law

It turns out, we can also study martingales as time goes backwards. In particular, we will recover yet another convergence theorem, as well as a very strong proof of the strong law of large numbers.

Definition 2.46 (Backward Martingales)**.** Let $(\mathcal{F}_n)_{n\le 0}$ be a **backward filtration** of sub-σ-algebras of $\mathcal{F}$ such that $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$ for every $n \le -1$. A process $(M_n)_{n\le 0}$ is a **backward martingale** if M_n is integrable and $\mathcal{F}_n$-measurable for every $n \le 0$, and

$$\mathbb{E}[\, M_{n+1} |\, \mathcal{F}_n] = M_n \qquad \text{almost surely for every } n \le -1. \tag{2.5.1}$$

Backward martingales are also called reverse martingales. We can equally use $n \le -1$, or another fixed upper endpoint.

The martingale identity is unchanged, but we can now study the process as $n \to -\infty$, when the σ-algebras become smaller. In words, as we gradually forget information, we can still recover a limit.

Theorem 2.47 (Backward Martingale Convergence)**.** *Every backward martingale $(M_n)_{n\le 0}$ converges almost surely to a finite random variable as $n \to -\infty$.*

Proof. Conditional Jensen's inequality therefore gives $|M_n|$ is a submartingale, and we have

$$\mathbb{E}|M_n| \le \mathbb{E}|M_0| =: C \qquad \text{for every } n \le 0. \tag{2.5.2}$$

Next, fix $a < b$ and $N \ge 1$. The finite block $M_{-N}, \dots, M_0$ is an ordinary martingale with respect to $\mathcal{F}_{-N}, \dots, \mathcal{F}_0$. Let $U_N(a,b)$ count the completed upcrossings of $[a,b]$ in this block.

The rest of the argument is the same as in the proof of Theorem 2.41, so we will only sketch it. By Lemma 2.40,

$$(b-a)\mathbb{E}U_N(a,b) \le \mathbb{E}(M_0 - a)^+ - \mathbb{E}(M_{-N} - a)^+ \le C + |a|. \tag{2.5.3}$$

The counts $U_N(a,b)$ increase with N. For every positive integer r, continuity of probability and Markov's inequality give

$$\mathbb{P}\left(\sup_N U_N(a,b) \ge r\right) = \lim_{N\to\infty} \mathbb{P}(U_N(a,b) \ge r) \le \frac{C + |a|}{r(b-a)}. \tag{2.5.4}$$

Similarly we let $r \to \infty$ to show the number of upcrossings of $[a,b]$ is finite almost surely, which also holds for every rational $a < b$ by countable additivity. A small caveat is that reading these

return crossings in reverse gives upcrossings in the reversed path, so the two upcrossing counts differ by at most one.

Thus $M_0, M_{-1}, \ldots$ also has finitely many upcrossings of each rational interval. By Lemma 2.39, M_n has a limit, possibly infinite, as $n \to -\infty$ almost surely.

To exclude an infinite limit, we use the same $R > 0$ Markov's inequality argument as Theorem 2.41. □

Unlike the forward martingale convergence theorem, this result does not require a separate bound on the first moments. The backward martingale property already supplies (2.5.2).

Example 2.48 (Sample Averages as a Backward Martingale)**.** Let $X_1, X_2, \ldots$ be i.i.d. integrable random variables. As in Chapter 1, write their sample average as $S_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. Define

$$M_{-n} = S_n, \qquad \mathcal{F}_{-n} = \sigma(S_n, X_{n+1}, X_{n+2}, \ldots), \qquad n \geq 1. \tag{2.5.5}$$

This information records the average of the first n observations and all the later observations, but not the individual values of the first n.

Since $S_{n+1} = (nS_n + X_{n+1})/(n+1)$, every variable generating $\mathcal{F}_{-(n+1)}$ is $\mathcal{F}_{-n}$-measurable. Thus $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$, as required for a filtration indexed by negative integers.

We claim that

$$\mathbb{E}[X_1 | \mathcal{F}_{-n}] = S_n = M_{-n}. \tag{2.5.6}$$

To check the definition of conditional expectation, take $A \in \mathcal{F}_{-n}$. Its indicator can therefore be written as a measurable function of the average and the later observations:

$$\mathbf{1}_A = h(S_n, X_{n+1}, X_{n+2}, \ldots). \tag{2.5.7}$$

Since the X_i's are iid. and S_n is permutation invariance with respect to $(X_i)_{i=1}^n$, we have that

$$(X_1, X_2, \ldots, X_n, \mathbf{1}_A) \stackrel{d}{=} (X_{\sigma(1)}, X_{\sigma(2)}, \ldots, X_{\sigma(n)}, \mathbf{1}_A), \tag{2.5.8}$$

for any permutation σ. As a result, we can choose $\sigma(i) = j$ and take the marginal of the i-th entry and the last entry to get

$$(X_i, \mathbf{1}_A) \stackrel{d}{=} (X_j, \mathbf{1}_A). \tag{2.5.9}$$

Taking expectations of the products shows that $\mathbb{E}[X_i \mathbf{1}_A]$ is the same for every $1 \leq i \leq n$, and hence

$$\mathbb{E}[X_1 \mathbf{1}_A] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i \mathbf{1}_A] = \mathbb{E}[S_n \mathbf{1}_A]. \tag{2.5.10}$$

Since S_n is $\mathcal{F}_{-n}$-measurable, this proves (2.5.6). The tower property now shows that $(M_k)_{k \leq -1}$ is a backward martingale. By Theorem 2.47, the sample averages $S_n = M_{-n}$ have an almost-sure finite limit as $n \to \infty$.

This gives convergence, but we still have to identify the limit. To start we will require a standard inequality.

Lemma 2.49 (Cauchy–Schwarz Inequality)**.** *Let $X, Y \in L^2$. Then*

$$\mathbb{E}|XY| \leq \sqrt{\mathbb{E}X^2 \mathbb{E}Y^2}. \tag{2.5.11}$$

In particular, taking $Y = 1$ gives $\mathbb{E}|X| \leq \sqrt{\mathbb{E}X^2}$.

We note that the Cauchy–Schwarz inequality can be applied on all inner product spaces, not just for random variables.

Proof. Firstly, $2|XY| \leq X^2 + Y^2$, so XY is integrable.

For every $t \in \mathbb{R}$, we have

$$0 \leq \mathbb{E}(|X| - t|Y|)^2 = \mathbb{E}X^2 - 2t\mathbb{E}|XY| + t^2\mathbb{E}Y^2. \tag{2.5.12}$$

If $\mathbb{E}Y^2 = 0$, then $Y = 0$ almost surely and the inequality holds trivially. Otherwise, taking $t = \mathbb{E}|XY|/\mathbb{E}Y^2$ gives (2.5.11). □

Theorem 2.50 (Strong Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}|X_1| < \infty$, and write $\mu = \mathbb{E}X_1$. Then*

$$S_n = \frac{1}{n}\sum_{i=1}^{n} X_i \longrightarrow \mu \qquad \textit{almost surely.} \tag{2.5.13}$$

Proof. By Example 2.48, $S_n \to S_\infty$ almost surely for some finite random variable S_∞. It remains to identify the limit.

For $K > 0$, define the clipped observations $X_i^{(K)} = \max\{-K, \min\{X_i, K\}\}$ and their average $S_n^{(K)} = \frac{1}{n}\sum_{i=1}^{n} X_i^{(K)}$. These observations are bounded and remain i.i.d., so Cauchy–Schwarz inequality gives us

$$\mathbb{E}|S_n^{(K)} - \mathbb{E}X_1^{(K)}| \leq \sqrt{\mathrm{Var}(S_n^{(K)})} = \sqrt{\frac{\mathrm{Var}(X_1^{(K)})}{n}}. \tag{2.5.14}$$

Next we can use the triangle inequality and identical distributions to obtain

$$\mathbb{E}|S_n - S_n^{(K)}| \leq \mathbb{E}|X_1 - X_1^{(K)}|, \qquad \text{and} \qquad |\mu - \mathbb{E}X_1^{(K)}| \leq \mathbb{E}|X_1 - X_1^{(K)}|. \tag{2.5.15}$$

Combining these bounds gives

$$\mathbb{E}|S_n - \mu| \leq \sqrt{\frac{\mathrm{Var}(X_1^{(K)})}{n}} + 2\mathbb{E}|X_1 - X_1^{(K)}|. \tag{2.5.16}$$

Since X_1 is integrable, the contribution from sufficiently large values can be made arbitrarily small. In particular,

$$\mathbb{E}|X_1 - X_1^{(K)}| \leq \mathbb{E}[|X_1|\mathbf{1}_{\{|X_1|>K\}}] \to 0 \quad \text{as } K \to \infty. \tag{2.5.17}$$

Given $\varepsilon > 0$, first choose K so that the second term in (2.5.16) is less than $\varepsilon/2$. For this fixed K, choose n large enough that the first term is less than $\varepsilon/2$. Thus $\mathbb{E}|S_n - \mu| \to 0$, and Markov's inequality gives $S_n \to \mu$ in probability.

For every $\varepsilon > 0$, the triangle inequality gives

$$\mathbb{P}(|S_\infty - \mu| > \varepsilon) \leq \mathbb{P}(|S_\infty - S_n| > \varepsilon/2) + \mathbb{P}(|S_n - \mu| > \varepsilon/2) \longrightarrow 0. \tag{2.5.18}$$

Thus $\mathbb{P}(|S_\infty - \mu| > 1/r) = 0$ for each positive integer r. Taking the union over r gives $\mathbb{P}(S_\infty \neq \mu) = 0$. Consequently $S_n \to \mu$ almost surely. □

Example 2.51 (Exchangeable Sequence)**.** Let $Y_1, Y_2, \ldots$ be i.i.d. random variables, and let Z be independent of the entire sequence. For a measurable function $f : \mathbb{R}^2 \to \mathbb{R}$, set

$$X_i = f(Y_i, Z), \qquad i \geq 1, \tag{2.5.19}$$

and assume $\mathbb{E}f(Y_1, Z)^2 < \infty$.

Each observation X_i uses the same random variable Z and a fresh independent random variable Y_i, therefore X_i's are not independent in general. However, permuting the Y_i leaves their joint distribution with Z unchanged, so it also leaves the joint distribution of the X_i unchanged. A sequence whose joint distribution is unchanged by finite permutations is called **exchangeable**. More precisely, for any $n > 0$ and permutation π that is a bijection on $[n]$, we have that

$$(X_1, \ldots, X_n) \stackrel{d}{=} (X_{\pi(1)}, \ldots, X_{\pi(n)})\,. \tag{2.5.20}$$

That being said, even without independence, the empirical average is still a backward martingale. The construction is the same as the independence case, where for any $n \geq 1$ and $S_n = \frac{1}{n}\sum_{i=1}^n X_i$, we let

$$\mathcal{F}_{-n} = \sigma(S_n, X_{n+1}, X_{n+2}, \ldots). \tag{2.5.21}$$

As before, $S_{n+1} = (nS_n + X_{n+1})/(n+1)$ gives $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$, hence this forms a backward filtration.

Permuting the first n observations leaves S_n and all later observations unchanged. Thus the symmetry argument in Example 2.48 applies and gives

$$\mathbb{E}[\,X_1 |\,\mathcal{F}_{-n}] = S_n =: M_{-n}. \tag{2.5.22}$$

The tower property makes $(M_k)_{k \leq -1}$ a backward martingale. By Theorem 2.47, $S_n \to S_\infty$ almost surely for some finite random variable S_∞.

It remains to identify the limit. Intuitively speaking, we are averaging over the randomness of Y_i but leaving Z fixed. In particular, conditional on Z, we should heuristically recover a type of law of large numbers and get

$$S_n | Z \xrightarrow{p} \mathbb{E}[X_1 | \sigma(Z)]\,. \tag{2.5.23}$$

To this goal, we will define

$$m(z) = \mathbb{E}f(Y_1, z), \tag{2.5.24}$$

where z is fixed and the expectation averages over Y_1. The moment assumption makes this expectation finite for almost every value selected by Z; set $m(z) = 0$ elsewhere. Independence gives $m(Z) = \mathbb{E}[\,X_i |\,\sigma(Z)]$ for every i. Conditional Jensen's inequality gives $\mathbb{E}m(Z)^2 \leq \mathbb{E}X_1^2 < \infty$.

We will also define $D_i := X_i - m(Z)$, and note that $\mathbb{E}D_i = \mathbb{E}[X_i - \mathbb{E}[X_i|\sigma(Z)]] = 0$. It's sufficient to show the law of large numbers applies to D_i instead of X_i, and in order to show this, we will also require the D_i's to be *uncorrelated*.

For $i \neq j$, Y_j is independent of the pair (Y_i, Z), so $\mathbb{E}[\,D_j |\,\sigma(Y_i, Z)] = m(Z) - m(Z) = 0$. Since D_i is $\sigma(Y_i, Z)$-measurable, we have

$$\mathbb{E}[D_i D_j] = \mathbb{E}[D_i\,\mathbb{E}[D_j|\sigma(Y_i, Z)]] = 0\,, \tag{2.5.25}$$

Expanding the square and using the identical distributions of the D_i therefore gives

$$\mathbb{E}(S_n - m(Z))^2 = \frac{1}{n}\mathbb{E}D_1^2 \longrightarrow 0. \tag{2.5.26}$$

Next we will use Chebyshev's inequality to get $S_n \to m(Z)$ in probability. Since $S_n \to S_\infty$ almost surely, uniqueness of limits in probability identifies $S_\infty = m(Z)$ almost surely. Consequently,

$$S_n \longrightarrow m(Z) = \mathbb{E}[\,X_1 |\, \sigma(Z)] \qquad \text{almost surely.} \tag{2.5.27}$$

This is a law of large numbers with the conditional mean as its limit. Note the limit here can be random, since the shared random variable Z is not averaged away.

Exercise 2.19 (Forgetting an Observation)**.** Let $\xi_1, \xi_2, \ldots$ be independent Bernoulli random variables with $\mathbb{P}(\xi_n = 1) = p$, where $0 < p < 1$. Set $\mathcal{F}_{-n} = \sigma(\xi_n, \xi_{n+1}, \ldots)$ and $M_{-n} = \mathbb{E}[\,\xi_1 |\, \mathcal{F}_{-n}]$ for $n \geq 1$.

(a) Show that $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$, so $(\mathcal{F}_k)_{k \leq -1}$ is a filtration.

(b) Compute M_{-1} and M_{-n} for $n \geq 2$. Check the backward martingale property and identify the almost-sure limit.

(c) Set $X_{-n} = \xi_n$ for $n \geq 1$. Is $(X_k)_{k \leq -1}$ itself a backward martingale with respect to $(\mathcal{F}_k)_{k \leq -1}$? Explain the distinction between this process and $(M_k)_{k \leq -1}$.

Exercise 2.20 (Empirical Frequencies)**.** Let $X_1, X_2, \ldots$ be i.i.d. real-valued random variables.

(a) For a Borel set B, apply the strong law to $\mathbf{1}_{\{X_i \in B\}}$ to show that

$$\frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{X_i \in B\}} \longrightarrow \mathbb{P}(X_1 \in B) \qquad \text{almost surely.} \tag{2.5.28}$$

(b) Let $B_1, B_2, \ldots$ be a countable family of Borel sets. Show that the convergence in (a) holds for all these sets simultaneously outside one null event.

(c) Take $B = (-\infty, x]$. Explain why the observed fraction of values at most x converges to the CDF $F_X(x)$, and why this holds simultaneously for every rational x.

(d) Define the empirical probability measure

$$\mu_n = \frac{1}{n} \sum_{i=1}^{n} \delta_{X_i}. \tag{2.5.29}$$

Show that, almost surely,

$$\mu_n \xrightarrow{d} \mu_X. \tag{2.5.30}$$

Hint: Use the simultaneous convergence from part (c). On the resulting probability-one event, choose rational $q < x < r$ and use monotonicity to extend convergence to every continuity point of F_X. Then use the CDF characterization of weak convergence.

Chapter 3

Markov Chains

3.1 Finite-State Markov Chains and Duality

For a general stochastic process $(X_0, X_1, \ldots, X_n)$, X_n will depend on the entire history of $(X_0, X_1, \ldots, X_{n-1})$. For example, if the X_i are words in a dictionary D (a finite set), then we can imagine that a full random sentence $(X_0, \ldots, X_n)$ contains a structure where all words depend on each other.

In fact, a modern language model is essentially a sequence of functions $(F_n)_{n\geq 1}$, each of which maps a sequence $(X_0, \ldots, X_{n-1}) \in D^n$ to a probability distribution on D. More precisely,

$$F_n : D^n \to \mathcal{P}(D)\,, \tag{3.1.1}$$

and we can let $\mu_n = F_n(X_0, \ldots, X_{n-1})$ be the output of the language model and then sample $X_n \sim \mu_n$. As a result, X_n is trivially dependent on the entire history.

However, the probabilistic structure of such processes is very challenging to model and analyze (i.e. hard to prove theorems). We need to look for a *simpler* model to start the analysis with. And one obvious idea to simplify is to let X_n *only depend on* the previous value X_{n-1}.

To make this idea precise, we will require the following definitions.

We begin by recalling Markov chains on a finite state space, where distributions are vectors and transitions are described by a matrix. The same matrix will act on both distributions and functions of the state. Understanding these two actions will help us extend the theory to continuous state spaces.

Definition 3.1 (Finite-State Markov Chain)**.** Let $E = \{1, \ldots, m\}$, and let $P = (P_{ij})_{i,j\in E}$ be a **transition matrix**, meaning that $P_{ij} \geq 0$ and $\sum_{j\in E} P_{ij} = 1$ for every $i \in E$. An E-valued process $(X_n)_{n\geq 0}$ is a **time-homogeneous Markov chain** with transition matrix P if

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, \ldots, X_n = i_n) = P_{i_n j} \tag{3.1.2}$$

for every $n \geq 0$, $j \in E$, and history $(i_0, \ldots, i_n)$ having positive probability.

The left side of (3.1.2) conditions on the whole observed history, while the right side depends only on the current state i_n. This is often called the **Markov property**. Time homogeneity just means that the same transition matrix applies at every step, where as generically you can let P depend on $n \in \mathbb{N}$.

Let $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$ be the information available at time n. Using the conditional-expectation notation from Chapter 2, the Markov property is equivalently

$$\mathbb{E}[\mathbf{1}_{\{X_{n+1}=j\}} \Big| \mathcal{F}_n] = P_{X_n j}, \qquad j \in E. \tag{3.1.3}$$

Indeed, on each history with positive probability, both sides give the conditional probability in (3.1.2).

At this point, we will introduce the concept of **duality**. To start, we can write the law of X_n as a row vector μ_n, with entries $\mu_{n,i} = \mathbb{P}(X_n = i)$. We can then identify a function $\varphi : E \to \mathbb{R}$ with the column vector of its values. The indicator $\mathbf{1}_{\{j\}}$ is the standard basis vector with a one in position j. These indicators form a basis for the space of functions on E, since every function has the expansion

$$\varphi = \sum_{j \in E} \varphi(j) \mathbf{1}_{\{j\}}. \tag{3.1.4}$$

For a fixed law of X_n, the expectation is a **linear functional** on this space, i.e. a linear function that maps to a real value. In this case, it maps each function φ to the number $\mathbb{E}\varphi(X_n)$. A linear functional is determined by its values on a basis, and in finite dimensions, it can be represented as an **inner product**.

Here those values are exactly the state probabilities:

$$\mathbb{E}\mathbf{1}_{\{j\}}(X_n) = \mathbb{P}(X_n = j) = \mu_{n,j}. \tag{3.1.5}$$

Applying linearity to (3.1.4), and writing $\varphi_i = \varphi(i)$ for the vector coordinates, gives

$$\mathbb{E}\varphi(X_n) = \sum_{i \in E} \mu_{n,i} \varphi_i = \underbrace{\mu_n}_{1 \times m} \underbrace{\varphi}_{m \times 1}, \tag{3.1.6}$$

where the final expression is a vector inner product. Thus the probability row vector μ_n is the coordinate representation of the expectation functional acting on column vectors of function values.

To see the corresponding dual basis, let δ_i be the point mass at state i, represented by the row vector with a one in position i. It acts on a function by evaluating its i-th coordinate: $\delta_i \varphi = \varphi(i)$. In particular,

$$\delta_i \mathbf{1}_{\{j\}} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{3.1.7}$$

The point masses $\delta_1, \ldots, \delta_m$ form the **dual basis** to the indicators $\mathbf{1}_{\{1\}}, \ldots, \mathbf{1}_{\{m\}}$. A general law combines these coordinate evaluations with its probability weights:

$$\mu_n = \sum_{i \in E} \mu_{n,i} \delta_i. \tag{3.1.8}$$

This is the **duality between laws and functions**: a law acts on functions as a linear functional, and expectation is their pairing.

We call φ a **test function** because we can use its expectation to test a law. For instance, an indicator $\varphi = \mathbf{1}_B$ gives the probability that the state belongs to B:

$$\mu_n \mathbf{1}_B = \mathbb{E}\mathbf{1}_B(X_n) = \mathbb{P}(X_n \in B). \tag{3.1.9}$$

Lemma 3.2 (Test Functions Determine the Law)**.** *Let μ and ν be two probability distributions on the finite state space E. Then*

$$\mu = \nu \quad \Longleftrightarrow \quad \mu\varphi = \nu\varphi \quad \text{for every } \varphi : E \to \mathbb{R}. \tag{3.1.10}$$

In particular, if $\mu \neq \nu$, there exists a test function whose expectations under the two laws are different.

Proof. If $\mu = \nu$, the expectations agree for every test function by (3.1.6). Conversely, if $\mu \neq \nu$, set $B = \{i \in E : \mu_i > \nu_i\}$. Since both distributions have total mass one, B is nonempty. The test function $\varphi = \mathbf{1}_B$ then gives

$$\mu\varphi - \nu\varphi = \sum_{i \in B} (\mu_i - \nu_i) > 0. \tag{3.1.11}$$

Thus agreement for every test function implies $\mu = \nu$, and the same calculation supplies a distinguishing test function when the laws differ. □

To the probability of the next state X_{n+1} starting from the law of X_n, we can take expectations of the conditional transition probability in (3.1.3) and get

$$\mu_{n+1,j} = \mathbb{E}\mathbb{P}(X_{n+1} = j | \mathcal{F}_n) = \mathbb{E}P_{X_n j} = \sum_{i \in E} \mu_{n,i} P_{ij}, \qquad \text{or} \qquad \mu_{n+1} = \mu_n P. \tag{3.1.12}$$

In other words, a **left multiplication** of μ_n onto P **updates the probability** to μ_{n+1}.

We now use a concrete example to see how the same matrix acts on test functions.

Example 3.3 (Expected Next Values)**.** Consider a chain on $E = \{0, 1\}$ driven by independent tosses of a biased coin. At each step, with probability p we move to the other state, and with probability $1 - p$ we stay at the current state, where $0 < p < 1$. Its transition matrix is

$$P = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}, \tag{3.1.13}$$

where rows and columns are ordered as $0, 1$. Let φ be a test function with values φ_0 and φ_1 at the two states. The arrows in Figure 3.1 show the possible transitions from each state.

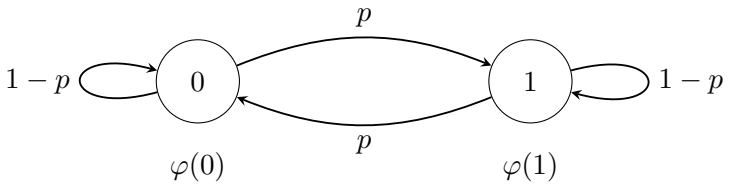


Figure 3.1: Outgoing arrows give the possible next states and their probabilities. To calculate the expected next value of a test function, weight its destination values by the probabilities on the outgoing arrows.

Starting from state 0, the expected next value of the function is $(1-p)\varphi_0 + p\varphi_1$. Starting from state 1, it is $p\varphi_0 + (1-p)\varphi_1$. These two answers form a new function on the state space, which is exactly the vector obtained by multiplying φ by P:

$$P\varphi = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix} \begin{pmatrix} \varphi_0 \\ \varphi_1 \end{pmatrix} = \begin{pmatrix} (1-p)\varphi_0 + p\varphi_1 \\ p\varphi_0 + (1-p)\varphi_1 \end{pmatrix}. \tag{3.1.14}$$

Thus $P\varphi$ records an expected next value for each possible current state.

We can change the question while keeping the same chain. For example, consider the test function $\mathbf{1}_{\{1\}}$, which records whether the state is 1. Averaging its zero-or-one values over the next transition gives

$$P\mathbf{1}_{\{1\}} = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} p \\ 1-p \end{pmatrix}. \tag{3.1.15}$$

The two entries of $P\mathbf{1}_{\{1\}}$ are the probabilities of reaching state 1 from states 0 and 1, respectively. The same matrix multiplication calculates the expected next value of any test function.

In general, the same matrix acts on a test function by averaging its possible next values. For a fixed starting state i, define

$$(P\varphi)(i) = \sum_{j \in E} P_{ij}\varphi_j. \tag{3.1.16}$$

This is the expected value of φ after one step from i. As i varies, these expected values form a new function $P\varphi$ on E. In other words, a **right multiplication** of φ to the transition probability P computes the **conditional expectation** of the next step given the current state.

By the basis expansion in (3.1.4), every test function is a linear combination of indicators on each state. Therefore the indicator calculation in Example 3.3 extends to every test function by the linearity property. Expanding $\varphi(X_{n+1})$ in indicators and using (3.1.3) gives

$$\begin{aligned} \mathbb{E}[\varphi(X_{n+1})|\,\mathcal{F}_n] &= \sum_{j \in E} \varphi_j \mathbb{E}[\mathbf{1}_{\{X_{n+1}=j\}} \Big|\, \mathcal{F}_n] \\ &= \sum_{j \in E} P_{X_n j}\varphi_j = (P\varphi)(X_n). \end{aligned} \tag{3.1.17}$$

We will also refer to the map $\varphi \mapsto P\varphi$ as the **transition operator**.

Proposition 3.4 (Duality Between Distributions and Test Functions)**.** *For every probability row vector μ and test function φ,*

$$(\mu P)\varphi = \mu(P\varphi). \tag{3.1.18}$$

Proof. We group the same finite sum first by the next state and then by the starting state:

$$\begin{aligned} (\mu P)\varphi &= \sum_{j \in E} \left(\sum_{i \in E} \mu_i P_{ij} \right) \varphi_j \\ &= \sum_{i \in E} \mu_i \left(\sum_{j \in E} P_{ij}\varphi_j \right) = \mu(P\varphi). \end{aligned} \tag{3.1.19}$$

□

On the left side of (3.1.18), we first find the next distribution and then average the test function. On the right side, we first calculate the expected next value from each starting state and then average over the starting distribution. This identity expresses the duality between the two actions of P. In matrix notation it is associativity; in probability it is total expectation:

$$\mathbb{E}\varphi(X_{n+1}) = \mathbb{E}\left[\mathbb{E}[\varphi(X_{n+1})|\,\mathcal{F}_n]\right] = \mathbb{E}(P\varphi)(X_n). \tag{3.1.20}$$

For several steps, the same matrix powers describe both actions. The laws evolve by $\mu_n = \mu_0 P^n$, while repeated conditioning gives

$$\mathbb{E}[\varphi(X_{n+k})|\,\mathcal{F}_n] = (P^k\varphi)(X_n), \qquad n,k \geq 0. \tag{3.1.21}$$

For example, when $k = 2$, the tower property and (3.1.17) give

$$\begin{aligned}\mathbb{E}[\varphi(X_{n+2})|\,\mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[\varphi(X_{n+2})|\,\mathcal{F}_{n+1}]|\,\mathcal{F}_n] \\ &= \mathbb{E}[\,(P\varphi)(X_{n+1})|\,\mathcal{F}_n] = (P^2\varphi)(X_n).\end{aligned} \tag{3.1.22}$$

Repeating this argument proves (3.1.21) for every k, with $P^0 = I$. Taking $\varphi = \mathbf{1}_{\{j\}}$ shows that $(P^k)_{ij}$ is the probability of reaching j in k steps from i.

Write $P_n = P^n$. The family $(P_n)_{n\geq 0}$ is the **discrete-time transition semigroup**, with

$$P_0 = I, \qquad P_{r+s} = P_r P_s, \qquad r,s \geq 0. \tag{3.1.23}$$

In terms of entries, this is the **Chapman–Kolmogorov identity**

$$(P^{r+s})_{ij} = \sum_{\ell\in E} (P^r)_{i\ell}(P^s)_{\ell j}. \tag{3.1.24}$$

We sum over the possible intermediate states after r steps. Duality at time n is therefore

$$\mathbb{E}\varphi(X_n) = (\mu_0 P^n)\varphi = \mu_0(P^n\varphi). \tag{3.1.25}$$

The probability of a particular path is its starting probability multiplied by the transition probabilities along that path. This gives the joint laws of the chain.

Proposition 3.5 (Joint Laws of a Finite-State Markov Chain)**.** *Let $(X_n)_{n\geq 0}$ be a Markov chain on a finite state space E, with initial distribution μ_0 and transition matrix P. For states $i_0,\dots,i_n \in E$ and $n \geq 1$,*

$$\mathbb{P}(X_0 = i_0,\dots,X_n = i_n) = \mu_{0,i_0}\prod_{k=1}^{n} P_{i_{k-1},i_k}. \tag{3.1.26}$$

In particular, the initial distribution and transition matrix determine all finite-dimensional distributions of the chain.

Proof. We condition on the history to remove one transition at a time. Set $A = \{X_0 = i_0,\dots,X_{n-1} = i_{n-1}\} \in \mathcal{F}_{n-1}$. By the Markov property and total expectation,

$$\begin{aligned}\mathbb{P}(A \cap \{X_n = i_n\}) &= \mathbb{E}\big[\mathbf{1}_A \mathbb{E}[\mathbf{1}_{\{X_n=i_n\}}\big|\,\mathcal{F}_{n-1}]\big] \\ &= \mathbb{E}[\mathbf{1}_A P_{X_{n-1},i_n}] = P_{i_{n-1},i_n}\mathbb{P}(A).\end{aligned} \tag{3.1.27}$$

The last equality uses $X_{n-1} = i_{n-1}$ on A. Repeating this step and using $\mathbb{P}(X_0 = i_0) = \mu_{0,i_0}$ gives (3.1.26). □

Summing the path probabilities over the intermediate states recovers the matrix-power formula for the probability of reaching a specified final state. The same path formula holds on a countable state space.

Recall that a probability row vector π is a **stationary distribution** if $\pi P = \pi$. By duality, this is equivalent to $\pi(P\varphi) = \pi\varphi$ for every test function φ. The converse follows by choosing the indicators of individual states. Starting with law π then gives $\mu_n = \pi$ at every time. Stationarity alone does not say that other initial distributions approach π.

Example 3.6 (A Two-Step Expectation)**.** Consider the biased-coin chain from Example 3.3. After two steps, we are back at the starting state if we stay twice or switch twice, with total probability $(1-p)^2 + p^2$. We are at the other state if we switch exactly once, with probability $2p(1-p)$. Applying the transition matrix twice to a general test function therefore gives

$$P^2\varphi = \begin{pmatrix} (1-p)^2 + p^2 & 2p(1-p) \\ 2p(1-p) & (1-p)^2 + p^2 \end{pmatrix} \begin{pmatrix} \varphi_0 \\ \varphi_1 \end{pmatrix}. \tag{3.1.28}$$

For example, take the test function $\mathbf{1}_{\{1\}}$ and start at $X_0 = 0$. The probability of being at state 1 after two steps is

$$\mathbb{P}(X_2 = 1) = (P^2\mathbf{1}_{\{1\}})(0) = 2p(1-p). \tag{3.1.29}$$

This is the probability that exactly one of the two tosses makes us switch states.

The same vector and matrix formulas extend to countable state spaces by using infinite sums. For bounded test functions, the sums defining $P\varphi$ and the pairing with a probability distribution are absolutely convergent, so the duality identity still holds. For continuous state spaces, probability vectors and these sums will be replaced by probability measures and integrals. Measurability will ensure that the resulting functions can be evaluated at a random state and averaged.

Exercise 3.1 (What Test Functions Determine)**.** Let $E = \{1, \dots, m\}$ with $m \geq 3$, and fix a test function φ. This exercise shows why its expectation alone does not determine a probability law.

(a) Use linear algebra to show that there is a nonzero row vector v with $\sum_{i \in E} v_i = 0$ and $v\varphi = 0$. Hint: these are two homogeneous linear equations in m unknowns.

(b) Let μ be the uniform probability vector on E. Show that, for sufficiently small $\varepsilon > 0$, the vector $\nu = \mu + \varepsilon v$ is a different probability law with $\nu\varphi = \mu\varphi$.

(c) Explain why this does not contradict Lemma 3.2. Which test functions are sufficient to distinguish every pair of probability laws?

Exercise 3.2 (A Chain from Independent Coin Tosses)**.** Let $\xi_1, \xi_2, \dots$ be independent Bernoulli random variables with parameter $p \in (0,1)$, independent of $X_0 \in \{0,1\}$. At each step, set $X_{n+1} = X_n$ if $\xi_{n+1} = 0$ and $X_{n+1} = 1 - X_n$ if $\xi_{n+1} = 1$.

(a) Prove that this is a Markov chain with transition matrix (3.1.13).

(b) Prove that the uniform law on $\{0,1\}$ is stationary. Explain why every X_n is uniform when X_0 is uniform.

(c) Show that when $p = 1/2$, the variables $X_1, X_2, \dots$ are independent and uniform. When $p \neq 1/2$ and X_0 is uniform, show that X_0 and X_1 are not independent.

Exercise 3.3 (Conditional Expectations Along a Markov Chain)**.** Let (X_n) be a Markov chain on a finite state space E with transition matrix P and natural filtration $(\mathcal{F}_n)$. Fix an integer $N \geq 1$ and a test function $\varphi : E \to \mathbb{R}$, and set $M_k = (P^{N-k}\varphi)(X_k)$ for $0 \leq k \leq N$.

(a) Show that each M_k is $\mathcal{F}_k$-measurable and integrable.

(b) Use the one-step Markov property and the tower property to verify

$$M_k = \mathbb{E}[\varphi(X_N) | \mathcal{F}_k], \qquad 0 \leq k \leq N. \tag{3.1.30}$$

Hint: start at $k = N$ and proceed to earlier times.

(c) Use part (b) and the tower property to prove that $(M_k)_{k=0}^N$ is a martingale.

Exercise 3.4 (The Same Stationary Distribution, Different Dynamics)**.** On $E = \{0, 1\}$, consider the two transition matrices

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{3.1.31}$$

(a) Show that the uniform law π on $\{0, 1\}$ is stationary for both matrices.

(b) Start each chain with law π and describe its sample paths. Use these descriptions to show that the chains have the same one-time laws but different joint laws.

(c) Instead start at $X_0 = 0$. Prove that neither chain's law converges to π. Explain what this shows about stationarity and convergence.

3.2 Continuous-State Markov Chains and Transition Kernels

On a finite state space, each row of the transition matrix is a probability distribution for the next state. The same idea applies on a continuous state space, but a probability vector is replaced by a probability measure. We will mainly consider a state space $E \subseteq \mathbb{R}^d$, with its Borel σ-algebra $\mathcal{B}(E)$. For a continuously distributed next state, the probability of each individual point can be zero, so we describe the probabilities of Borel sets instead.

Definition 3.7. A **transition kernel** is a function $P : E \times \mathcal{B}(E) \to [0, 1]$ such that

(1) for each $x \in E$, $B \mapsto P(x, B)$ is a probability measure on E, and

(2) for each $B \in \mathcal{B}(E)$, $x \mapsto P(x, B)$ is measurable.

We interpret $P(x, B)$ as the probability of moving into B in one step when the current state is x. In particular, we can imagine sampling $Y \sim P(x, \cdot)$, which means that $\mathbb{P}(Y \in B) = P(x, B)$. The first condition gives the next-state law from each fixed starting state. The second condition lets us evaluate these probabilities at a random current state: $P(X_n, B)$ must be a measurable random variable before we can take its expectation.

We can now state the Markov property in terms of conditional probabilities.

Definition 3.8 (Markov Chain)**.** Let $(X_n)_{n\geq 0}$ take values in E, and let $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$. We call $(X_n)_{n\geq 0}$ a **time-homogeneous Markov chain** with transition kernel P if

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n) = \mathbb{P}(X_{n+1} \in B \mid \sigma(X_n)) = P(X_n, B) \tag{3.2.1}$$

almost surely, for every $n \geq 0$ and every $B \in \mathcal{B}(E)$.

Here it's important to observe that the left side uses the whole observed history, while the right side uses only X_n. This is the **Markov property**. The same kernel describes the entire conditional distribution of the next state.

Example 3.9 (Continuous States with Two Possible Next Values)**.** Let $\xi_1, \xi_2, \ldots$ be independent Bernoulli random variables with parameter $p \in (0,1)$, independent of $X_0 \in [0,1]$. Define $X_{n+1} = (X_n + \xi_{n+1})/2$. As in the finite-state coin example, each step uses a fresh toss independent of the past. From a current state x, the next state is $x/2$ or $(x+1)/2$, so

$$P(x, \cdot) = (1-p)\delta_{x/2} + p\delta_{(x+1)/2}. \tag{3.2.2}$$

Conditioning on the history fixes X_n and leaves the fresh toss with its original law, giving (3.2.1). The state space is continuous even though each next-state law is supported on two points. In particular, a transition kernel need not have a density.

Example 3.10 (AR(1) Process)**.** Let $\xi_1, \xi_2, \ldots$ be independent $\mathcal{N}(0,1)$ random variables, independent of X_0. For a fixed $a \in \mathbb{R}$, we define the autoregressive process of order 1 (AR(1)) by

$$X_{n+1} = aX_n + \xi_{n+1}, \qquad n \geq 0. \tag{3.2.3}$$

The history $X_0, \ldots, X_n$ is determined by $X_0, \xi_1, \ldots, \xi_n$, so the new noise ξ_{n+1} is independent of $\mathcal{F}_n$. Given the current information, X_n is known and the next value is Gaussian with mean aX_n and variance one. Thus this is a Markov chain with transition kernel

$$P(x, B) = \int_B \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(y-ax)^2}{2}\right) dy. \tag{3.2.4}$$

The same idea applies when the update is a measurable function of the current state and fresh independent noise. The earlier noise affects the next step only through the current state.

A transition kernel tells us how to move from one state to the next. To describe the whole chain, we also need to specify where it starts. Let μ_0 be the law of X_0. By the Markov property and total expectation, for every $B \in \mathcal{B}(E)$,

$$\mathbb{P}(X_1 \in B) = \mathbb{E}P(X_0, B) = \int_E P(x, B)\, \mu_0(dx) =: (\mu_0 P)(B). \tag{3.2.5}$$

Thus $\mu_0 P$ is the distribution after one step when the starting distribution is μ_0. This extends multiplication of a probability row vector by the transition matrix.

The pairing between laws and test functions also has the same interpretation as before. For a probability measure μ and a bounded measurable function $\varphi : E \to \mathbb{R}$, write

$$\mu(\varphi) = \int_E \varphi(x)\, \mu(dx). \tag{3.2.6}$$

If X has law μ, then this is $\mathbb{E}\varphi(X)$. The notation $\mu(\varphi)$ replaces the row-vector and column-vector product $\mu\varphi$ from Section 3.1.

Remark 3.11 (Expectation as an Inner Product). The law tells us how to weight the values of a test function when taking an expectation. On a finite state space or if μ is a discrete measure, this is the usual inner product of vectors:

$$\mu(\varphi) = \sum_{i \in E} \mu_i \varphi_i = \langle \mu, \varphi \rangle. \tag{3.2.7}$$

If μ has a density f, the sum is replaced by an integral:

$$\mu(\varphi) = \int_E f(x)\varphi(x)\,dx = \langle f, \varphi\rangle_{L^2(E,dx)}, \tag{3.2.8}$$

where the inner-product notation applies when $f, \varphi \in L^2(E, dx)$. Thus expectation is an inner product of functions, as in the previous chapter, with the density supplying the weights. Integration against a probability measure extends this relationship to laws that do not have a density.

Two laws agree if and only if their expectations agree for every bounded measurable test function. Indeed, taking $\varphi = \mathbf{1}_B$ recovers the probability of every Borel set B. If the laws differ, the indicator of a set on which they assign different probabilities distinguishes them.

To average the next value of a test function from a fixed state x, we replace the finite sum by an integral:

$$(P\varphi)(x) = \int_E \varphi(y)\,P(x, dy). \tag{3.2.9}$$

This is the **transition operator**, and recall from the finite state space case, this is the conditional expectation of $\varphi(X_{n+1})$ if X_n starts at x.

Here x is the current state and y is the possible next state being averaged over. The output $P\varphi$ is again a function of the current state. For bounded measurable φ, integration against a kernel can be constructed from approximations by a linear combination of indicator functions in Definition 3.8.

See Appendix A.1 for more details, but we will not go into the technicalities here except to say that $P\varphi$ is measurable and bounded in absolute value, hence we can evaluate conditional expectations of $(P\varphi)(X_n)$.

Since $P(X_n, \cdot)$ is the conditional law of X_{n+1} given $\mathcal{F}_n$, integrating a bounded measurable test function against this law gives

$$\mathbb{E}[\varphi(X_{n+1})|\,\mathcal{F}_n] = (P\varphi)(X_n). \tag{3.2.10}$$

As mentioned before, the Markov property specifies conditional expectations of indicators, and we can extend this to all bounded measurable functions via an approximation argument in Appendix A.1. Taking $\varphi = \mathbf{1}_B$ recovers the conditional probabilities in the definition.

For the two-map chain in Example 3.9, the operator is

$$(P\varphi)(x) = (1-p)\varphi(x/2) + p\varphi((x+1)/2). \tag{3.2.11}$$

For the AR(1) process, it can be written as

$$(P\varphi)(x) = \mathbb{E}\varphi(ax + \xi_1) = \int_{\mathbb{R}} \varphi(ax+z)\frac{e^{-z^2/2}}{\sqrt{2\pi}}\,dz. \tag{3.2.12}$$

Here we hold x fixed, evaluate the function at the next state, and average over the fresh noise. The result is still a function of the current state x.

Proposition 3.12 (Duality for a Transition Kernel)**.** *For a probability measure μ and a bounded measurable test function φ,*

$$(\mu P)(\varphi) = \mu(P\varphi). \tag{3.2.13}$$

Proof. We use the basic integration rule for a kernel: averaging the kernel measures and then integrating a bounded measurable function is the same as averaging their integrals. Thus

$$(\mu P)(\varphi) = \int_E \left(\int_E \varphi(y)\, P(x, dy) \right) \mu(dx) = \mu(P\varphi). \tag{3.2.14}$$

□

Here on the left, we first update the distribution and then average the function. On the right, we first calculate the expected next value from each starting state and then average over the initial distribution. This is the same duality as in Section 3.1, with integrals replacing the finite sums.

For several steps, compose the transition operator as before, writing $P_n = P^n$ and $P_0 = I$. The same tower-property calculation used in the finite-state case gives

$$\mathbb{E}[\,\varphi(X_{n+k}) |\, \mathcal{F}_n] = (P^k \varphi)(X_n), \qquad n, k \geq 0. \tag{3.2.15}$$

The calculation is valid because each successive $P^k\varphi$ is still bounded and measurable. Taking indicators gives the k-step transition kernel $P_k(x, B) = (P^k \mathbf{1}_B)(x)$. In particular,

$$P_2(x, B) = \int_E P(y, B)\, P(x, dy). \tag{3.2.16}$$

We average the probability of reaching B from an intermediate state y over the law of that intermediate state. More generally, the Chapman–Kolmogorov identity becomes

$$P_{r+s}(x, B) = \int_E P_s(y, B)\, P_r(x, dy), \qquad r, s \geq 0. \tag{3.2.17}$$

The laws therefore evolve by

$$\mathbb{P}(X_n \in B) = (\mu_0 P_n)(B) = (\mu_0 P^n)(B). \tag{3.2.18}$$

These are the individual-time distributions. The same initial distribution and transition kernel also determine how the observations at different times depend on one another.

Proposition 3.13 (Joint Laws of a Markov Chain)**.** *Let $(X_n)_{n\geq 0}$ be a Markov chain with initial distribution μ_0 and transition kernel P. For $n \geq 1$ and Borel sets $B_0, \ldots, B_n \subseteq E$,*

$$\begin{aligned} &\mathbb{P}(X_0 \in B_0, \ldots, X_n \in B_n) \\ &\quad = \int_{B_0} \mu_0(dx_0) \int_{B_1} P(x_0, dx_1) \cdots \int_{B_n} P(x_{n-1}, dx_n). \end{aligned} \tag{3.2.19}$$

In particular, μ_0 and P determine every finite-dimensional distribution of the chain.

Proof. We condition on the history before the last observation and then repeat this step backwards.

First write the joint probability as the expectation of a product of indicators:

$$\begin{aligned}\mathbb{P}(X_0 \in B_0, \dots, X_n \in B_n) &= \mathbb{E}\left[\prod_{k=0}^{n} \mathbf{1}_{\{X_k \in B_k\}}\right] \\ &= \mathbb{E}\left[\mathbb{E}[\prod_{k=0}^{n} \mathbf{1}_{\{X_k \in B_k\}} \,\middle|\, \mathcal{F}_{n-1}]\right] \\ &= \mathbb{E}\left[\prod_{k=0}^{n-1} \mathbf{1}_{\{X_k \in B_k\}} \mathbb{E}[\mathbf{1}_{\{X_n \in B_n\}} \,\middle|\, \mathcal{F}_{n-1}]\right] \\ &= \mathbb{E}\left[\prod_{k=0}^{n-1} \mathbf{1}_{\{X_k \in B_k\}} P(X_{n-1}, B_n)\right].\end{aligned} \tag{3.2.20}$$

The second equality uses total expectation. In the third equality, the indicators involving $X_0, \dots, X_{n-1}$ are $\mathcal{F}_{n-1}$-measurable, so they can be taken out of the conditional expectation. The final equality uses the Markov property to replace the conditional probability of $X_n \in B_n$ by $P(X_{n-1}, B_n)$.

The factor $\mathbf{1}_{B_{n-1}}(X_{n-1})P(X_{n-1}, B_n)$ is a bounded measurable function of X_{n-1}. For $n \geq 2$, applying (3.2.10) gives

$$\begin{aligned}\mathbb{E}[\mathbf{1}_{B_{n-1}}(X_{n-1})P(X_{n-1}, B_n) \,|\, \mathcal{F}_{n-2}] &= \int_{B_{n-1}} P(x_{n-1}, B_n) P(X_{n-2}, dx_{n-1}) \\ &= \int_{B_{n-1}} P(X_{n-2}, dx_{n-1}) \int_{B_n} P(x_{n-1}, dx_n).\end{aligned} \tag{3.2.21}$$

We now condition the product inside the expectation in (3.2.20) on $\mathcal{F}_{n-2}$ and take out the earlier indicators to get

$$\begin{aligned}&\mathbb{P}(X_0 \in B_0, \dots, X_n \in B_n) \\ &\quad = \mathbb{E}\left[\prod_{k=0}^{n-2} \mathbf{1}_{\{X_k \in B_k\}} \int_{B_{n-1}} P(X_{n-2}, dx_{n-1}) \int_{B_n} P(x_{n-1}, dx_n)\right].\end{aligned} \tag{3.2.22}$$

Each repetition replaces one more observation by an integral against its transition kernel. Continue until only a function of X_0 remains, and then integrate against μ_0 to obtain (3.2.19). For $n = 1$, the first conditioning step already leaves only the expectation over X_0. □

When the initial distribution has density f_0 and $P(x, dy) = p(x, y)\, dy$ for a jointly measurable transition density p, the joint density takes the familiar product form

$$f_{0:n}(x_0, \dots, x_n) = f_0(x_0) \prod_{k=1}^{n} p(x_{k-1}, x_k). \tag{3.2.23}$$

Each factor describes one stage of sampling: first choose the initial value, then sample each new value using the previous one.

We can now ask whether there is a starting distribution that is preserved by this sampling procedure.

Definition 3.14. A probability measure π on E is a **stationary distribution** for P if $\pi P = \pi$, that is,

$$\int_E P(x, B)\, \pi(dx) = \pi(B) \qquad \text{for every } B \in \mathcal{B}(E). \tag{3.2.24}$$

Intuitively speaking, this is saying that if we started at $X_0 \sim \pi$, then after one step of the Markov chain, our next step X_1 is also distributed as π. Furthermore, we can take n steps and we still have $\pi P^n = \pi$, so every X_n has the same law.

By duality and the fact that test functions determine the law, stationarity is equivalently

$$\pi(P\varphi) = \pi(\varphi) \qquad \text{for every bounded measurable } \varphi. \tag{3.2.25}$$

Thus stationarity means that every expected test value is preserved by one transition.

Example 3.15 (A Stationary Distribution on an Interval)**.** Return to the chain $X_{n+1} = (X_n + \xi_{n+1})/2$ from Example 3.9, with $p = 1/2$. Let λ be the uniform law on $[0, 1]$. For a bounded measurable test function φ, the two affine changes of variables give

$$\begin{aligned} \lambda(P\varphi) &= \frac{1}{2}\int_0^1 \varphi(x/2)\, dx + \frac{1}{2}\int_0^1 \varphi((x+1)/2)\, dx \\ &= \int_0^{1/2} \varphi(y)\, dy + \int_{1/2}^1 \varphi(y)\, dy = \lambda(\varphi). \end{aligned} \tag{3.2.26}$$

Thus λ is stationary. With this initial law, the conditional mean $\mathbb{E}[\, X_1 |\, \sigma(X_0)] = X_0/2 + 1/4$ depends on X_0. Independence would make it equal to the constant $\mathbb{E}X_1 = 1/2$, so the consecutive states are dependent.

Example 3.16 (A Stationary Distribution for the AR(1) Process)**.** Return to $X_{n+1} = aX_n + \xi_{n+1}$ from Example 3.10, and suppose $|a| < 1$. Start from a fixed value $X_0 = x$. Iterating the recursion gives

$$X_n = a^n x + \sum_{k=1}^{n} a^{n-k} \xi_k, \qquad n \geq 1. \tag{3.2.27}$$

This is a sum of independent Gaussian variables, so

$$X_n \sim \mathcal{N}\left(a^n x, \frac{1 - a^{2n}}{1 - a^2}\right). \tag{3.2.28}$$

Taking $n \to \infty$, since $|a| < 1$, we obtain

$$X_n \xrightarrow{d} \pi = \mathcal{N}\left(0, \frac{1}{1 - a^2}\right). \tag{3.2.29}$$

To check that π is stationary, take $X_0 \sim \pi$ independent of the standard Gaussian noises. Then $X_1 = aX_0 + \xi_1$ also has law π, so $\pi P = \pi$. Starting with π preserves the distribution at every time; starting from x makes the distribution approach π as time increases. Sample paths still fluctuate, despite their distributions staying the same when we start with π.

Exercise 3.5 (Kernels from Fresh Noise)**.** Let $\xi_1, \xi_2, \dots$ be independent random variables with common law ρ on $\mathbb{R}^r$, independent of $X_0 \in E$. Let $f : E \times \mathbb{R}^r \to E$ be measurable and set $X_{n+1} = f(X_n, \xi_{n+1})$. Define

$$P(x, B) = \int_{\mathbb{R}^r} \mathbf{1}_B(f(x, z))\, \rho(dz). \tag{3.2.30}$$

(a) Show that $B \mapsto P(x, B)$ is a probability measure for every fixed x.

(b) Show that $x \mapsto P(x, B)$ is measurable for every Borel set B. You may use the integration fact that integrating a bounded jointly measurable function over z against ρ gives a measurable function of x.

(c) Verify the conditional-expectation formula

$$\mathbb{E}[\varphi(X_{n+1}) \,|\, \mathcal{F}_n] = \int_{\mathbb{R}^r} \varphi(f(X_n, z))\, \rho(dz) = (P\varphi)(X_n) \tag{3.2.31}$$

for bounded measurable φ. Explain why the past affects the next step only through X_n.

Exercise 3.6 (A Deterministic Kernel)**.** Let $T : E \to E$ be measurable and define $P(x, \cdot) = \delta_{T(x)}$.

(a) Prove that P is a transition kernel and that $(P\varphi)(x) = \varphi(T(x))$.

(b) Show that $(\mu P)(B) = \mu(T^{-1}(B))$ for every probability law μ and Borel set B.

(c) Prove that π is stationary for P if and only if

$$\int_E \varphi(T(x))\, \pi(dx) = \int_E \varphi(x)\, \pi(dx) \tag{3.2.32}$$

for every bounded measurable test function φ. Explain the relation between preserving a law and preserving all of its expected test values.

Exercise 3.7 (Convergence Using the Same Coin Tosses)**.** Consider the chain $X_{n+1} = (X_n + \xi_{n+1})/2$ on $[0, 1]$, where the fresh tosses have parameter $p = 1/2$. Let λ be its uniform stationary distribution from Example 3.15.

(a) Starting from $X_0 = x$ and $Y_0 = y$, drive both chains with the same tosses. Prove that $X_n - Y_n = 2^{-n}(x - y)$ for every $n \geq 0$.

(b) Now allow arbitrary $X_0, Y_0 \in [0, 1]$, independent of the tosses. Use uniform continuity to show that, for every bounded continuous test function φ,

$$|\mathbb{E}\varphi(X_n) - \mathbb{E}\varphi(Y_n)| \longrightarrow 0. \tag{3.2.33}$$

Hint: the two states are always at distance at most 2^{-n}.

(c) Choose Y_0 with law λ. Deduce that the law of X_n converges weakly to λ from any initial law on $[0, 1]$, and that λ is the unique stationary distribution.

Exercise 3.8 (Combining Transition Kernels)**.** Let P and Q be transition kernels on E and let $\alpha \in [0, 1]$.

(a) Prove that $R(x, B) = \alpha P(x, B) + (1 - \alpha)Q(x, B)$ is a transition kernel. Interpret one transition using R as a random choice of transition rule.

(b) Define

$$(PQ)(x, B) = \int_E Q(y, B)\, P(x, dy). \tag{3.2.34}$$

Prove that PQ is a transition kernel, and show that $(PQ)\varphi = P(Q\varphi)$ and $\mu(PQ) = (\mu P)Q$. Interpret the order of the two transitions.

(c) If π is stationary for both P and Q, prove that it is stationary for both R and PQ.

3.3 Generators and Kolmogorov Equations

Generators play a fundamental role in the study of Markov chains and Markov processes. In discrete time, we define $L = P - I$, so knowing the generator is equivalent to knowing the transition kernel. However, this viewpoint extends to continuous-time processes, where the generator is a much more explicit and convenient object to work with compared to the transition kernel.

Several properties of a Markov chain can be studied through its generator. In this section, we focus on how the generator describes the evolution of expectations. The martingale construction arises as a consequence: subtracting the accumulated expected changes leaves a process with conditionally mean-zero increments.

Definition 3.17 (Discrete Generator)**.** For a bounded measurable function $\varphi : E \to \mathbb{R}$, the **discrete generator** of the chain is the operator $L = P - I$, given by

$$(L\varphi)(x) = (P\varphi)(x) - \varphi(x) = \int_E (\varphi(y) - \varphi(x))\, P(x, dy). \tag{3.3.1}$$

Thus $(L\varphi)(x)$ is the expected change in φ over one step starting from x. On a finite state space, this is the matrix $L = P - I$ acting on the vector of function values:

$$(L\varphi)_i = \sum_{j \in E} P_{ij}(\varphi_j - \varphi_i). \tag{3.3.2}$$

We also use the generator formula (3.3.1) for unbounded φ whenever $\int_E |\varphi(y)|P(x, dy) < \infty$ for every x.

Recall the semigroup property lets us split an evolution into one step followed by the remaining steps. For a bounded measurable test function φ, we can write

$$P_{n+1}\varphi = P(P_n\varphi). \tag{3.3.3}$$

Observe that if we take the difference between the next step and the current, we get

$$P_{n+1}\varphi - P_n\varphi = (P - I)P_n\varphi = LP_n\varphi\,. \tag{3.3.4}$$

Essentially, the generator L gives us yet another way to character the evolution of $P_n\varphi$ via its differences. The same can be applied to probability distributions, which gives us the **Kolmogorov equations** and we will derive them next.

Before we start, we will define $\mathbb{E}_x$ for expectation under the same transition kernel P, with the initial state chosen deterministically as $X_0 = x$. For a bounded measurable test function φ, this means

$$\mathbb{E}_x\, \varphi(X_n) := \int_E \varphi(y)\, P_n(x, dy) = (P_n\varphi)(x). \tag{3.3.5}$$

Note the starting point x is part of the specification of the chain, therefore we do not need to condition on a potentially zero probability event of $\{X_0 = x\}$.

However, conditioning is a precise interpretation when the initial law $\mathcal{L}(X_0)$ and the transition kernel P have densities. In particular, let $\mu_0 = \mathcal{L}(X_0) = f_0(x)dx$, and $P(x, dy) = p(x, y)dy$. Then the joint-density formula (3.2.23) gives us

$$\frac{f_{0:1}(x, y)}{f_0(x)} = p(x, y), \qquad f_0(x) > 0. \tag{3.3.6}$$

Thus the conditional density of X_1 given $X_0 = x$ is $p(x, \cdot)$, and integrating φ against it gives $\mathbb{E}_x \varphi(X_1)$. Just to emphasize once again, the kernel definition lets us use $\mathbb{E}_x$ for any starting state, whether or not densities are available.

For a probability law μ, the notation μL describes the change in the law after one step:

$$(\mu L)(B) = \int_E P(x, B)\, \mu(dx) - \mu(B) = (\mu P)(B) - \mu(B), \qquad B \in \mathcal{B}(E). \tag{3.3.7}$$

On a finite state space, this is the usual row-vector product $\mu L = \mu(P - I)$.

Theorem 3.18 (Kolmogorov Equations)**.** *Let $(X_n)_{n\geq 0}$ be a Markov chain on E with generator $L = P - I$, and let $\mu_n = \mathcal{L}(X_n)$. For a bounded measurable function $\varphi : E \to \mathbb{R}$, define*

$$u_n(x) := \mathbb{E}_x\, \varphi(X_n) = (P_n \varphi)(x). \tag{3.3.8}$$

Then, for every $n \geq 0$,

$$\begin{cases} u_{n+1} - u_n = L u_n, & u_0 = \varphi \quad \textit{(backward)}, \\ \mu_{n+1} - \mu_n = \mu_n L, & \mu_0 = \mathcal{L}(X_0) \quad \textit{(forward)}. \end{cases} \tag{3.3.9}$$

Proof. Both equations follow by subtracting the present value from the one-step update. By (3.3.3), $u_{n+1} = P u_n$, so

$$u_{n+1} - u_n = (P - I)u_n = L u_n. \tag{3.3.10}$$

Similarly, $\mu_{n+1} = \mu_n P$ gives

$$\mu_{n+1} - \mu_n = \mu_n(P - I) = \mu_n L. \tag{3.3.11}$$

□

The reason why the equation for u_n is called a backward equation is because it propagation the future expected value of $\varphi(X_n)$ back to $X_0 = x$. Historically, this equation was sometimes written literally backwards in time, with a final time N and the incremental time $n' = N - n$.

Both the forward and backward equations can be understood by taking expectations of test functions. More precisely, kernel duality gives us

$$\mu_{n+1}(\varphi) - \mu_n(\varphi) = \mu_n(P\varphi) - \mu_n(\varphi) = \mu_n(L\varphi). \tag{3.3.12}$$

Later, we will see continuous-time versions in which time derivatives replace the one-step differences.

Remark 3.19. Recall that to understand the joint distribution of an entire Markov chain, it is equivalent to understand the transition kernel P. With the generator approach, we have shown that it is *equivalent to understand the Kolmogorov equations.*

As it turns out, we can an even more precise characterization of $\varphi(X_n)$. While the Kolmogorov equations describe the expected evolution, along an actual trajectory, each change in $\varphi(X_n)$ remains random.

Observe that for a bounded test function φ, the Markov property gives us

$$\mathbb{E}[\varphi(X_{n+1}) - \varphi(X_n) | \mathcal{F}_n] = (P\varphi)(X_n) - \varphi(X_n) = (L\varphi)(X_n). \tag{3.3.13}$$

Since this term is predictable, we can actually separate it out and isolate a martingale.

Proposition 3.20 (Martingale Decomposition)**.** *Let $(X_n)_{n\geq 0}$ be a Markov chain with transition kernel P, and let $\varphi : E \to \mathbb{R}$ be bounded and measurable. Then we can write*

$$\varphi(X_n) - \varphi(X_0) = \sum_{k=0}^{n-1} (L\varphi)(X_k) + M_n^{\varphi}, \qquad n \geq 0, \tag{3.3.14}$$

where $(M_n^{\varphi})_{n\geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$, with $M_0^{\varphi} = 0$.

Before we start the proof, note that if we take the expected value $\mathbb{E}_x$, we recover exactly the Kolmogorov backward equation. Therefore, the martingale term M_n^{φ} is an additional piece of information on the trajectories of of $\varphi(X_n)$.

Proof. Define M_n^{φ} by (3.3.14) and check that its increments have conditional mean zero. The process is adapted, and boundedness of φ and $L\varphi$ makes every M_n^{φ} integrable. Its next increment is

$$M_{n+1}^{\varphi} - M_n^{\varphi} = \varphi(X_{n+1}) - \varphi(X_n) - (L\varphi)(X_n). \tag{3.3.15}$$

The last term is $\mathcal{F}_n$-measurable, and (3.3.13) gives

$$\mathbb{E}[\, M_{n+1}^{\varphi} - M_n^{\varphi} \,|\, \mathcal{F}_n] = (L\varphi)(X_n) - (L\varphi)(X_n) = 0. \tag{3.3.16}$$

□

The sum in (3.3.14) is predictable: its increment from time n to $n+1$ is $(L\varphi)(X_n)$, which is known at time n. Taking expectations and using $M_0^{\varphi} = 0$ also gives

$$\mathbb{E}\varphi(X_n) = \mathbb{E}\varphi(X_0) + \sum_{k=0}^{n-1} \mathbb{E}(L\varphi)(X_k). \tag{3.3.17}$$

The martingale term vanishes under expectation, while the accumulated expected changes are averaged over the visited states. This turns an equation for the observed trajectory into an equation for its expected evolution. In particular, starting from $X_0 = x$ and writing $u_n = P_n\varphi$, we obtain

$$u_n - \varphi = \sum_{k=0}^{n-1} P_k(L\varphi). \tag{3.3.18}$$

Subtracting this identity at consecutive times gives

$$u_{n+1} - u_n = P_n(L\varphi) = Lu_n, \tag{3.3.19}$$

The last equality uses $L = P - I$ and $P_n = P^n$, so L and P_n commute. Thus taking expectations of the martingale decomposition recovers the summed evolution equation, whose one-step difference is the backward Kolmogorov equation.

A bounded measurable function φ is called **harmonic** for the chain if $L\varphi = 0$, or equivalently $P\varphi = \varphi$. In this case, (3.3.13) says that $(\varphi(X_n))$ is already a martingale. More generally, $L\varphi \geq 0$ makes it a submartingale, and $L\varphi \leq 0$ makes it a supermartingale.

Exercise 3.9 (Stationary Distributions and the Generator)**.** Let P be a transition kernel and $L = P - I$ its generator.

(a) Show that L determines the kernel through

$$P(x, B) = (L\mathbf{1}_B)(x) + \mathbf{1}_B(x). \tag{3.3.20}$$

(b) Prove that a probability law π is stationary if and only if $\pi(L\varphi) = 0$ for every bounded measurable φ. Interpret this as a balance of expected one-step changes under the stationary distribution.

(c) Suppose π is stationary and φ is bounded and measurable, with $L\varphi \leq 0$ everywhere. Prove that $L\varphi = 0$ almost surely under π.

Exercise 3.10 (Expected Changes up to a Stopping Time)**.** Let φ be bounded and measurable, and let $\tau \leq N$ be a stopping time for a Markov chain, where N is a fixed integer.

(a) For the martingale in Proposition 3.20, verify

$$M^{\varphi}_{(k+1)\wedge\tau} - M^{\varphi}_{k\wedge\tau} = \mathbf{1}_{\{\tau>k\}}\left(M^{\varphi}_{k+1} - M^{\varphi}_{k}\right). \tag{3.3.21}$$

Use this identity to show that the stopped process is a martingale.

(b) Deduce the stopped expectation identity

$$\mathbb{E}\varphi(X_\tau) = \mathbb{E}\varphi(X_0) + \mathbb{E}\left[\sum_{k=0}^{\tau-1}(L\varphi)(X_k)\right]. \tag{3.3.22}$$

Explain how this extends (3.3.17).

3.4 Convergence to Stationarity

We can also use the generator to study convergence to a stationary distribution π. To start, we will introduce the $L^2(\pi)$ inner product

$$\langle\varphi, \psi\rangle_\pi = \int_E \varphi(x)\psi(x)\,\pi(dx), \qquad \|\varphi\|^2_\pi = \langle\varphi, \varphi\rangle_\pi. \tag{3.4.1}$$

The functions with finite norm form the function space $L^2(\pi)$. Thus we can think of test functions in $L^2(\pi)$ as vectors, and the generator L acts on them like a matrix.

On a finite state space $E = \{1, \ldots, m\}$, suppose $\pi_i > 0$ for every i. Viewing test functions as column vectors, the inner product is

$$\langle\varphi, \psi\rangle_\pi = \varphi^{\mathsf{T}}\,\mathrm{diag}(\pi)\psi. \tag{3.4.2}$$

where $\mathrm{diag}(\pi)$ is the diagonal matrix with entries $\pi_1, \ldots, \pi_m$. In particular, when π is uniform, this becomes $\langle\varphi, \psi\rangle_\pi = \frac{1}{m}\varphi^{\mathsf{T}}\psi$. Therefore intuitively speaking, $L^2(\pi)$ is a "twisted" version of the Euclidean geometry.

Definition 3.21 (Reversibility)**.** We say that the generator L is **reversible** with respect to π if, for every $\varphi, \psi \in L^2(\pi)$,

$$\langle\varphi, L\psi\rangle_\pi = \langle L\varphi, \psi\rangle_\pi. \tag{3.4.3}$$

Equivalently, this property of L is also called self-adjoint in $L^2(\pi)$. On a finite state space with π being the uniform distribution, this translates to

$$\varphi^\top L\psi = \varphi^\top L\psi\,, \tag{3.4.4}$$

in other words, $L = L^\top$. Therefore, we can interpreted reversibility or the self-adjoint property as *matrix symmetry* but weighted by π.

Definition 3.22 (Positive Semidefiniteness)**.** We say that the transition operator P is **positive semidefinite** in $L^2(\pi)$ if, for every $\varphi \in L^2(\pi)$,

$$\langle \varphi, P\varphi\rangle_\pi \geq 0. \tag{3.4.5}$$

On a finite state space with uniform π, this means that

$$\varphi^\mathsf{T}(I + L)\varphi \geq 0 \qquad \text{for every } \varphi \in \mathbb{R}^m. \tag{3.4.6}$$

For a reversible matrix L, this is equivalent to saying every eigenvalue of L is at least -1.

Definition 3.23 (Spectral Gap)**.** We say that L has a **spectral gap** of at least $\lambda > 0$ in $L^2(\pi)$ if

$$-\langle \varphi, L\varphi\rangle_\pi \geq \lambda\|\varphi\|_\pi^2 \qquad \text{whenever } \varphi \in L^2(\pi),\ \pi(\varphi) = 0. \tag{3.4.7}$$

On a finite state space with uniform π, this means that

$$-\varphi^\top L\varphi \geq \lambda\varphi^\top\varphi \qquad \text{whenever } \varphi \in \mathbb{R}^m,\ \pi\varphi = 0. \tag{3.4.8}$$

To interpret this condition, we first note that constant functions, i.e. $\mathbf{1} \in \mathbb{R}^m$, is an eigenvector where $L\mathbf{1} = 0$. Therefore $\varphi : \pi(\varphi) = 0$ is orthogonal to $\mathbf{1}$. Furthermore if L is symmetric, then the gap condition says the smallest nonzero eigenvalue of $-L$ is at least λ.

For both positive semidefinite and spectral gap conditions with non-uniform π, we should think about the eigenstructure as the $L^2(\pi)$ "twisted" version of the usual eigenstructure, just as we had thought about reversibility as a "twisted" symmetry.

Together, these three conditions give an exponential rate of convergence through the backward Kolmogorov equation. Given that we will be studying mean zero test functions, i.e. $\pi(\varphi) = 0$, we will first discuss why they are relevant to convergence to stationarity.

Consider $\mu_n = \mathcal{L}(X_n)$ and take a bounded test function φ. Duality and the fact that P_n preserves constants give

$$\begin{aligned}\mu_n(\varphi) - \pi(\varphi) &= \mu_0(P_n\varphi) - \mu_0(\pi(\varphi)\mathbf{1}) \\ &= \mu_0\left(P_n(\varphi - \pi(\varphi)\mathbf{1})\right).\end{aligned} \tag{3.4.9}$$

Here $\mathbf{1}$ is the constant function equal to one, with $P_n\mathbf{1} = \mathbf{1}$ and $\mu_0(\mathbf{1}) = 1$. The centered function $\varphi - \pi(\varphi)\mathbf{1}$ has mean zero under π. The constant part stays fixed under P_n, so we seek a contraction bound for the centered part. After proving this bound, we will return to (3.4.9) to relate it to convergence of the laws μ_n.

Let us first consider the finite-state case with a uniform stationary distribution, where reversibility becomes matrix symmetry. The proof shows how the spectral gap controls the decay of each nonconstant eigenvector.

Lemma 3.24 (Contraction in the Finite Uniform Case)**.** *Let $E = \{1, 2, \dots, m\}$, and consider a Markov chain (X_n) on E with transition matrix P and uniform stationary distribution π, so that $\pi_i = 1/m$ for every $i \in E$. Assume that P is symmetric and positive semidefinite, and let $L = P - I$. Suppose that, for some $\lambda \in (0, 1)$,*

$$-\varphi^\top L\varphi \geq \lambda\|\varphi\|^2 \qquad \text{whenever } \mathbf{1}^\top\varphi = 0. \tag{3.4.10}$$

Then, for every $\varphi \in \mathbb{R}^m$ with $\mathbf{1}^\top\varphi = 0$ and every $n \geq 0$,

$$\|P^n\varphi\| \leq (1-\lambda)^n\|\varphi\|. \tag{3.4.11}$$

Proof. Since P is symmetric and $P\mathbf{1} = \mathbf{1}$, it has an orthonormal eigenbasis $v_1, \dots, v_m$ with $v_1 = \mathbf{1}/\sqrt{m}$. Write $Pv_j = \mu_j v_j$. For $j \geq 2$, the vector v_j has mean zero. Positive semidefiniteness gives $\mu_j \geq 0$, and the gap condition gives $1 - \mu_j \geq \lambda$. Thus $0 \leq \mu_j \leq 1 - \lambda$.

Every mean-zero vector can be written as $\varphi = \sum_{j=2}^m a_j v_j$. By orthonormality, for $n \geq 1$,

$$\|P^n\varphi\|^2 = \sum_{j=2}^m a_j^2\mu_j^{2n} \leq (1-\lambda)^{2n}\sum_{j=2}^m a_j^2 = (1-\lambda)^{2n}\|\varphi\|^2. \tag{3.4.12}$$

Taking square roots proves the claim; for $n = 0$, the bound is equality. □

Example 3.25 (A Two-State Chain)**.** Let $E = \{1, 2\}$ and consider the chain with transition matrix and generator

$$P = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad L = P - I = \begin{pmatrix} -1/4 & 1/4 \\ 1/4 & -1/4 \end{pmatrix}. \tag{3.4.13}$$

The matrix P is symmetric, and its row and column sums are one, so $\pi = (1/2, 1/2)$ is stationary. Its eigenvectors $(1, 1)^\mathsf{T}$ and $(1, -1)^\mathsf{T}$ have eigenvalues 1 and $1/2$, respectively. Both eigenvalues are nonnegative, so P is positive semidefinite.

Every mean-zero vector has the form $\varphi = a(1, -1)^\mathsf{T}$. Therefore $L\varphi = -\varphi/2$, which gives

$$-\varphi^\mathsf{T} L\varphi = \frac{1}{2}\|\varphi\|^2. \tag{3.4.14}$$

Thus all the assumptions of Lemma 3.24 hold with $\lambda = 1/2$. In this example the contraction bound is exact:

$$P^n\varphi = 2^{-n}\varphi, \qquad \|P^n\varphi\| = 2^{-n}\|\varphi\|, \qquad n \geq 0. \tag{3.4.15}$$

The constant component stays fixed, while the mean-zero component is halved at every step.

To see convergence of the distributions explicitly, let $\mu_0 = (q, 1-q)$ for any $q \in [0, 1]$. As a row vector, this decomposes as

$$\mu_0 = \left(\frac{1}{2}, \frac{1}{2}\right) + \left(q - \frac{1}{2}\right)(1, -1). \tag{3.4.16}$$

Since $(1, -1)P = \frac{1}{2}(1, -1)$ and $\pi P = \pi$, multiplying by P^n gives

$$\mu_n = \mu_0 P^n = \left(\frac{1}{2} + 2^{-n}\left(q - \frac{1}{2}\right), \frac{1}{2} - 2^{-n}\left(q - \frac{1}{2}\right)\right) \longrightarrow \left(\frac{1}{2}, \frac{1}{2}\right). \tag{3.4.17}$$

Thus both state probabilities converge to $1/2$, with their deviations from $1/2$ halved at each step.

The following theorem proves the same contraction in $L^2(\pi)$ directly from inner-product inequalities.

Theorem 3.26 (Convergence to Stationarity)**.** *Let (X_n) be a Markov chain with generator L and stationary distribution π. Suppose that L is reversible, P is positive semidefinite, and that L has a spectral gap of at least $\lambda \in (0,1)$. Then, for every $\varphi \in L^2(\pi)$ with $\pi(\varphi) = 0$ and every $n \geq 0$,*

$$\|P_n\varphi\|_\pi \leq (1-\lambda)^n \|\varphi\|_\pi. \tag{3.4.18}$$

Proof. The key estimate is the one-step contraction

$$\|P\varphi\|_\pi \leq (1-\lambda)\|\varphi\|_\pi \qquad \text{whenever } \pi(\varphi) = 0. \tag{3.4.19}$$

We obtain this estimate from the assumptions in the first two steps, then iterate it in the third.

Step 1: For every mean-zero φ, positive semidefiniteness gives us a lower bound and spectral gap gives us an upper bound

$$0 \leq \langle \varphi, P\varphi\rangle_\pi = \|\varphi\|_\pi^2 + \langle \varphi, L\varphi\rangle_\pi \leq (1-\lambda)\|\varphi\|_\pi^2. \tag{3.4.20}$$

Step 2: Using Cauchy–Schwarz and the bound from Step 1, we have

$$\begin{aligned} |\langle \psi, P\varphi\rangle_\pi| &\leq \sqrt{\langle \psi, P\psi\rangle_\pi \langle \varphi, P\varphi\rangle_\pi} \\ &\leq (1-\lambda)\|\psi\|_\pi \|\varphi\|_\pi \end{aligned} \tag{3.4.21}$$

for any mean-zero $\varphi, \psi \in L^2(\pi)$.

Now choose $\psi = P\varphi$, which has mean zero because stationarity gives $\pi(P\varphi) = \pi(\varphi) = 0$. Therefore

$$\|P\varphi\|_\pi^2 = \langle P\varphi, P\varphi\rangle_\pi \leq (1-\lambda)\|P\varphi\|_\pi \|\varphi\|_\pi. \tag{3.4.22}$$

Dividing $\|P\varphi\|_\pi$ on both sides gives us the bound below (if $\|P\varphi\|_\pi = 0$ we also recover the bound trivially)

$$\|P\varphi\|_\pi \leq (1-\lambda)\|\varphi\|_\pi \qquad \text{whenever } \pi(\varphi) = 0. \tag{3.4.23}$$

Step 3: Set $u_n = P_n\varphi$, where $\pi(\varphi) = 0$. Stationarity gives $\pi(u_n) = 0$, and the backward Kolmogorov equation gives

$$u_{n+1} = u_n + Lu_n = Pu_n. \tag{3.4.24}$$

Every u_n therefore satisfies the contraction bound (3.4.23). Applying it to u_{n-1} and using $u_0 = \varphi$ gives

$$\begin{aligned} \|u_n\|_\pi &= \|Pu_{n-1}\|_\pi \\ &\leq (1-\lambda)\|u_{n-1}\|_\pi \leq \cdots \leq (1-\lambda)^n \|\varphi\|_\pi, \qquad n \geq 1. \end{aligned} \tag{3.4.25}$$

This proves (3.4.18), with equality when $n = 0$. □

We now apply Theorem 3.26 to the centered test function in (3.4.9). The theorem says that the $L^2(\pi)$ norm of $P_n(\varphi - \pi(\varphi))$ decreases exponentially. When the initial distribution has a square-integrable density with respect to π, Cauchy–Schwarz transfers this bound to the difference of expectations on the left. The following exercise develops this argument.

In particular, taking $\varphi = \mathbf{1}_B$ compares the probabilities $\mu_n(B)$ and $\pi(B)$. By the duality between distributions and test functions, we should therefore interpret this result as saying that μ_n converges to the stationary distribution π exponentially fast.

The exercise below will make the notions of convergence precise.

Exercise 3.11 (Chi-Square and Total Variation Convergence)**.** Suppose P satisfies the assumptions of Theorem 3.26, with stationary distribution π and gap λ. Let $f_0 \in L^2(\pi)$ be nonnegative with $\pi(f_0) = 1$, and suppose the initial law is

$$\mu_0(B) = \int_B f_0(x)\,\pi(dx), \qquad B \in \mathcal{B}(E). \tag{3.4.26}$$

Thus f_0 is the density of μ_0 with respect to π. Set $f_n = P_n f_0$. For a probability distribution μ with density f with respect to π, the **chi-square divergence** from π is

$$\chi^2(\mu\|\pi) = \int_E (f-1)^2\,d\pi = \|f-1\|_\pi^2. \tag{3.4.27}$$

The **total variation distance** between μ and π is

$$d_{\mathrm{TV}}(\mu,\pi) = \sup_{B\in\mathcal{B}(E)} |\mu(B) - \pi(B)|. \tag{3.4.28}$$

(a) Use reversibility and duality to prove that $f_n \geq 0$, $\pi(f_n) = 1$, and

$$\mu_n(B) = \int_B f_n(x)\,\pi(dx) \qquad \text{for every } B \in \mathcal{B}(E). \tag{3.4.29}$$

(b) Apply Theorem 3.26 to $f_0 - 1$ to prove

$$\chi^2(\mu_n\|\pi) \leq (1-\lambda)^{2n}\chi^2(\mu_0\|\pi). \tag{3.4.30}$$

(c) Verify the bound

$$d_{\mathrm{TV}}(\mu_n,\pi) \leq \frac{1}{2}\sqrt{\chi^2(\mu_n\|\pi)}\,, \tag{3.4.31}$$

and deduce exponential convergence to π in total variation distance. Hint: since $\pi(f_n - 1) = 0$, the positive and negative parts of $f_n - 1$ have equal integrals. Then apply Cauchy–Schwarz.

Chapter 4

Brownian Motion

4.1 From Random Walks to Brownian Motion

This course so far have only considered discrete time stochastic processes, and from this point onward, we will start to consider processes indexed by a nonnegative real number $t \geq 0$. The reader might be interested in why this is a good modeling choice for stochastic processes, aren't all processes we can measure "discrete" to some extent?

Indeed, moving to continuous time is primarily a *modeling choice*. In other words, the mathematics of continuous time stochastic processes are *nicer* to work with. At the same time, if we can demonstrate that continuous time processes remain *good approximations* of discrete time processes, then it is an entirely justifiable approach for analysis.

The simplest model we can start with is the simple random walk, where we sample independent steps accumulate into a random trajectory. We will see that this process can be very well approximated by a canonical continuous time process called the *Brownian motion*.

Let $\xi_1, \xi_2, \ldots$ be independent random variables taking the values -1 and 1 with equal probability, and set $S_n = \sum_{k=1}^{n} \xi_k$, with $S_0 = 0$. For a positive integer N, take N steps per unit time and scale each step by $1/\sqrt{N}$

$$B_t^{(N)} = \frac{1}{\sqrt{N}} S_{\lfloor Nt \rfloor}, \qquad t \geq 0. \tag{4.1.1}$$

Note this is a **piecewise constant interpolation** of the discrete time random walk values.

The scaling comes from the variances being stable (or the CLT scaling)

$$\mathbb{E}B_t^{(N)} = 0, \qquad \mathrm{Var}(B_t^{(N)}) = \frac{\lfloor Nt \rfloor}{N} \longrightarrow t. \tag{4.1.2}$$

For each fixed $t > 0$, the Central Limit Theorem (Theorem 1.33) tells us that $B_t^{(N)} \xrightarrow{d} \mathcal{N}(0, t)$. Given that disjoint time intervals use disjoint blocks of the independent steps, this means that their increments are also independent and Gaussian.

This suggests a limiting continuous-time process whose increments are independent and Gaussian, with variance proportional to elapsed time, which we will define next.

Definition 4.1 (Standard Brownian Motion)**.** A real-valued process $(B_t)_{t \geq 0}$ is a **standard Brownian motion**, also called a **Wiener process**, if:

(1) $B_0 = 0$ almost surely;
(2) for every $0 = t_0 < t_1 < \cdots < t_m$, the increments $B_{t_1} - B_{t_0}, \ldots, B_{t_m} - B_{t_{m-1}}$ are independent;
(3) $B_t - B_s \sim \mathcal{N}(0, t-s)$ whenever $0 \le s < t$;
(4) almost surely, the sample path $t \mapsto B_t$ is continuous.

The increment distribution depends only on $t - s$, so Brownian motion has **stationary increments**, i.e. adding a constant c to both t and c does not change the distribution of the increment.

We will not do a full construction of the Brownian motion in this course, given that it is rather technical and not very enlightening to see. We will motivate the existence by studying finite dimensional distributions of random walks, and show it converges to a Gaussian process matching that of the Brownian motion.

Proposition 4.2 (Gaussian Laws and Covariance)**.** *Standard Brownian motion is a centered Gaussian process with*

$$\mathbb{E}B_t = 0, \qquad \mathrm{Cov}(B_s, B_t) = \min\{s, t\}, \qquad s, t \ge 0. \tag{4.1.3}$$

Proof. We first express the observations as sums of independent Gaussian increments, then compute their covariances. For $0 = t_0 < t_1 < \cdots < t_m$, each B_{t_j} is a partial sum of the independent Gaussian variables $B_{t_k} - B_{t_{k-1}}$. Thus $(B_{t_1}, \ldots, B_{t_m})$ is jointly Gaussian, and every coordinate has mean zero. If $0 \le s < t$, the increment $B_t - B_s$ is independent of B_s and has mean zero, so

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s(B_s + B_t - B_s)] = \mathbb{E}B_s^2 = s. \tag{4.1.4}$$

The case $s = t$ is the variance formula, and symmetry gives the result when $t < s$. □

Note that the independent increments do not make the values at different times independent. The values B_s and B_t share all the increments accumulated before time s, therefore their covariance is non-zero when $s < t$. Together, the mean and covariance determine every finite-dimensional Gaussian law, as in Chapter 1.

For example, if Z_1, Z_2 are independent standard Gaussian variables, then

$$(B_1, B_2) \overset{d}{=} (Z_1, Z_1 + Z_2), \qquad \mathrm{Cov}\begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}. \tag{4.1.5}$$

The second observation contains the first observation plus one fresh independent increment.

We now return to the scaled random walk and prove convergence of its observations at any fixed finite collection of times. This is called **finite-dimensional convergence**.

Proposition 4.3 (Finite-Dimensional Convergence of the Random Walk)**.** *Let $S_n = \sum_{k=1}^{n} \xi_k$, with $S_0 = 0$, where the ξ_k are independent and take the values -1 and 1 with equal probability. Define the scaled random walk by*

$$B_t^{(N)} = \frac{1}{\sqrt{N}} S_{\lfloor Nt \rfloor}, \qquad t \ge 0, \tag{4.1.6}$$

and let B be standard Brownian motion. For every $0 < t_1 < \cdots < t_m$,

$$\left(B_{t_1}^{(N)}, \ldots, B_{t_m}^{(N)}\right) \xrightarrow{d} (B_{t_1}, \ldots, B_{t_m}). \tag{4.1.7}$$

Proof. We apply the central limit theorem to the independent blocks of steps between observation times, then take their partial sums. Set $t_0 = 0$, $\Delta t_j = t_j - t_{j-1}$, and $N_j = \lfloor N t_j \rfloor$, with the dependence of N_j on N understood. Then $\Delta N_j = N_j - N_{j-1}$ is the number of steps in the j-th time interval. Write $\Delta B_{t_j} = B_{t_j} - B_{t_{j-1}}$ for the Brownian increment and

$$\Delta B_{t_j}^{(N)} := B_{t_j}^{(N)} - B_{t_{j-1}}^{(N)} = \frac{1}{\sqrt{N}} \sum_{k=N_{j-1}+1}^{N_j} \xi_k. \tag{4.1.8}$$

This block contains ΔN_j independent fair steps. Since $\Delta N_j / N \to \Delta t_j > 0$, the central limit theorem gives

$$\Delta B_{t_j}^{(N)} \xrightarrow{d} \mathcal{N}(0, \Delta t_j). \tag{4.1.9}$$

These are exactly the laws of the Brownian increments ΔB_{t_j}. For each N, the walk increments are independent because their blocks are disjoint, and the Brownian increments are also independent. Thus the joint laws converge:

$$\left(\Delta B_{t_1}^{(N)}, \ldots, \Delta B_{t_m}^{(N)}\right) \xrightarrow{d} \left(\Delta B_{t_1}, \ldots, \Delta B_{t_m}\right). \tag{4.1.10}$$

Taking partial sums is a continuous linear map, and the partial sums recover the observations of each process. This gives the claimed finite-dimensional convergence. □

Proposition 4.3 shows that the scaled random walk approaches Brownian motion when observed at any fixed finite collection of times. However, this does not tell me whether or not the *entire path* converges to the Brownian motion. To study convergence on the path-space $C([0,T])$, we need an analogue of the Central Limit Theorem on this space, and Brownian motion plays the role of the *Gaussian* limit for these random functions.

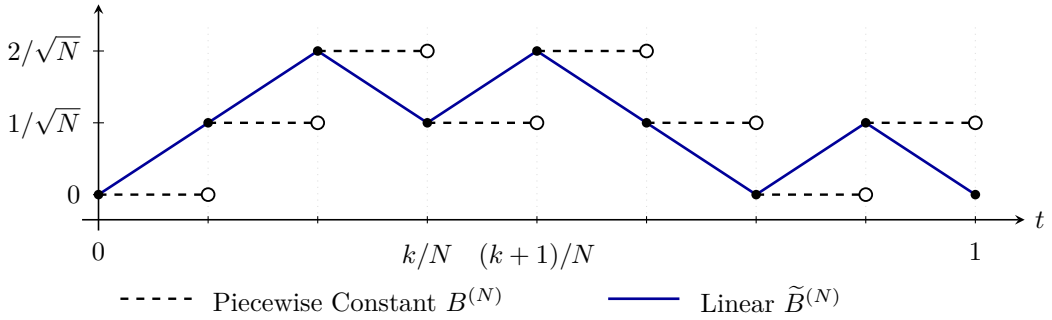


Figure 4.1: Piecewise constant and linear interpolations of the same scaled random walk, illustrated with $N = 8$ on $[0, 1]$. Both agree at the grid times; open circles mark the excluded right endpoints of the constant pieces. Linear interpolation produces a continuous path and changes the value by at most $1/\sqrt{N}$ at every time. The labeled interval has $k = 3$.

To state this precisely, we will need to fix a time interval $[0, T]$, and embed the discrete time random walk into the path space $C([0, T])$ first. Note the previous construction of $B_t^{(N)} = \frac{1}{\sqrt{N}} S_{\lfloor Nt \rfloor}$ creates a continuous time (piecewise constant) process, but is *not continuous*. The piecewise constant interpolation can still converge, but requires more technical tools, and we will discuss this a bit more after the next result.

For simplicity, we will instead consider a **linear interpolation** of the random walk. More precisely, on each interval between consecutive grid times, we define the convex combination

$$\widetilde{B}_t^{(N)} = (1-\lambda)\frac{S_k}{\sqrt{N}} + \lambda\frac{S_{k+1}}{\sqrt{N}}, \qquad t \geq 0, \tag{4.1.11}$$

where $k = \lfloor Nt \rfloor$ and $\lambda = Nt - \lfloor Nt \rfloor \in [0,1)$.

The two interpolations are compared in Figure 4.1. For the fair steps used so far, this interpolation changes the walk by at most $1/\sqrt{N}$. We can now regard both the interpolated walk and Brownian motion as random elements of $C([0,T])$, the space of continuous functions on this interval.

Recall that in order to define convergence in distribution, we need to define test functions that are continuous and bounded. And in order to define continuity, we need a **metric** on the space of $C([0,T])$ for some $T > 0$. To this goal, we will equip our space with the **uniform metric** (or distance)

$$d(f,g) = \sup_{0 \leq t \leq T} |f(t) - g(t)|, \qquad f, g \in C([0,T]). \tag{4.1.12}$$

Two paths are close in this distance when their values stay close throughout the interval.

We are now ready to state a general version of random walk converging to Brownian motion.

Theorem 4.4 (Donsker's Theorem)**.** *Let $\xi_1, \xi_2, \ldots$ be i.i.d. with $\mathbb{E}\,\xi_1 = 0$ and $\mathbb{E}\,\xi_1^2 = 1$. Set $S_n = \sum_{k=1}^n \xi_k$, with $S_0 = 0$. For each positive integer N, define the linear interpolation by*

$$\widetilde{B}_t^{(N)} = (1-\lambda)\frac{S_k}{\sqrt{N}} + \lambda\frac{S_{k+1}}{\sqrt{N}}, \qquad t \geq 0, \tag{4.1.13}$$

where $k = \lfloor Nt \rfloor$ and $\lambda = Nt - \lfloor Nt \rfloor \in [0,1)$. Then, for every $T > 0$,

$$\widetilde{B}^{(N)} \xrightarrow{d} B \quad \text{in } C([0,T]) \tag{4.1.14}$$

with the uniform distance, where B is standard Brownian motion.

Observe that the convergence is **universal**, in the sense that the limit does not depend on the distribution of the ξ_k's (as long as they are zero mean with variance 1). In other words, all random walks with proper rescaling converge to this limit. As a result, Donsker's Theorem is also known as the *Functional Central Limit Theorem*.

We will not prove Donsker's Theorem here, but the main barrier remaining is technical: we need to control the level of continuity of the interpolated paths uniformly along the sequence. Intuitively speaking, a sequence of continuous functions can converge pointwise to a discontinuous limit, so continuity of each individual path is not enough. We need additional probabilistic control of their fluctuations between nearby times. These details are beyond the scope of this course, and we will take the theorem as given.

Remark 4.5 (Piecewise Constant Paths). The piecewise constant interpolations $B_t^{(N)} = S_{\lfloor Nt \rfloor}/\sqrt{N}$ actually also converge to the Brownian motion limit, it just required more technical work. These paths belong to $D([0,T])$, the space of right-continuous functions with left limits.

Under the assumptions of Theorem 4.4, the processes $B^{(N)}$ converge in distribution to B in this space with the **Skorohod J_1 topology**. Intuitively, this topology handles the technicalities of discontinuous paths. This approach can be very helpful, especially when processes we are working with have jumps.

Exercise 4.1 (Recognizing Brownian Motion from Its Gaussian Laws)**.** Let $(X_t)_{t\geq 0}$ be a centered Gaussian process with almost surely continuous paths and covariance $\mathrm{Cov}(X_s, X_t) = \min\{s, t\}$.

(a) Show that $X_0 = 0$ almost surely and $X_t - X_s \sim \mathcal{N}(0, t-s)$ for $s < t$.
(b) Prove that increments over disjoint time intervals are independent. Use the fact that uncorrelated coordinates of a Gaussian vector are independent.
(c) Conclude that X is standard Brownian motion.

Exercise 4.2 (The Same One-Time Laws)**.** Let $Z \sim \mathcal{N}(0,1)$ and define $X_t = \sqrt{t}\, Z$ for $t \geq 0$.

(a) Show that X has continuous paths and that X_t has the same law as B_t for every fixed t.
(b) Show that X does not have independent increments and therefore is not Brownian motion. You may compare X_1 and $X_2 - X_1$.
(c) Explain why one-time convergence of the scaled random walks would not by itself identify the Brownian finite-dimensional limit.

Exercise 4.3 (Convergence of the Random-Walk Maximum)**.** Let S_k and $\widetilde{B}^{(N)}$ be as in Theorem 4.4. Use Donsker's theorem to prove

$$\frac{1}{\sqrt{N}} \max_{0\leq k\leq N} S_k \xrightarrow{d} \max_{0\leq t\leq 1} B_t. \tag{4.1.15}$$

The following parts guide the argument; you do not need to identify the distribution of the Brownian maximum.

(a) Show that the maximum of $\widetilde{B}^{(N)}$ on $[0,1]$ equals the maximum of its grid values:

$$\max_{0\leq t\leq 1} \widetilde{B}^{(N)}_t = \frac{1}{\sqrt{N}} \max_{0\leq k\leq N} S_k. \tag{4.1.16}$$

Use the convex-average formula on each grid interval.
(b) For $f \in C([0,1])$, define $M(f) = \max_{0\leq t\leq 1} f(t)$. Prove that $|M(f) - M(g)| \leq d(f,g)$, and conclude that M is continuous for the uniform distance.
(c) Let $\varphi : \mathbb{R} \to \mathbb{R}$ be bounded and continuous. Show that $F(f) = \varphi(M(f))$ is a bounded continuous function on $C([0,1])$. Apply Donsker's theorem to F and use part (a) to obtain (4.1.15).

Exercise 4.4 (A Narrow Spike Between Observation Times)**.** For $n \geq 1$, define the continuous function on $[0,1]$

$$f_n(t) = \max\{1 - n^2|t - 1/n|, 0\}. \tag{4.1.17}$$

(a) Prove that $f_n(t) \to 0$ for every fixed t, but $\sup_{0\leq t\leq 1} |f_n(t)| = 1$ for every n.
(b) Regard the f_n as deterministic random paths. Show that their finite-dimensional distributions converge to those of the zero path.
(c) Verify that $F(f) = \min\{1, \sup_{0\leq t\leq 1} |f(t)|\}$ is bounded and continuous for the uniform distance. Use this test function to show that the laws of f_n do not converge to the zero-path law in $C([0,1])$.

4.2 Transition Kernels and the Markov Property

Similar to a Markov chain, Brownian motion maintains a type of Markov property, now with a continuous time index. We will make this notion precise in this section.

Let us consider the natural filtration

$$\mathcal{F}_t = \sigma(B_r : 0 \le r \le t), \qquad t \ge 0. \tag{4.2.1}$$

It records the observations up to time t, and $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s \le t$. Independent increments imply that $B_{s+t} - B_s$ is independent of $\mathcal{F}_s$: it is independent of every finite collection of observations before time s, and hence of the σ-algebra they generate.

If we start a Brownian motion at x, it's equivalent to consider the process $X_t = x + B_t$. Since $X_t \sim \mathcal{N}(x, t)$ for $t > 0$, we can recover the transition density and kernel

$$p_t(x, y) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{(y-x)^2}{2t}\right), \qquad P_t(x, A) = \int_A p_t(x, y)\, dy, \tag{4.2.2}$$

where $A \in \mathcal{B}(\mathbb{R})$. At time zero, we set $P_0(x, A) = \mathbf{1}_A(x)$. Unlike a discrete-time chain, Brownian motion has a transition kernel for every elapsed time t.

Proposition 4.6 (Markov Property of Brownian Motion)**.** *For every $s, t \ge 0$ and every $A \in \mathcal{B}(\mathbb{R})$,*

$$\mathbb{P}(B_{s+t} \in A \mid \mathcal{F}_s) = \mathbb{P}(B_{s+t} \in A \mid \sigma(B_s)) = P_t(B_s, A) \tag{4.2.3}$$

almost surely.

Proof. For $t > 0$, the random variable $P_t(B_s, A)$ is $\mathcal{F}_s$-measurable. We check that it has the defining averaging property of conditional expectation. For every $H \in \mathcal{F}_s$, the increment $B_{s+t} - B_s$ is independent of $(\mathbf{1}_H, B_s)$ and has law $\mathcal{N}(0, t)$. Thus, integrating over the independent increment gives

$$\begin{aligned} \mathbb{E}[\mathbf{1}_H \mathbf{1}_A(B_{s+t})] &= \int_{\mathbb{R}} \mathbb{E}[\mathbf{1}_H \mathbf{1}_A(B_s + z)] p_t(0, z)\, dz \\ &= \mathbb{E}[\mathbf{1}_H P_t(B_s, A)]. \end{aligned} \tag{4.2.4}$$

This proves that $P_t(B_s, A)$ is the conditional probability given $\mathcal{F}_s$. Since it is a function of B_s, the tower property gives the same conditional probability given $\sigma(B_s)$. When $t = 0$, all three expressions equal $\mathbf{1}_A(B_s)$. □

Thus knowing the whole path up to time s gives the same prediction of B_{s+t} as knowing only B_s. The kernel depends on the elapsed time t, rather than the time s at which we start observing the increment. This is the continuous-time version of time homogeneity.

As in Chapter 3, the kernel also acts on test functions. For a bounded measurable $\varphi : \mathbb{R} \to \mathbb{R}$ and $t > 0$, define

$$(P_t \varphi)(x) = \int_{\mathbb{R}} \varphi(y) p_t(x, y)\, dy = \mathbb{E}\, \varphi(x + B_t), \tag{4.2.5}$$

and set $P_0 \varphi = \varphi$. Integrating a test function against the conditional law in (4.2.3) gives

$$\mathbb{E}[\, \varphi(B_{s+t}) |\, \mathcal{F}_s] = (P_t \varphi)(B_s). \tag{4.2.6}$$

The extension from indicators to bounded measurable functions is the same approximation argument used in Chapter 3.

Independent Gaussian increments also give the semigroup identity

$$P_s P_t = P_{s+t}, \qquad s, t \geq 0. \tag{4.2.7}$$

To see this for $s, t > 0$, let $Z \sim \mathcal{N}(0, s)$ and $W \sim \mathcal{N}(0, t)$ be independent. Then $Z + W \sim \mathcal{N}(0, s + t)$, so

$$(P_s P_t \varphi)(x) = \mathbb{E}\, \varphi(x + Z + W) = (P_{s+t}\varphi)(x). \tag{4.2.8}$$

If either time is zero, the identity follows from $P_0 = I$. Evolving for time s and then time t therefore gives the same law as evolving for time $s + t$.

For a random initial value X_0 with law μ_0, independent of B, the process $X_t = X_0 + B_t$ has law $\mu_t = \mu_0 P_t$. Here $\mu_0 P_t$ is the distribution obtained by starting with law μ_0 and evolving for time t:

$$(\mu_0 P_t)(A) = \int_{\mathbb{R}} P_t(x, A)\, \mu_0(dx), \qquad A \in \mathcal{B}(\mathbb{R}). \tag{4.2.9}$$

We average the probability of reaching A from x over the initial distribution of x. This extends the finite-state operation of multiplying a probability row vector by a transition matrix. The duality between laws and test functions is unchanged:

$$\mu_t(\varphi) = \mu_0(P_t \varphi). \tag{4.2.10}$$

We can now write continuous-time versions of the Kolmogorov equations from Chapter 3. For Brownian motion, these follow directly from the Gaussian transition density.

Theorem 4.7 (Kolmogorov Equations for Brownian Motion)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be bounded and continuous, and set $u(t, x) = (P_t \varphi)(x)$. Then u satisfies the* ***backward Kolmogorov equation***

$$\partial_t u(t, x) = \frac{1}{2} \partial_{xx} u(t, x), \qquad t > 0, \qquad u(0, x) = \varphi(x). \tag{4.2.11}$$

For any initial probability law μ_0, the law $\mu_t = \mu_0 P_t$ has density

$$f_t(y) = \int_{\mathbb{R}} p_t(x, y)\, \mu_0(dx), \qquad t > 0. \tag{4.2.12}$$

This density satisfies the ***forward Kolmogorov equation***

$$\partial_t f_t(y) = \frac{1}{2} \partial_{yy} f_t(y), \qquad t > 0. \tag{4.2.13}$$

Its initial condition is $\mu_t \xrightarrow{d} \mu_0$ as $t \downarrow 0$.

Proof. We use the heat-equation identity for the Gaussian kernel:

$$\partial_t p_t(x, y) = \frac{1}{2} \partial_{xx} p_t(x, y) = \frac{1}{2} \partial_{yy} p_t(x, y), \qquad t > 0. \tag{4.2.14}$$

Differentiating inside the integral defining u gives the backward equation:

$$\partial_t u(t, x) = \int_{\mathbb{R}} \varphi(y) \partial_t p_t(x, y)\, dy = \frac{1}{2} \int_{\mathbb{R}} \varphi(y) \partial_{xx} p_t(x, y)\, dy = \frac{1}{2} \partial_{xx} u(t, x). \tag{4.2.15}$$

For the density formula, the definition of $\mu_t = \mu_0 P_t$ gives

$$\begin{aligned}\mu_t(A) &= \int_{\mathbb{R}} \int_A p_t(x,y)\, dy\, \mu_0(dx) \\ &= \int_A \left(\int_{\mathbb{R}} p_t(x,y)\, \mu_0(dx) \right) dy = \int_A f_t(y)\, dy.\end{aligned} \tag{4.2.16}$$

Differentiating this density gives the forward equation:

$$\partial_t f_t(y) = \int_{\mathbb{R}} \partial_t p_t(x,y)\, \mu_0(dx) = \frac{1}{2} \int_{\mathbb{R}} \partial_{yy} p_t(x,y)\, \mu_0(dx) = \frac{1}{2} \partial_{yy} f_t(y). \tag{4.2.17}$$

For $t > 0$, the Gaussian kernel bounds justify differentiating inside these integrals.

Brownian path continuity gives $x + B_t \to x$ almost surely as $t \downarrow 0$. Since φ is bounded and continuous, taking expectations gives $u(t,x) \to \varphi(x)$. Similarly, starting from $X_0 \sim \mu_0$ independent of B, we have $X_0 + B_t \to X_0$ almost surely, which gives $\mu_t \xrightarrow{d} \mu_0$. □

Both equations are the **heat equation**, but they describe different objects. The backward equation evolves the expected value of a test function as a function of the starting position x, while the forward equation evolves the density at the final position y. Later, Itô's formula will give the corresponding equations for general SDEs, where an explicit transition density is usually unavailable.

We extend the martingale definition from Chapter 2 to continuous time.

Definition 4.8 (Continuous-Time Martingale)**.** Let $(\mathcal{F}_t)_{t \geq 0}$ be a filtration. An adapted process $(M_t)_{t \geq 0}$ is a **martingale** if $\mathbb{E}|M_t| < \infty$ for every $t \geq 0$ and

$$\mathbb{E}[\, M_t |\, \mathcal{F}_s] = M_s \qquad \text{almost surely whenever } 0 \leq s \leq t. \tag{4.2.18}$$

The same independent-increment calculation that gives the Markov property produces two useful martingales.

Proposition 4.9 (Basic Brownian Martingales)**.** *With respect to the natural filtration of B, both $(B_t)_{t \geq 0}$ and $(B_t^2 - t)_{t \geq 0}$ are martingales.*

Proof. Both processes are adapted and integrable because Gaussian variables have finite first and second moments. We use the mean and variance of the independent increment to check the two conditional expectations.

For $0 \leq s \leq t$, the increment $B_t - B_s$ is independent of $\mathcal{F}_s$ and has mean zero. Hence

$$\mathbb{E}[\, B_t |\, \mathcal{F}_s] = B_s + \mathbb{E}[\, B_t - B_s |\, \mathcal{F}_s] = B_s. \tag{4.2.19}$$

For the second process, expand $B_t^2 = (B_s + (B_t - B_s))^2$. The mixed term has conditional mean zero, while the squared increment has conditional expectation $t - s$. Therefore

$$\mathbb{E}[\, B_t^2 - t \Big|\, \mathcal{F}_s] = B_s^2 + (t - s) - t = B_s^2 - s. \tag{4.2.20}$$

These are the martingale identities for the two processes. □

Brownian motion has no expected change given the past, but its square has an expected increase of $t - s$ over a time interval of length $t - s$. Subtracting the elapsed time removes this increase.

Finally, time shifts, sign changes, and a simultaneous scaling of time and space preserve the Brownian definition. The scaling matches a time factor c with a spatial factor $\sqrt{c}$, just as in Section 4.1.

Proposition 4.10 (Brownian Symmetries)**.** *If B is standard Brownian motion, then each of the following is also standard Brownian motion:*

(1) $(B_{h+t} - B_h)_{t\geq 0}$ *for every fixed* $h \geq 0$*;*
(2) $(-B_t)_{t\geq 0}$*;*
(3) $(c^{-1/2}B_{ct})_{t\geq 0}$ *for every* $c > 0$*.*

Proof. We check the properties in Definition 4.1. Each transformed process starts at zero and has continuous paths.

For the time shift, increments over disjoint time intervals are increments of B over disjoint intervals shifted by h. They are therefore independent centered Gaussians, with variance equal to the original interval lengths. For the sign change, independence is preserved, and a centered Gaussian increment has the same law as its negative.

For scaling, set $\widetilde{B}_t = c^{-1/2}B_{ct}$. Its increments are independent centered Gaussians, and for $s < t$,

$$\operatorname{Var}(\widetilde{B}_t - \widetilde{B}_s) = c^{-1}(ct - cs) = t - s. \tag{4.2.21}$$

Thus each process satisfies the Brownian definition. □

Exercise 4.5 (Combining Independent Brownian Motions)**.** Let B and W be independent standard Brownian motions, and let $a, b \in \mathbb{R}$ satisfy $a^2 + b^2 = 1$.

(a) Prove that $aB + bW$ is standard Brownian motion.
(b) Set $U = (B + W)/\sqrt{2}$ and $V = (B - W)/\sqrt{2}$. Prove that U and V are independent standard Brownian motions. To check independence, consider arbitrary finite collections of observations from both processes and use their joint Gaussian law.

4.3 Quadratic Variation

We start by considering an exercise in classical calculus. If f is continuous and g is continuously differentiable on $[0, T]$, we can consider integrals of the type

$$\int_0^T f(t)g'(t)\,dt. \tag{4.3.1}$$

Furthermore, we can approximate this integral using a Riemann sum. In particular, consider the uniform grid

$$\Delta t = \frac{T}{N}, \qquad t_j = j\Delta t, \qquad t_j^* = \frac{t_{j-1} + t_j}{2}, \quad j = 1, \ldots, N\,, \tag{4.3.2}$$

and let $\Delta g(t_j) = g(t_j) - g(t_{j-1})$ be the increment of g, this allows us to define the left endpoint, midpoint, and right endpoint Riemann sums

$$\begin{aligned} L_N &= \sum_{j=1}^{N} f(t_{j-1})\Delta g(t_j), \\ M_N &= \sum_{j=1}^{N} f(t_j^*)\Delta g(t_j), \\ R_N &= \sum_{j=1}^{N} f(t_j)\Delta g(t_j). \end{aligned} \tag{4.3.3}$$

Without going into the details, a first order Taylor expansion argument allows us to approximate $\Delta g(t_j)$ with $g'(s_j)\Delta t$ with error $o(\Delta t)$ for any $s_j \in [t_{j-1}, t_j]$. Consequently, the Riemann-sum gives us that all three choices of Riemann sums give us the same limit

$$\begin{aligned} L_N &= \sum_{j=1}^{N} f(t_{j-1})g'(t_{j-1})\Delta t + o(1) \longrightarrow \int_0^T f(t)g'(t)\,dt, \\ M_N &= \sum_{j=1}^{N} f(t_j^*)g'(t_j^*)\Delta t + o(1) \longrightarrow \int_0^T f(t)g'(t)\,dt, \\ R_N &= \sum_{j=1}^{N} f(t_j)g'(t_j)\Delta t + o(1) \longrightarrow \int_0^T f(t)g'(t)\,dt. \end{aligned} \tag{4.3.4}$$

In particular, if $f = g$ and $g(0) = 0$, all three limits are $g(T)^2/2$ by the ordinary chain rule.

Now let us repeat the same calculation with $f(t) = g(t) = B_t$, a standard Brownian motion. The finite sums in (4.3.3) still make sense, even though we have not defined an integral against Brownian motion. Write $\Delta B_{t_j} = B_{t_j} - B_{t_{j-1}}$.

For the left endpoint, expanding the change in the square gives

$$B_{t_j}^2 - B_{t_{j-1}}^2 = 2B_{t_{j-1}}\Delta B_{t_j} + (\Delta B_{t_j})^2. \tag{4.3.5}$$

Summing over j and using $B_0 = 0$, we obtain

$$L_N = \sum_{j=1}^{N} B_{t_{j-1}}\Delta B_{t_j} = \frac{B_T^2}{2} - \frac{1}{2}\sum_{j=1}^{N}(\Delta B_{t_j})^2. \tag{4.3.6}$$

For the right endpoint, the corresponding expansion is $B_{t_j}^2 - B_{t_{j-1}}^2 = 2B_{t_j}\Delta B_{t_j} - (\Delta B_{t_j})^2$. Thus

$$R_N = \sum_{j=1}^{N} B_{t_j}\Delta B_{t_j} = \frac{B_T^2}{2} + \frac{1}{2}\sum_{j=1}^{N}(\Delta B_{t_j})^2. \tag{4.3.7}$$

Note each of the increments ΔB_{t_j} are independent $\mathcal{N}(0, \Delta t)$, so their squares have mean $\frac{T}{N}$. We can then use the Law of Large Numbers to get

$$\sum_{j=1}^{N}(\Delta B_{t_j})^2 = \frac{T}{N}\sum_{j=1}^{N}\left(\frac{\Delta B_{t_j}}{\sqrt{\Delta t}}\right)^2 \longrightarrow T\,. \tag{4.3.8}$$

This implies that we have the following limits

$$\begin{aligned} L_N &\longrightarrow \frac{B_T^2 - T}{2}, \\ M_N &\longrightarrow \frac{B_T^2}{2}, \\ R_N &\longrightarrow \frac{B_T^2 + T}{2}, \end{aligned} \tag{4.3.9}$$

where you will show in Exercise 4.7 the midpoint sum has limit $B_T^2/2$.

The differences in these limits will be a key concept to understanding integration with respect to Brownian motion. In particular, the left- and right-endpoint sums differ by

$$R_N - L_N = \sum_{j=1}^{N} (\Delta B_{t_j})^2 \longrightarrow T \quad \text{in probability.} \tag{4.3.10}$$

We will define this limiting quantity more generally below.

Definition 4.11 (Quadratic Variation)**.** For a continuous real-valued process X, its **quadratic variation** is defined by

$$[X]_T := \lim_{N\to\infty} \sum_{j=1}^{N} (X_{t_j} - X_{t_{j-1}})^2 \tag{4.3.11}$$

whenever this sequences converges in probability, and t_j is defined by the uniform grid of $[0, T]$. If the limit exists for each $T > 0$, the family $([X]_t)_{t\geq 0}$ is the quadratic variation process, with $[X]_0 = 0$.

For calculations, we write $[X]_T^{(N)}$ as shorthand for the finite sum in (4.3.11). In general, $[X]_T^{(N)}$ does not necessarily have a limit, and even if it does, the limit could be zero. For Brownian motion, the Law of Large Numbers calculation also gives us the following result.

Proposition 4.12 (Quadratic Variation of Brownian Motion)**.** *For a standard Brownian motion,* $[B]_t = t$ *for every* $t \geq 0$*.*

As a consequence of a non-zero quadratic variation, we can actually learn about the property of non-differentiability of Brownian motion as well.

Proposition 4.13 (Brownian Paths Are Not Continuously Differentiable)**.** *For any* $T > 0$*, the Brownian motion's realized path* $(B_t(\omega))_{t\in[0,T]}$ *is almost surely not continuously differentiable.*

Proof. Let $g \in C^1([0, T])$. Recall the earlier left endpoint sum L_N and right endpoint sum R_N converge to the same limit $\int_0^T g(t)g'(t)\,dt$ by (4.3.4).

Their difference therefore satisfies

$$[g]_T^{(N)} = \sum_{j=1}^{N} (\Delta g(t_j))^2 = R_N - L_N \longrightarrow 0. \tag{4.3.12}$$

However, since the Brownian motion has a non-zero quadratic variation, the contrapositive implies the Brownian motion paths are not continuously differentiable. □

In fact, Brownian paths are nowhere differentiable with probability one, but we will not prove this stronger result here. While the quadratic variation is deterministic for Brownian motion, in general, the quadratic variation can be a stochastic process.

Example 4.14 (Random Quadratic Variation)**.** Let A be independent of B, with $\mathbb{P}(A = 1) = \mathbb{P}(A = 2) = 1/2$, and define $X_t = AB_t$. For $t > 0$, the law of X_t is an equal mixture of $\mathcal{N}(0, t)$ and $\mathcal{N}(0, 4t)$, so X is not Brownian motion. Each increment is multiplied by A, giving the exact identity $[X]_T^{(N)} = A^2[B]_T^{(N)}$. Independence and Proposition 4.12 give

$$\mathbb{E}\left[([X]_T^{(N)} - A^2T)^2\right] = \mathbb{E}A^4\,\frac{2T^2}{N} \longrightarrow 0. \tag{4.3.13}$$

Hence $[X]_t = A^2t$. This quadratic variation is itself a stochastic process: each path is an increasing straight line with random slope 1 or 4.

To prepare for SDEs with several sources of noise, we also consider Brownian motion in several dimensions.

Definition 4.15 (Multidimensional Brownian Motion)**.** A **standard Brownian motion in** $\mathbb{R}^d$ is a process $B_t = (B_t^1, \dots, B_t^d)^\top$ whose coordinates are independent standard Brownian motions.

Its increments are independent over disjoint time intervals, with $B_t - B_s \sim \mathcal{N}(0, (t-s)I_d)$ for $s < t$. To compare the fluctuations of two processes, we replace squared increments by products of increments.

Definition 4.16 (Quadratic Covariation)**.** For two continuous real-valued processes X and Y, their **quadratic covariation** is defined by

$$[X, Y]_T := \lim_{N\to\infty} \sum_{j=1}^{N} (X_{t_j} - X_{t_{j-1}})(Y_{t_j} - Y_{t_{j-1}}) \tag{4.3.14}$$

whenever this sequence converges in probability, and t_j is defined by the uniform grid of $[0, T]$. If the limit exists for each $T > 0$, the family $([X, Y]_t)_{t\geq 0}$ is the quadratic covariation process, with $[X, Y]_0 = 0$.

We write $[X, Y]_T^{(N)}$ for the finite sum in (4.3.14). In particular, $[X, X]_t = [X]_t$. Covariation measures how the small increments of the two processes accumulate together; unlike quadratic variation, it can be negative.

Proposition 4.17 (Brownian Covariation)**.** *For a standard d-dimensional Brownian motion,*

$$[B^i, B^j]_T = \begin{cases} T, & i = j, \\ 0, & i \neq j. \end{cases} \tag{4.3.15}$$

Proof. We compute the second moment using independence of the coordinates and of increments on disjoint intervals. For $i \neq j$, each product $\Delta B_{t_k}^i \Delta B_{t_k}^j$ has mean zero and second moment $(\Delta t)^2$. The products are independent across k, so

$$\mathbb{E}\left[\left([B^i, B^j]_T^{(N)}\right)^2\right] = N(\Delta t)^2 = \frac{T^2}{N} \longrightarrow 0. \tag{4.3.16}$$

This proves the off-diagonal limits in L^2; the diagonal limits follow from Proposition 4.12. □

Independent Brownian coordinates have zero covariation, but two Brownian motions can have nonzero covariation when they share a source of noise.

Example 4.18 (Correlated Brownian Motions)**.** Let B^1 and B^2 be independent standard Brownian motions, and let $\rho \in [-1, 1]$. The process

$$W_t = \rho B_t^1 + \sqrt{1-\rho^2}\, B_t^2 \tag{4.3.17}$$

is also a standard Brownian motion: it starts at zero, has continuous paths and independent increments, and its increments are centered Gaussian with variance $(\rho^2 + 1 - \rho^2)(t-s) = t-s$. Expanding the finite covariation sums gives

$$[B^1, W]_T^{(N)} = \rho [B^1]_T^{(N)} + \sqrt{1-\rho^2}\, [B^1, B^2]_T^{(N)} \longrightarrow \rho T \quad \text{in } L^2. \tag{4.3.18}$$

Hence $[B^1, W]_t = \rho t$, whereas $[B^1]_t = [W]_t = t$. For $t > 0$, we also have $\mathrm{Cov}(B_t^1, W_t) = \rho t$, so their correlation is ρ. The covariation is nonzero whenever $\rho \neq 0$, and it is negative when $\rho < 0$.

Quadratic variation and covariation will supply the second-order terms in Itô's formula. We next turn to the construction of the Itô integral, using the left-endpoint convention that our Brownian calculation has already distinguished.

Exercise 4.6 (Adding a Smooth Drift)**.** Let $g \in C^1([0,\infty))$ be deterministic, let $\sigma \in \mathbb{R}$, and set $X_t = g(t) + \sigma B_t$. Fix $T > 0$ and use the uniform grid from (4.3.2).

(a) With $C = \max_{0 \le t \le T} |g'(t)|$, show that $[g]_T^{(N)} \le C^2 T^2 / N$.

(b) Apply Cauchy–Schwarz to the finite sum defining $[g, B]_T^{(N)}$ to show that

$$\mathbb{E}\left[\left([g, B]_T^{(N)} \right)^2 \right] \le \frac{C^2 T^3}{N}. \tag{4.3.19}$$

(c) Expand the squared increments of X and deduce that $[X]_T^{(N)} \to \sigma^2 T$ in L^2. Explain why the smooth drift does not contribute to quadratic variation.

Exercise 4.7 (Time-Midpoint Sums)**.** Fix $T > 0$ and consider the actual time-midpoint sum $M_N = \sum_{j=1}^N B_{t_j^*} \Delta B_{t_j}$ on the uniform grid of $[0, T]$.

(a) By expanding the change in the square on each half-interval, show that

$$M_N - \frac{B_T^2}{2} = \frac{1}{2} \sum_{j=1}^N \left((B_{t_j^*} - B_{t_{j-1}})^2 - (B_{t_j} - B_{t_j^*})^2 \right). \tag{4.3.20}$$

(b) Each half-interval increment is $\mathcal{N}(0, \Delta t/2)$. Show that each of the two squared-increment sums in part (a) has mean $T/2$ and variance $T^2/(2N)$.

(c) Deduce that $M_N \to B_T^2/2$ in L^2. Using independence of all $2N$ half-interval increments, also calculate $\mathbb{E}[(M_N - B_T^2/2)^2]$.

(d) Show that averaging the endpoint values gives the exact identity

$$\sum_{j=1}^N \frac{B_{t_{j-1}} + B_{t_j}}{2}\, \Delta B_{t_j} = \frac{B_T^2}{2}. \tag{4.3.21}$$

Explain why this sum and the actual time-midpoint sum have the same limit but are different random variables for fixed N.

Exercise 4.8 (Uncorrelated Brownian Motions Can Be Dependent)**.** Let A be independent of B, with $\mathbb{P}(A = 1) = \mathbb{P}(A = -1) = 1/2$, and set $W_t = AB_t$.

(a) Show that W is standard Brownian motion. For independent increments, observe that multiplying a vector of independent centered Gaussian increments by either fixed sign leaves its joint law unchanged.

(b) Show that $\mathrm{Cov}(B_s, W_t) = 0$ for every $s, t \geq 0$.

(c) For $t > 0$, use $|W_t| = |B_t|$ to show that B_t and W_t are dependent. One way is to compare the events $\{|B_t| \leq \sqrt{t}\}$ and $\{|W_t| \leq \sqrt{t}\}$, whose common probability lies strictly between zero and one. Why does this not contradict the independence of uncorrelated jointly Gaussian variables?

(d) Show that $[B, W]_T^{(N)} \to AT$ in L^2. Explain how this covariation can be nonzero almost surely even though the covariance in part (b) is zero.

Appendix A

Technical Background

A.1 Approximation by Simple Functions

On a finite state space, every function is a linear combination of state indicators. On a continuous state space, a measurable function need not have such a finite representation, but it can be approximated by functions that do. This lets us extend identities for probabilities to identities for expectations.

A **simple function** is a finite linear combination of indicators of measurable sets:

$$s(x) = \sum_{j=1}^{m} a_j \mathbf{1}_{B_j}(x). \tag{A.1}$$

Equivalently, it is a measurable function that takes only finitely many values. For example, rounding a bounded nonnegative function down to the nearest multiple of $1/2$ gives a simple function. Using finer levels improves the approximation.

Proposition A.1 (Monotone Approximation)**.** *For every nonnegative measurable function $\varphi : E \to [0, \infty)$, there are nonnegative simple functions s_k such that $s_k(x) \uparrow \varphi(x)$ for every $x \in E$.*

Proof. We truncate large values and round the remaining values down to progressively finer levels. Specifically, set

$$s_k(x) = 2^{-k} \left\lfloor 2^k \min\{\varphi(x), 2^k\} \right\rfloor, \qquad k \geq 1. \tag{A.2}$$

Each s_k is measurable and takes only finitely many values, so it is simple. Raising the truncation level and halving the grid spacing can only increase the rounded value, so $s_k \leq s_{k+1} \leq \varphi$. For each fixed x, eventually $\varphi(x) < 2^k$ and $0 \leq \varphi(x) - s_k(x) < 2^{-k}$. Thus $s_k(x) \uparrow \varphi(x)$. □

We use the **monotone convergence theorem** as a basic integration result: if $0 \leq g_k \uparrow g$, then, for any measure ν,

$$\int g_k \, d\nu \uparrow \int g \, d\nu. \tag{A.3}$$

The limiting integral is allowed to be infinite. In particular, $0 \leq V_k \uparrow V$ for random variables implies $\mathbb{E}V_k \uparrow \mathbb{E}V$.

There is also a conditional version. If $0 \le V_k \uparrow V$ and V is integrable, then

$$\mathbb{E}[V_k | \mathcal{G}] \uparrow \mathbb{E}[V | \mathcal{G}] \quad \text{almost surely.} \tag{A.4}$$

To see this, conditional monotonicity gives an increasing sequence bounded above by $\mathbb{E}[V | \mathcal{G}]$. Write its limit as W. Then W is $\mathcal{G}$-measurable and integrable. For every $A \in \mathcal{G}$, ordinary monotone convergence and the definition of conditional expectation give

$$\mathbb{E}[W\mathbf{1}_A] = \lim_{k\to\infty} \mathbb{E}[\mathbb{E}[V_k | \mathcal{G}]\mathbf{1}_A] = \lim_{k\to\infty} \mathbb{E}[V_k \mathbf{1}_A] = \mathbb{E}[V\mathbf{1}_A]. \tag{A.5}$$

Thus W satisfies the defining properties of $\mathbb{E}[V | \mathcal{G}]$, proving the conditional version.

We now apply this approximation to a transition kernel P. For a simple function s as in (A.1), linearity gives

$$(Ps)(x) = \sum_{j=1}^{m} a_j P(x, B_j). \tag{A.6}$$

Each term is measurable in x by the definition of a transition kernel. For a nonnegative measurable φ, choose the increasing simple approximations above. Monotone convergence with respect to the measure $P(x, \cdot)$ gives $Ps_k(x) \uparrow P\varphi(x)$. Hence $P\varphi$ is measurable as a pointwise limit of measurable functions. For a bounded measurable φ, add a constant c so that $\varphi + c \ge 0$. Since $P(x, E) = 1$, we have $P(\varphi + c) = P\varphi + c$, so $P\varphi$ is measurable in this case as well. Also, $|P\varphi(x)| \le \sup_{y\in E} |\varphi(y)|$.

Finally, let (X_n) be a Markov chain with transition kernel P and natural filtration $(\mathcal{F}_n)$. The conditional-probability definition is an identity for indicators:

$$\mathbb{E}[\mathbf{1}_B(X_{n+1}) | \mathcal{F}_n] = P(X_n, B) = (P\mathbf{1}_B)(X_n). \tag{A.7}$$

By linearity, (A.6) extends this identity to every simple function:

$$\mathbb{E}[s(X_{n+1}) | \mathcal{F}_n] = (Ps)(X_n). \tag{A.8}$$

Now suppose φ is bounded, measurable, and nonnegative, and take $s_k \uparrow \varphi$. Conditional monotone convergence applies because $\varphi(X_{n+1})$ is bounded and hence integrable. Together with monotone convergence for the kernel integral, it gives

$$\begin{gathered} \mathbb{E}[s_k(X_{n+1}) | \mathcal{F}_n] \uparrow \mathbb{E}[\varphi(X_{n+1}) | \mathcal{F}_n], \\ (Ps_k)(X_n) \uparrow (P\varphi)(X_n). \end{gathered} \tag{A.9}$$

The two sequences are equal almost surely for every k, so their limits agree almost surely. For a general bounded measurable φ, apply this result to $\varphi + c \ge 0$ and subtract c from both sides. We obtain the test-function form of the Markov property:

$$\mathbb{E}[\varphi(X_{n+1}) | \mathcal{F}_n] = (P\varphi)(X_n). \tag{A.10}$$

A.2 Approximation by Simple Processes

The construction of the Itô integral uses approximation in the process norm from **??**. For continuous adapted processes, we can obtain this approximation by truncating large values and then sampling at the left endpoints of a uniform grid.

Proposition A.2 (Approximation of Continuous Adapted Processes)**.** *Fix $T > 0$ and let H be a continuous real-valued process adapted to $(\mathcal{F}_t)_{0\le t\le T}$, with $\|H\|_T^2 = \mathbb{E}\int_0^T H_t^2\,dt < \infty$. There exists a sequence of simple processes $H^{(n)}$, as in* **??***, such that*

$$\|H^{(n)} - H\|_T \longrightarrow 0. \tag{A.11}$$

Proof. We first approximate a bounded process on a uniform grid, then use truncation to handle an unbounded process. Suppose $|H_t| \le M$ for a deterministic constant M. On the grid $t_j = jT/N$, define $S_t^{(N)} = H_{t_{j-1}}$ for $t \in (t_{j-1}, t_j]$. The coefficients are bounded and known at the left endpoints, so $S^{(N)}$ is a simple process. Continuity of each path on $[0, T]$ gives convergence of these step approximations as the grid becomes finer. Since $|S_t^{(N)} - H_t|^2 \le 4M^2$, dominated convergence gives

$$\|S^{(N)} - H\|_T^2 = \mathbb{E}\int_0^T (S_t^{(N)} - H_t)^2\,dt \longrightarrow 0. \tag{A.12}$$

For a general H, clip its values to $[-M, M]$ by setting $H_t^{[M]} = \max\{-M, \min\{H_t, M\}\}$. This process remains continuous and adapted. Also, $H_t^{[M]} \to H_t$ and $|H_t^{[M]} - H_t|^2 \le H_t^2$. The assumption $\|H\|_T < \infty$ therefore lets us apply dominated convergence again to obtain $\|H^{[M]} - H\|_T \to 0$ as $M \to \infty$.

Given $\varepsilon > 0$, choose M so that this truncation error is less than $\varepsilon/2$. The bounded case supplies a simple process S with $\|S - H^{[M]}\|_T < \varepsilon/2$. The triangle inequality gives

$$\|S - H\|_T \le \|S - H^{[M]}\|_T + \|H^{[M]} - H\|_T < \varepsilon. \tag{A.13}$$

Choosing such a simple process for each $\varepsilon = 1/n$ proves the result. □

Appendix B

Solutions to Exercises

Solutions to Chapter 1

Solution to Exercise 1.1. The key is that the report distinguishes two groups of worlds, not the individual worlds within each group.

(a) An event determined by the report must contain all of A or none of A, and all of A^c or none of A^c. Thus it must be one of $\varnothing, A, A^c, \Omega$. Conversely, the report determines whether each of these events occurred. This collection contains Ω and is closed under complements. Any countable union of its members is again a union of some of the two groups A, A^c, so it remains in $\mathcal{F}$. Hence $\mathcal{F}$ is a σ-algebra.

(b) The axioms give $\mathbb{P}(\varnothing) = 0$ and $\mathbb{P}(\Omega) = 1$. Together with $\mathbb{P}(A) = 0.3$ and $\mathbb{P}(A^c) = 1 - 0.3 = 0.7$, this specifies $\mathbb{P}$ on all four events in its domain.

(c) No: $\{\omega_1\} \notin \mathcal{F}$, so its probability is not defined in this model. An enlarged model could assign probabilities to individual worlds, but the mass 0.3 could be split between ω_1 and ω_2 in many ways.

Solution to Exercise 1.2. All of the conditions hold exactly when none of them fails. For each n, the complement A_n^c belongs to $\mathcal{F}$. Their countable union therefore belongs to $\mathcal{F}$, and so does its complement. De Morgan's identity now gives

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c \in \mathcal{F}. \tag{B.1}$$

Solution to Exercise 1.3.

(a) Inclusion-exclusion gives $\mathbb{P}(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$.

(b) Neither occurs on $(A \cup B)^c$, whose probability is $1 - 0.7 = 0.3$.

(c) Remove the event that both occur from the event that at least one occurs. The resulting probability is $0.7 - 0.2 = 0.5$.

No independence assumption is needed for these calculations.

Solution to Exercise 1.4. The information distinguishes A from A^c, but not the two worlds within either group.

(a) The function X is constant on each group. For every Borel set B, its preimage $X^{-1}(B)$ is therefore one of $\varnothing, A, A^c, \Omega$, so X is measurable. In contrast, $Y^{-1}(\{0\}) = \{\omega_1\} \notin \mathcal{F}$, so Y is not measurable.

(b) The random variable is X, and $X^{-1}([9,11]) = A^c$. Therefore $\mathbb{P}(X \in [9,11]) = \mathbb{P}(A^c) = 0.7$.

Solution to Exercise 1.5.

(a) For the first vector, $(S_0^{(A)})^{-1}([9,11]) = (S_1^{(A)})^{-1}([9,11]) = \{\omega_1\}$. For the second vector, $(S_0^{(B)})^{-1}([9,11]) = \{\omega_1\}$ and $(S_1^{(B)})^{-1}([9,11]) = \{\omega_2\}$. Every one of these four events has probability $1/2$.

(b) For $R = [9,11] \times [9,11]$, the preimages are $(S^{(A)})^{-1}(R) = \{\omega_1\}$ and $(S^{(B)})^{-1}(R) = \varnothing$. Their probabilities are therefore $1/2$ and 0, respectively.

(c) No. All four one-day probabilities in part (a) are $1/2$, but the probability that both prices lie in $[9,11]$ is $1/2$ for $S^{(A)}$ and 0 for $S^{(B)}$. Thus these one-day probabilities do not determine this two-day probability.

Solution to Exercise 1.6.

(a) The cylinder set is

$$B_{\text{end}} = \{f \in C([0,1]) : f(0) \in [9,11],\ f(1) \in [9,11]\}. \tag{B.2}$$

Both f_1 and f_2 have value 10 at both endpoints. Hence $S^{-1}(B_{\text{end}}) = \Omega$ and $\mathbb{P}(S \in B_{\text{end}}) = 1$.

(b) No. The path f_1 stays in the interval throughout, whereas $f_2(1/2) = 12 > 11$. Therefore

$$\{\omega \in \Omega : S(\omega)(t) \in [9,11] \text{ for every } t \in [0,1]\} = \{\omega_1\}, \tag{B.3}$$

and this event has probability 0.7.

Solution to Exercise 1.7.

(a) The four possible values of (X, Y) are distinct, so their masses can be read directly from the table:

$$\mu_{X,Y} = 0.1\delta_{(0,0)} + 0.2\delta_{(0,1)} + 0.3\delta_{(1,0)} + 0.4\delta_{(1,1)}. \tag{B.4}$$

(b) Adding the masses with the same first coordinate gives μ_X; adding those with the same second coordinate gives μ_Y. Thus

$$\mu_X = 0.3\delta_0 + 0.7\delta_1, \qquad \mu_Y = 0.4\delta_0 + 0.6\delta_1. \tag{B.5}$$

(c) The Borel region $\{(x,y) \in \mathbb{R}^2 : x \le y\}$ contains the atoms $(0,0)$, $(0,1)$, and $(1,1)$. Therefore $\mathbb{P}(X \le Y) = 0.1 + 0.2 + 0.4 = 0.7$.

Solution to Exercise 1.8.

(a) The function is nonnegative, and its support is the triangle $0 \le x \le y \le 1$. Its total integral is

$$\int_0^1 \int_x^1 2\, dy\, dx = \int_0^1 2(1-x)\, dx = 1. \tag{B.6}$$

Thus $f_{X,Y}$ is a joint probability density.

(b) For a fixed $x \in [0,1]$, the possible values of y range from x to 1. For a fixed $y \in [0,1]$, the possible values of x range from 0 to y. Therefore the marginal densities are

$$f_X(x) = \begin{cases} 2(1-x), & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases} \qquad f_Y(y) = \begin{cases} 2y, & 0 \le y \le 1, \\ 0, & \text{otherwise}. \end{cases} \tag{B.7}$$

Equivalently, for every $B \in \mathcal{B}(\mathbb{R})$,

$$\mu_X(B) = \int_{B \cap [0,1]} 2(1-x)\,dx, \qquad \mu_Y(B) = \int_{B \cap [0,1]} 2y\,dy. \tag{B.8}$$

(c) Within the support, the condition $x + y \le 1$ gives $0 \le x \le 1/2$ and $x \le y \le 1 - x$. Hence

$$\mathbb{P}(X + Y \le 1) = \int_0^{1/2} \int_x^{1-x} 2\,dy\,dx = \frac{1}{2}. \tag{B.9}$$

Solution to Exercise 1.9.

(a) The total mass is

$$q + (1-q) \int_0^\infty r e^{-rx}\,dx = q + (1-q) = 1. \tag{B.10}$$

(b) Since X is nonnegative, integration by parts, together with $xe^{-rx} \to 0$ as $x \to \infty$, gives

$$\int_{\mathbb{R}} |x|\,\mu_X(dx) = q \cdot 0 + (1-q) \int_0^\infty x r e^{-rx}\,dx = \frac{1-q}{r} < \infty. \tag{B.11}$$

Thus X is integrable, and the same calculation gives $\mathbb{E}X = (1-q)/r$.

Solution to Exercise 1.10.

(a) Each card in its original position contributes one to both N and the sum of the indicators, while every other card contributes zero. Hence

$$N = \sum_{i=1}^n I_i. \tag{B.12}$$

(b) Among the $n!$ possible orders, exactly $(n-1)!$ place card i in position i. Therefore

$$\mathbb{P}(I_i = 1) = \frac{(n-1)!}{n!} = \frac{1}{n}, \qquad \mathbb{E}I_i = \frac{1}{n}. \tag{B.13}$$

(c) By linearity of expectation,

$$\mathbb{E}N = \sum_{i=1}^n \mathbb{E}I_i = \sum_{i=1}^n \frac{1}{n} = 1. \tag{B.14}$$

Independence of the indicators is not needed.

Solution to Exercise 1.11.

(a) If $N(\omega) = j$, then exactly the first j indicators in the sum are one and the remaining indicators are zero. The sum therefore equals $j = N(\omega)$ in every possible world.

(b) Taking expectations in the pointwise identity and using linearity and (1.4.9) gives

$$\mathbb{E}N = \sum_{k=1}^{m} \mathbb{E}\mathbf{1}_{\{N \geq k\}} = \sum_{k=1}^{m} \mathbb{P}(N \geq k). \tag{B.15}$$

(c) The three tail probabilities are $\mathbb{P}(N \geq 1) = 0.9$, $\mathbb{P}(N \geq 2) = 0.7$, and $\mathbb{P}(N \geq 3) = 0.4$. Thus

$$\mathbb{E}N = 0.9 + 0.7 + 0.4 = 2. \tag{B.16}$$

Solution to Exercise 1.12.

(a) Adding the entries in each row and column gives

$$\mu_X = 0.3\delta_0 + 0.7\delta_1, \qquad \mu_Y = 0.4\delta_0 + 0.6\delta_1. \tag{B.17}$$

(b) The four entries in the table satisfy

$$0.12 = 0.3 \cdot 0.4, \quad 0.18 = 0.3 \cdot 0.6, \quad 0.28 = 0.7 \cdot 0.4, \quad 0.42 = 0.7 \cdot 0.6. \tag{B.18}$$

Thus every joint point mass is the product of the corresponding marginal point masses. Since X and Y only take the values zero and one, the required factorization for arbitrary Borel sets follows by adding the appropriate point masses. Therefore X and Y are independent.

Solution to Exercise 1.13.

(a) Independence gives

$$\mathbb{P}(I_1 = 0, \ldots, I_n = 0) = (1-p)^n. \tag{B.19}$$

(b) The event that at least one random variable equals one is the complement of the event in part (a), so its probability is $1 - (1-p)^n$.

(c) For each i, the probability that $I_i = 1$ and all the other random variables equal zero is $p(1-p)^{n-1}$. These n events are disjoint, so the probability that exactly one random variable equals one is $np(1-p)^{n-1}$.

Solution to Exercise 1.14.

(a) The correlation coefficient is

$$\rho(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}} = \frac{3}{\sqrt{4 \cdot 9}} = \frac{1}{2}. \tag{B.20}$$

(b) From the definition of covariance and linearity of expectation,

$$\operatorname{Cov}(X, Z_a) = \operatorname{Cov}(X, Y - aX) = \operatorname{Cov}(X, Y) - a\operatorname{Var}(X) = 3 - 4a. \tag{B.21}$$

Thus $\operatorname{Cov}(X, Z_a) = 0$ when $a = 3/4$.

(c) The resulting random variable is $Z = Y - \frac{3}{4}X$, and part (b) gives $\operatorname{Cov}(X, Z) = 0$. Moreover,

$$\begin{aligned} \operatorname{Var}(Z) &= \operatorname{Var}\left(Y - \frac{3}{4}X\right) \\ &= \operatorname{Var}(Y) + \frac{9}{16}\operatorname{Var}(X) - 2\left(\frac{3}{4}\right)\operatorname{Cov}(X, Y) = \frac{27}{4}. \end{aligned} \tag{B.22}$$

Since $Y = \frac{3}{4}X + Z$ and X and Z are uncorrelated,

$$\operatorname{Var}(Y) = \operatorname{Var}\left(\frac{3}{4}X + Z\right) = \frac{9}{16}\operatorname{Var}(X) + \operatorname{Var}(Z) = \frac{9}{4} + \frac{27}{4} = 9. \tag{B.23}$$

Solution to Exercise 1.15.

(a) Linearity of expectation and independence give

$$\mathbb{E}S_n = 10, \qquad \operatorname{Var}(S_n) = \frac{9}{n}. \tag{B.24}$$

(b) Chebyshev's inequality gives

$$\mathbb{P}(|S_n - 10| \geq 1) \leq \frac{\operatorname{Var}(S_n)}{1^2} = \frac{9}{n}. \tag{B.25}$$

(c) The bound is at most 0.05 exactly when $9/n \leq 0.05$, or $n \geq 180$. Thus the smallest such integer is 180.

(d) For every $\varepsilon > 0$, Chebyshev's inequality gives

$$0 \leq \mathbb{P}(|S_n - 10| \geq \varepsilon) \leq \frac{9}{n\varepsilon^2} \longrightarrow 0. \tag{B.26}$$

This is the conclusion of the weak law of large numbers with mean 10.

Solution to Exercise 1.16.

(a) Since $X_n = 1 - X$,

$$\mathbb{P}(X_n = 0) = \mathbb{P}(X = 1) = \frac{1}{2}, \qquad \mathbb{P}(X_n = 1) = \mathbb{P}(X = 0) = \frac{1}{2}. \tag{B.27}$$

Thus X_n and X have the same law for every n, so their distribution functions are equal and $X_n \xrightarrow{d} X$.

(b) Because X takes only the values zero and one, $|X_n - X| = |1 - 2X| = 1$ in every possible world. In particular,

$$\mathbb{P}\left(|X_n - X| \geq \frac{1}{2}\right) = 1 \tag{B.28}$$

for every n. Hence X_n does not converge in probability to X.

Solution to Exercise 1.17.

(a) Since $\mathbb{E}X_i = p$ and $\mathrm{Var}(X_i) = p(1-p)$, linearity gives the mean, while independence and Corollary 1.22 give the variance:

$$\mathbb{E}S_n = p, \qquad \mathrm{Var}(S_n) = \frac{p(1-p)}{n}. \tag{B.29}$$

The weak law therefore says that $S_n \xrightarrow{\mathbb{P}} p$.

(b) The central limit theorem gives

$$\frac{\sqrt{n}(S_n - p)}{\sqrt{p(1-p)}} \xrightarrow{d} Z, \tag{B.30}$$

where Z has the standard normal law.

(c) Standardizing the event gives

$$\begin{aligned}\mathbb{P}\left(S_n \le p + \frac{c}{\sqrt{n}}\right) &= \mathbb{P}\left(\frac{\sqrt{n}(S_n - p)}{\sqrt{p(1-p)}} \le \frac{c}{\sqrt{p(1-p)}}\right) \\ &\longrightarrow \Phi\left(\frac{c}{\sqrt{p(1-p)}}\right).\end{aligned} \tag{B.31}$$

Solutions to Chapter 2

Solution to Exercise 2.1.

(a) The available information distinguishes only A from A^c, so $\mathcal{G} = \{\varnothing, A, A^c, \Omega\}$.

(b) The conditional average of X on A is $(0+2)/2 = 1$, while its conditional average on A^c is $(4+6+8+10)/4 = 7$. Hence

$$\mathbb{E}[X \,|\, \mathcal{G}] = \mathbf{1}_A + 7\mathbf{1}_{A^c}. \tag{B.32}$$

(c) The testing identity is automatic on $\varnothing$. On A and A^c, respectively,

$$\mathbb{E}\mathbf{1}_A = \frac{1}{3} = \mathbb{E}[X\mathbf{1}_A], \qquad \mathbb{E}[7\mathbf{1}_{A^c}] = \frac{14}{3} = \mathbb{E}[X\mathbf{1}_{A^c}]. \tag{B.33}$$

Adding the two identities verifies the test on Ω. It also gives

$$\mathbb{E}\left[\mathbb{E}[X \,|\, \mathcal{G}]\right] = \frac{1}{3} + \frac{14}{3} = 5 = \mathbb{E}X. \tag{B.34}$$

Solution to Exercise 2.2.

(a) Conditional on A, only one of the three possible die values is even. Conditional on A^c, two of the three possible values are even. Therefore

$$\mathbb{P}(B \mid \mathcal{G}) = \frac{1}{3}\mathbf{1}_A + \frac{2}{3}\mathbf{1}_{A^c}. \tag{B.35}$$

(b) On A, both $\mathbb{E}[\mathbb{P}(B \mid \mathcal{G})\mathbf{1}_A]$ and $\mathbb{E}[\mathbf{1}_B\mathbf{1}_A]$ equal $1/6$. On A^c, both corresponding expectations equal $2/6$. Thus the testing identity holds on the two generating cells, and hence on every event in $\mathcal{G}$.

(c) Averaging the two conditional probabilities gives $(1/3)(1/2) + (2/3)(1/2) = 1/2 = \mathbb{P}(B)$.

Solution to Exercise 2.3.

(a) Since U is independent of $\mathcal{G} = \sigma(V)$, independent information does not change its expectation. Hence $\mathbb{E}[U|\,\mathcal{G}] = \mathbb{E}U = 1/2$ almost surely.

(b) If $V = 0$, the two possible values of $U + V$ are zero and one, whose average is $1/2$. If $V = 1$, the two possible values are one and two, whose average is $3/2$. Consequently,

$$\mathbb{E}[U + V|\,\mathcal{G}] = \frac{1}{2} + V. \tag{B.36}$$

(c) The observed term V remains random because its value is known from $\mathcal{G}$. The independent term U is replaced by its ordinary mean $1/2$.

Solution to Exercise 2.4.

(a) Since $-|X| \le X \le |X|$, monotonicity and linearity give

$$-\,\mathbb{E}[\,|X||\,\mathcal{G}] \le \mathbb{E}[X|\,\mathcal{G}] \le \mathbb{E}[\,|X||\,\mathcal{G}]. \tag{B.37}$$

The outer two quantities are negatives of one another, so the displayed bounds imply (2.1.31).

(b) Taking ordinary expectations in part (a) and using the law of total expectation gives

$$\mathbb{E}|\mathbb{E}[X|\,\mathcal{G}]| \le \mathbb{E}\,[\mathbb{E}[\,|X||\,\mathcal{G}]] = \mathbb{E}|X|, \tag{B.38}$$

which is (2.1.32).

Solution to Exercise 2.5.

(a) Linearity and the fact that conditioning a $\mathcal{G}$-measurable random variable on $\mathcal{G}$ leaves it unchanged give

$$\mathbb{E}[R|\,\mathcal{G}] = \mathbb{E}[X|\,\mathcal{G}] - \mathbb{E}[X|\,\mathcal{G}] = 0. \tag{B.39}$$

(b) The pull-out property and the law of total expectation give

$$\mathbb{E}[RY] = \mathbb{E}\,[\mathbb{E}[RY|\,\mathcal{G}]] = \mathbb{E}\,[Y\mathbb{E}[R|\,\mathcal{G}]] = 0. \tag{B.40}$$

Solution to Exercise 2.6.

(a) Each S_k is a function of $\xi_1, \ldots, \xi_k$, so $\sigma(S_0, \ldots, S_n) \subseteq \mathcal{F}_n$. Conversely, $\xi_k = S_k - S_{k-1}$ for $1 \le k \le n$, which gives the reverse inclusion.

(b) The first part follows because S_n is $\mathcal{F}_n$-measurable. Since S_{n-1} is $\mathcal{F}_{n-1}$-measurable, so is $H_n = \mathbf{1}_{\{S_{n-1}=0\}}$, and therefore $(H_n)_{n\ge1}$ is predictable. The event $\{\xi_{n+1} = 1\}$ is independent of $\mathcal{F}_n$ and has probability $1/2$. If it belonged to $\mathcal{F}_n$, independence from itself would force its probability to be zero or one. Thus ξ_{n+1} is not $\mathcal{F}_n$-measurable.

(c) Since S_n is $\mathcal{F}_n$-measurable, while ξ_{n+1} is independent of $\mathcal{F}_n$ with mean zero and square one,

$$\begin{aligned}
\mathbb{E}[\,M_{n+1}|\,\mathcal{F}_n] &= \mathbb{E}[(S_n + \xi_{n+1})^2 - (n+1)\Big|\,\mathcal{F}_n] \\
&= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1}|\,\mathcal{F}_n] + \mathbb{E}[\xi_{n+1}^2\Big|\,\mathcal{F}_n] - (n+1) \\
&= S_n^2 - n = M_n.
\end{aligned} \tag{B.41}$$

Solution to Exercise 2.7.

(a) The process is adapted, and independence gives $\mathbb{E}M_n = \prod_{k=1}^n \mathbb{E}X_k = 1$. If $A \in \mathcal{F}_n$, then $M_n \mathbf{1}_A$ is $\mathcal{F}_n$-measurable and is therefore independent of X_{n+1}. Hence

$$\mathbb{E}[M_{n+1}\mathbf{1}_A] = \mathbb{E}[M_n X_{n+1}\mathbf{1}_A] = \mathbb{E}[M_n\mathbf{1}_A]\mathbb{E}X_{n+1} = \mathbb{E}[M_n\mathbf{1}_A]. \tag{B.42}$$

Thus M_n satisfies the testing identity for $\mathbb{E}[\,M_{n+1}|\,\mathcal{F}_n]$, and so it is a martingale.

(b) Set $X_k = e^{\theta\xi_k}/\phi(\theta)$. These random variables are independent, nonnegative, and have mean one. Their product is

$$\prod_{k=1}^n X_k = \frac{\exp(\theta\sum_{k=1}^n \xi_k)}{\phi(\theta)^n}. \tag{B.43}$$

Part (a) therefore proves the claim.

Solution to Exercise 2.8.

(a) Independence gives $\mathbb{E}[\xi_{n+1}|\,\mathcal{F}_n] = \mathbb{E}\xi_{n+1} = \mu$. Therefore,

$$\begin{aligned}\mathbb{E}[\,S_{n+1} - (n+1)\mu|\,\mathcal{F}_n] &= S_n - (n+1)\mu + \mathbb{E}[\xi_{n+1}|\,\mathcal{F}_n] \\ &= S_n - n\mu.\end{aligned} \tag{B.44}$$

Thus $(S_n - n\mu)_{n\geq 0}$ is a martingale.

(b) The same calculation gives

$$\mathbb{E}[\,S_{n+1}|\,\mathcal{F}_n] = S_n + \mu. \tag{B.45}$$

Hence $(S_n)_{n\geq 0}$ is a submartingale when $\mu \geq 0$, a supermartingale when $\mu \leq 0$, and a martingale when $\mu = 0$.

(c) Here $\mu = p - (1-p) = 2p - 1$, so the centered martingale is $(S_n - n(2p-1))_{n\geq 0}$. The original random walk is a submartingale when $p \geq 1/2$, a supermartingale when $p \leq 1/2$, and a martingale when $p = 1/2$.

Solution to Exercise 2.11.

(a) For $\tau_1 = \inf\{n \geq 0 : S_n = 2\}$,

$$\{\tau_1 \leq n\} = \bigcup_{k=0}^n \{S_k = 2\} \in \mathcal{F}_n. \tag{B.46}$$

Thus τ_1 is a stopping time.

(b) If $\tau_2 = \inf\{n \geq 0 : S_n \notin \{-1, 0, 1\}\}$, then

$$\{\tau_2 \leq n\} = \bigcup_{k=0}^n \{S_k \notin \{-1,0,1\}\} \in \mathcal{F}_n, \tag{B.47}$$

so τ_2 is a stopping time.

(c) Since S_1 cannot equal zero,

$$\{L \leq 0\} = \{S_2 \neq 0\} = \{\xi_1 = \xi_2\}. \tag{B.48}$$

This event has probability $1/2$, so it does not belong to the trivial $\mathcal{F}_0$. Hence L is not a stopping time.

Solution to Exercise 2.12.

(a) For $n = 0$, the event $\{\tau + 1 \le 0\}$ is empty. For $n \ge 1$,

$$\{\tau + 1 \le n\} = \{\tau \le n - 1\} \in \mathcal{F}_{n-1} \subseteq \mathcal{F}_n. \tag{B.49}$$

Therefore $\tau + 1$ is a stopping time.

(b) For every $n \ge 1$,

$$\{\tau \le n\} = \bigcup_{k=1}^{n} \{\xi_k = 1\} \in \mathcal{F}_n, \tag{B.50}$$

so τ is a stopping time. However, $\{\tau - 1 \le 0\} = \{\xi_1 = 1\}$ has probability $1/2$ and therefore does not belong to the trivial $\mathcal{F}_0$. Thus $\tau - 1$ is not a stopping time.

(c) At time n, determining whether $\tau + 1 \le n$ only requires us to know whether $\tau \le n - 1$. Determining whether $\tau - 1 \le n$ may require knowing whether $\tau \le n + 1$, which uses one additional step of information.

Solution to Exercise 2.13.

(a) Conditional Jensen's inequality and the martingale property give

$$\mathbb{E}[\,|M_k|\,|\,\mathcal{F}_{k-1}] \ge |\mathbb{E}[\,M_k\,|\,\mathcal{F}_{k-1}]| = |M_{k-1}|. \tag{B.51}$$

(b) The first time that the adapted process $|M_n|$ reaches λ is a stopping time, and its minimum with N is bounded by N. It reaches λ by time N exactly when $|M_\tau| \ge \lambda$.

(c) On each path, the sum in (2.3.28) includes exactly the increments after τ, so it equals $|M_N| - |M_\tau|$. Since $\{\tau < k\} = \{\tau \le k - 1\} \in \mathcal{F}_{k-1}$, taking out what is known and using (a) gives

$$\begin{aligned} \mathbb{E}[|M_N| - |M_\tau|] &= \sum_{k=1}^{N} \mathbb{E}\left[\mathbf{1}_{\{\tau<k\}} \left(\mathbb{E}[\,|M_k|\,|\,\mathcal{F}_{k-1}] - |M_{k-1}|\right)\right] \\ &\ge 0. \end{aligned} \tag{B.52}$$

(d) By (b), $|M_\tau| \ge \lambda$ on the event that the maximum reaches λ. Since $|M_\tau|$ is nonnegative, (c) yields

$$\lambda\, \mathbb{P}\left(\max_{0\le n\le N} |M_n| \ge \lambda\right) \le \mathbb{E}|M_\tau| \le \mathbb{E}|M_N|. \tag{B.53}$$

Dividing by $\lambda > 0$ proves the claim.

Solution to Exercise 2.14.

(a) Pointwise, $n \wedge \tau = \sum_{k=1}^{n} \mathbf{1}_{\{\tau \ge k\}}$. Taking expectations gives the finite tail-sum identity.

(b) The stopped square-martingale identity and the barrier bounds give

$$\sum_{k=1}^{n} \mathbb{P}(\tau \ge k) = \mathbb{E}[n \wedge \tau] = \mathbb{E}S^2_{n\wedge\tau} \le \max\{a^2, b^2\}. \tag{B.54}$$

Thus all partial sums of the nonnegative tail series are bounded. The infinite tail-sum identity implies $\mathbb{E}\tau \le \max\{a^2, b^2\} < \infty$, so τ is finite almost surely.

(c) The squares agree on $\{\tau \le n\}$, and both lie between zero and $\max\{a^2, b^2\}$. Hence

$$\left|\mathbb{E}S_\tau^2 - \mathbb{E}S_{n\wedge\tau}^2\right| \le \max\{a^2, b^2\}\,\mathbb{P}(\tau > n) \longrightarrow 0. \tag{B.55}$$

The tail-sum identities now show that $\mathbb{E}\tau = \lim_{n\to\infty}\mathbb{E}[n \wedge \tau] = \mathbb{E}S_\tau^2$. The bounded stopped walk has mean zero at exit, and its only terminal values are $-a$ and b. Thus their probabilities are $b/(a+b)$ and $a/(a+b)$, respectively, and

$$\mathbb{E}\tau = \mathbb{E}S_\tau^2 = a^2\frac{b}{a+b} + b^2\frac{a}{a+b} = ab. \tag{B.56}$$

Solution to Exercise ??.

(a) The completed crossings end at times $1, 3, 5, \ldots$. Hence

$$U_N(1/4, 3/4) = \left\lfloor \frac{N+1}{2} \right\rfloor. \tag{B.57}$$

(b) The sequence $x_n = n$ is increasing, so it can complete at most one upcrossing of any fixed interval. It tends to ∞, which is one of the extended limits allowed by Lemma 2.39.

Solution to Exercise 2.16.

(a) The factors are $1/2$ or $3/2$, so $M_n \ge 0$. Independence gives $\mathbb{E}[1 + \xi_n/2 \,|\, \mathcal{F}_{n-1}] = 1$, and therefore $\mathbb{E}[M_n \,|\, \mathcal{F}_{n-1}] = M_{n-1}$. Thus M_n is a martingale, and $\mathbb{E}M_n = \mathbb{E}M_0 = 1$.

(b) Every nonnegative martingale is a nonnegative supermartingale, so Corollary 2.43 gives an almost-sure finite limit M_∞.

(c) Direct calculation gives

$$\rho^2 = \frac{2+\sqrt{3}}{4} < 1, \qquad \mathbb{E}\sqrt{M_n} = \prod_{k=1}^{n}\mathbb{E}\sqrt{1+\xi_k/2} = \rho^n. \tag{B.58}$$

For every $\varepsilon > 0$, Markov's inequality gives

$$\mathbb{P}(M_n \ge \varepsilon) \le \frac{\mathbb{E}\sqrt{M_n}}{\sqrt{\varepsilon}} = \frac{\rho^n}{\sqrt{\varepsilon}} \longrightarrow 0. \tag{B.59}$$

Since $M_n \to M_\infty$ almost surely, its limit must be zero: on $\{M_\infty > \varepsilon\}$, the values M_n are eventually larger than $\varepsilon/2$, while the probability of this latter event tends to zero. Thus $\mathbb{P}(M_\infty > \varepsilon) = 0$ for every $\varepsilon > 0$. We have $M_\infty = 0$ almost surely even though $\mathbb{E}M_n = 1$ at every finite time.

Solution to Exercise 2.17. Independence and $\mathbb{E}\xi_{n+1} = 0$ give $\mathbb{E}[S_{n+1} \,|\, \mathcal{F}_n] = S_n$ for $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$, so (S_n) is a martingale. On every path, $|S_{n+1} - S_n| = 1$ for every n. The successive differences of a convergent real sequence must tend to zero, so (S_n) cannot converge on any path. If $\sup_n \mathbb{E}|S_n|$ were finite, then Corollary 2.43 would imply almost-sure convergence, a contradiction. Therefore this supremum is infinite.

Solution to Exercise 2.9.

(a) Since $\mathcal{F}_{n+1} \subseteq \mathcal{F}_{n+2}$, the sequence $(\mathcal{G}_n)$ is a filtration. Each S_n is $\mathcal{F}_n$-measurable and hence also $\mathcal{G}_n$-measurable.

(b) Independence gives $\mathbb{E}[\,S_{n+1}|\,\mathcal{F}_n] = S_n$. But S_{n+1} is already $\mathcal{G}_n$-measurable, so $\mathbb{E}[\,S_{n+1}|\,\mathcal{G}_n] = S_{n+1} \neq S_n$. Thus the walk is a martingale only with respect to the first filtration.

(c) The payoff $H_n = \xi_n$ is $\mathcal{G}_{n-1} = \mathcal{F}_n$-measurable. It is not $\mathcal{F}_{n-1}$-measurable: otherwise its conditional mean given $\mathcal{F}_{n-1}$ would be ξ_n, whereas independence makes that conditional mean zero. Here $X_n = \sum_{k=1}^{n} \xi_k^2 = n$. With the original information, the stakes are not predictable. With the enlarged information, the walk is not a martingale. Neither filtration satisfies both hypotheses of the fair-game betting example.

Solution to Exercise 2.10.

(a) For $j < k$, D_j is $\mathcal{F}_{k-1}$-measurable. Cauchy–Schwarz gives $\mathbb{E}|D_j D_k| \leq \sqrt{\mathbb{E}D_j^2 \mathbb{E}D_k^2} < \infty$. Thus the taking-out property applies, and

$$\mathbb{E}[D_j D_k] = \mathbb{E}\,[D_j \mathbb{E}[\,D_k|\,\mathcal{F}_{k-1}]] = 0. \tag{B.60}$$

(b) Since $M_n = \sum_{k=1}^{n} D_k$, expanding the finite sum gives

$$\mathbb{E}M_n^2 = \sum_{k=1}^{n} \mathbb{E}D_k^2 + 2 \sum_{1 \leq j < k \leq n} \mathbb{E}[D_j D_k] = \sum_{k=1}^{n} \mathbb{E}D_k^2. \tag{B.61}$$

(c) We have $\mathbb{E}D_1 = 0$ and $\mathbb{E}[\,D_2|\,\mathcal{F}_1] = (2 + \xi_1)\mathbb{E}\xi_2 = 0$. Part (a) gives $\mathbb{E}[D_1 D_2] = 0$, so their covariance is zero. However, $\{|D_2| = 3\} = \{D_1 = 1\}$, an event of probability $1/2$. The probability of their intersection is $1/2$, not $1/4$, so D_1 and D_2 are not independent.

Solution to Exercise 2.15.

(a) Substituting $r = (1-p)/p$ gives $pr + (1-p)r^{-1} = (1-p) + p = 1$. Independence therefore gives $\mathbb{E}[\,r^{S_{n+1}} \,\big|\, \mathcal{F}_n] = r^{S_n}\mathbb{E}r^{\xi_{n+1}} = r^{S_n}$. Adaptedness holds, and at each fixed time the process takes only finitely many values, so it is integrable.

(b) If the walk survives a block of $a + b$ steps, that block cannot consist entirely of positive increments. Such a block has probability p^{a+b}, independently of the preceding increments. Applying this bound successively gives $\mathbb{P}(\tau > m(a+b)) \leq (1 - p^{a+b})^m$. This tends to zero, so τ is finite almost surely.

(c) Since $-a \leq S_{n\wedge\tau} \leq b$, $r^{S_{n\wedge\tau}} \leq \max\{r^{-a}, r^b\}$. Optional stopping gives $\mathbb{E}r^{S_\tau} = 1$. Writing $u = \mathbb{P}(S_\tau = b)$, we have $1 = r^{-a}(1-u) + r^b u$. Solving gives

$$u = \frac{1 - r^{-a}}{r^b - r^{-a}} = \frac{1 - r^a}{1 - r^{a+b}}. \tag{B.62}$$

The denominator is nonzero because $r > 0$ and $r \neq 1$.

Solution to Exercise 2.18.

(a) The bound gives $\sup_n \mathbb{E}|M_n| \leq K$. The martingale convergence theorem gives an almost-sure finite limit M_∞, and taking the limit in $|M_n| \leq K$ gives $|M_\infty| \leq K$ almost surely.

(b) On $\{|M_n - M_\infty| \le \varepsilon\}$ the difference is at most ε; on its complement it is at most $2K$. Taking expectations proves the stated bound. For any desired error $\delta > 0$, choose $\varepsilon = \delta/2$ and then take n large enough that the probability term contributes less than $\delta/2$. Thus $\mathbb{E}|M_n - M_\infty| \to 0$.

(c) Since $\mathbb{E}M_n = \mathbb{E}M_0$,

$$|\mathbb{E}M_\infty - \mathbb{E}M_0| = |\mathbb{E}[M_\infty - M_n]| \le \mathbb{E}|M_\infty - M_n| \longrightarrow 0. \tag{B.63}$$

Without a common bound, rare large values can retain a nonzero contribution to the expectation even as their probability tends to zero. For example, the product martingale $M_n = \prod_{k=1}^{n}(1 + \xi_k/2)$ with independent fair payoffs has $\mathbb{E}M_n = 1$ for every n, but its almost-sure limit is zero.

Solution to Exercise ??.

(a) The upcrossing inequality gives $(b-a)\mathbb{E}U_N(a,b) \le \mathbb{E}(M_N - a)^+ - \mathbb{E}(M_0 - a)^+$. Since $0 \le (M_N - a)^+ \le 1 - a$, dropping the second term proves the bound.

(b) At least q crossings occur in total exactly when the qth crossing is completed by some finite time N. The events $\{U_N(a,b) \ge q\}$ increase with N. Continuity of probability and Markov's inequality therefore give

$$\mathbb{P}(U_\infty(a,b) \ge q) = \lim_{N\to\infty} \mathbb{P}(U_N(a,b) \ge q) \le \frac{1-a}{q(b-a)}. \tag{B.64}$$

(c) The event of infinitely many crossings is contained in $\{U_\infty(a,b) \ge q\}$ for every q. Letting q increase shows that its probability is zero. The bound also quantifies how unlikely many repeated changes from at most a to at least b are; a wider gap gives a smaller bound.

Solution to Exercise 2.19.

(a) Every variable generating $\mathcal{F}_{-(n+1)}$ also generates $\mathcal{F}_{-n}$, so $\mathcal{F}_{-(n+1)} \subseteq \mathcal{F}_{-n}$.

(b) Since ξ_1 is $\mathcal{F}_{-1}$-measurable, $M_{-1} = \xi_1$. For $n \ge 2$, ξ_1 is independent of $\mathcal{F}_{-n}$, so $M_{-n} = \mathbb{E}\xi_1 = p$. Thus $\mathbb{E}[\,M_{-1}\,|\,\mathcal{F}_{-2}] = p = M_{-2}$, and the backward martingale identity holds at earlier times because both sides equal p. The almost-sure limit is p.

(c) Independence gives $\mathbb{E}[\,X_{-n}\,|\,\mathcal{F}_{-(n+1)}] = p$, which is not ξ_{n+1} because $0 < p < 1$ and ξ_{n+1} takes only the values zero and one. Since $X_{-(n+1)} = \xi_{n+1}$, the backward martingale identity fails. The process $(X_k)_{k\le -1}$ changes which observation it records. In contrast, $(M_k)_{k\le -1}$ always predicts the same observation ξ_1, using less information as $k \to -\infty$.

Solution to Exercise 2.20.

(a) The indicators $\mathbf{1}_{\{X_i \in B\}}$ are i.i.d. Bernoulli random variables with mean $q = \mathbb{P}(X_1 \in B)$ and variance $q(1-q) \le 1/4$. The strong law therefore gives the stated convergence, including the cases $q = 0$ and $q = 1$.

(b) For each j, let N_j be the null event where convergence fails for B_j. The countable union $\bigcup_{j\ge 1} N_j$ has probability zero. Outside this union, convergence holds for every B_j.

(c) For $B = (-\infty, x]$, the mean of the indicator is $\mathbb{P}(X_1 \le x) = F_X(x)$. There are countably many rational x, so part (b) gives the simultaneous conclusion.

(d) Since $\delta_{X_i}(B) = \mathbf{1}_{\{X_i \in B\}}$, μ_n is a probability measure and part (c) gives one probability-one event on which $\mu_n((-\infty, q]) \to F_X(q)$ for every rational q. Fix a sample path in this event and a continuity point x of F_X. For rational $q < x < r$, monotonicity gives

$$\mu_n((-\infty, q]) \le \mu_n((-\infty, x]) \le \mu_n((-\infty, r]). \tag{B.65}$$

Taking lower and upper limits in n bounds them below by $F_X(q)$ and above by $F_X(r)$. Letting $q \uparrow x$ and $r \downarrow x$ through rational values, continuity at x gives $\mu_n((-\infty, x]) \to F_X(x)$. Thus the empirical CDF converges at every continuity point on this same event. By Proposition 1.31, $\mu_n \xrightarrow{d} \mu_X$ on every such sample path, proving almost-sure weak convergence.

Solutions to Chapter 3

Solution to Exercise 3.1.

(a) The linear map $v \mapsto (\sum_i v_i, v\varphi)$ maps $\mathbb{R}^m$ into $\mathbb{R}^2$, so its rank is at most two. Since $m \ge 3$, the rank–nullity theorem gives a nonzero vector in its kernel.

(b) For $0 < \varepsilon < 1/(m \max_i |v_i|)$, every coordinate $\nu_i = 1/m + \varepsilon v_i$ is positive. Also $\sum_i \nu_i = 1 + \varepsilon \sum_i v_i = 1$, so ν is a probability law. Since $\varepsilon > 0$ and $v \neq 0$, it differs from μ. Nevertheless, $\nu\varphi = \mu\varphi + \varepsilon v\varphi = \mu\varphi$.

(c) The lemma requires agreement for every test function, while this construction gives agreement for one fixed function. The state indicators are sufficient to distinguish every pair of laws, since their expectations recover the individual coordinates.

Solution to Exercise 3.2.

(a) Each X_k, for $k \le n$, is a function of $X_0, \xi_1, \dots, \xi_n$. Thus the fresh toss ξ_{n+1} is independent of $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$. Conditioning on $\mathcal{F}_n$ fixes the current state, so for any test function,

$$\mathbb{E}[\varphi(X_{n+1}) | \mathcal{F}_n] = (1-p)\varphi(X_n) + p\varphi(1 - X_n) = (P\varphi)(X_n). \tag{B.66}$$

This verifies the Markov property with the stated matrix.

(b) If the current state is uniform, then both X_n and $1 - X_n$ are uniform. The independent coin selects between these two updates, so the next state remains uniform. This proves stationarity, and induction gives a uniform law at every time when X_0 is uniform.

(c) When $p = 1/2$, the conditional probability of either next state is $1/2$, regardless of the past. For $A \in \mathcal{F}_n$ and $j \in \{0, 1\}$,

$$\mathbb{P}(A \cap \{X_{n+1} = j\}) = \mathbb{E}\left[\mathbf{1}_A \mathbb{E}[\mathbf{1}_{\{X_{n+1}=j\}} \middle| \mathcal{F}_n]\right] = \frac{1}{2}\mathbb{P}(A). \tag{B.67}$$

Hence X_{n+1} is uniform and independent of $\mathcal{F}_n$. Applying this at successive times proves that $X_1, X_2, \dots$ are independent and uniform. If $p \neq 1/2$ and X_0 is uniform, stationarity gives $\mathbb{P}(X_1 = 1) = 1/2$, but $\mathbb{P}(X_1 = 1 \mid X_0 = 0) = p$. Since $\mathbb{P}(X_0 = 0) > 0$, these probabilities would agree under independence, so X_0 and X_1 are not independent.

Solution to Exercise 3.3.

(a) Each $P^{N-k}\varphi$ is a deterministic function on the finite state space. Its evaluation at X_k is therefore $\mathcal{F}_k$-measurable. Since stochastic matrices average function values, $|M_k| \leq \max_{i\in E} |\varphi_i|$, which proves integrability.

(b) At $k = N$, the identity holds because $M_N = \varphi(X_N)$ is $\mathcal{F}_N$-measurable. Suppose it holds at time $k+1$. The tower property and the one-step Markov property give

$$\begin{aligned} \mathbb{E}[\,\varphi(X_N)|\,\mathcal{F}_k] &= \mathbb{E}[\,(P^{N-k-1}\varphi)(X_{k+1})\Big|\,\mathcal{F}_k] \\ &= (P(P^{N-k-1}\varphi))(X_k) = (P^{N-k}\varphi)(X_k) = M_k. \end{aligned} \tag{B.68}$$

Descending induction proves the identity for every $0 \leq k \leq N$.

(c) For $k < N$, part (b) and the tower property give

$$\begin{aligned} \mathbb{E}[\,M_{k+1}|\,\mathcal{F}_k] &= \mathbb{E}[\,\mathbb{E}[\,\varphi(X_N)|\,\mathcal{F}_{k+1}]|\,\mathcal{F}_k] \\ &= \mathbb{E}[\,\varphi(X_N)|\,\mathcal{F}_k] = M_k. \end{aligned} \tag{B.69}$$

Together with part (a), this verifies the martingale property.

Solution to Exercise 3.4.

(a) The identity matrix preserves every distribution, and the matrix Q exchanges the two coordinates. Hence both preserve the uniform law.

(b) Under P, the chain stays at its initial state. Under Q, the chain changes state at every step. Starting with the uniform law gives uniform one-time laws for both chains, but the event $\{X_1 = X_0\}$ occurs surely for the first and never for the second. Their joint laws are therefore different.

(c) Starting from zero, the law under P is δ_0 at every time. Under Q, it is δ_0 at even times and δ_1 at odd times. The first law stays different from π, while the second alternates and has no limit. Thus a stationary distribution need not attract a chain started from a fixed state.

Solution to Exercise 3.5.

(a) For fixed x, the expression $P(x, B)$ is the probability that $f(x, \xi_1)$ belongs to B. Thus $P(x, \cdot)$ is the law of this random variable and is a probability measure.

(b) The function $(x, z) \mapsto \mathbf{1}_B(f(x, z))$ is bounded and jointly measurable. The permitted integration fact therefore makes its integral against ρ measurable as a function of x. These two parts verify the kernel conditions.

(c) The history $X_0, \ldots, X_n$ is a function of $X_0, \xi_1, \ldots, \xi_n$, so ξ_{n+1} is independent of $\mathcal{F}_n$. For $A \in \mathcal{F}_n$, independence lets us average over the fresh noise while holding the past fixed:

$$\mathbb{E}[\mathbf{1}_A \varphi(X_{n+1})] = \mathbb{E}\left[\mathbf{1}_A \int_{\mathbb{R}^r} \varphi(f(X_n, z))\,\rho(dz)\right]. \tag{B.70}$$

The expression inside the integral is bounded, and part (b)'s integration argument shows that the resulting function of X_n is $\mathcal{F}_n$-measurable. The defining property of conditional expectation proves the stated formula. Its dependence on the observed history is only through X_n.

Solution to Exercise 3.6.

(a) For fixed x, $P(x,\cdot)$ is a point mass and hence a probability measure. For fixed Borel B, $P(x,B) = \mathbf{1}_B(T(x))$ is measurable because T is measurable. Integrating against the point mass gives $(P\varphi)(x) = \varphi(T(x))$.

(b) By the definition of the updated law,

$$(\mu P)(B) = \int_E \mathbf{1}_B(T(x))\,\mu(dx) = \mu(T^{-1}(B)). \tag{B.71}$$

(c) Kernel duality gives $(\pi P)(\varphi) = \pi(P\varphi) = \int_E \varphi(T(x))\,\pi(dx)$. Thus stationarity implies the stated equality. Conversely, equality for every bounded measurable φ implies $\pi P = \pi$, since test functions determine the law. Preserving the distribution is therefore equivalent to preserving all of its expected test values.

Solution to Exercise 3.7.

(a) The common noise cancels when we subtract the two updates:

$$X_{n+1} - Y_{n+1} = \frac{1}{2}(X_n - Y_n). \tag{B.72}$$

Induction gives $X_n - Y_n = 2^{-n}(x-y)$.

(b) The same pathwise identity holds for random initial states, so $|X_n - Y_n| \le 2^{-n}$. A continuous function on the compact interval $[0,1]$ is uniformly continuous. Therefore

$$|\mathbb{E}\varphi(X_n) - \mathbb{E}\varphi(Y_n)| \le \sup_{\substack{u,v\in[0,1]\\ |u-v|\le 2^{-n}}} |\varphi(u) - \varphi(v)| \longrightarrow 0. \tag{B.73}$$

(c) By stationarity, every Y_n has law λ. Part (b) gives $\mathbb{E}\varphi(X_n) \to \lambda(\varphi)$ for every bounded continuous φ, which is weak convergence. If the initial law of X_0 is another stationary distribution ν, then every X_n has law ν. The resulting constant sequence of laws converges both to ν and to λ, so uniqueness of weak limits gives $\nu = \lambda$.

Solution to Exercise 3.8.

(a) For each x, $R(x,\cdot)$ is a convex combination of probability measures and is therefore a probability measure. For each Borel B, $R(x,B)$ is a linear combination of measurable functions of x. Thus R is a kernel. It describes choosing P with probability α and Q with probability $1-\alpha$, then using the chosen kernel for one transition.

(b) For fixed x, the measure $(PQ)(x,\cdot)$ is obtained by first using $P(x,\cdot)$ and then Q. Its total mass is one, and monotone convergence applied to disjoint unions proves countable additivity. For fixed B, the function $y \mapsto Q(y,B)$ is bounded and measurable, so integration against P makes $(PQ)(x,B)$ measurable in x. Hence PQ is a transition kernel. Applying kernel duality with initial law $P(x,\cdot)$ gives

$$((PQ)\varphi)(x) = \int_E (Q\varphi)(y)\, P(x,dy) = (P(Q\varphi))(x). \tag{B.74}$$

For any initial law μ, duality then gives

$$\begin{aligned}(\mu(PQ))(\varphi) &= \mu(P(Q\varphi)) = (\mu P)(Q\varphi) \\ &= ((\mu P)Q)(\varphi).\end{aligned} \tag{B.75}$$

Equality for all bounded measurable test functions proves $\mu(PQ) = (\mu P)Q$. Thus the kernel P acts first and Q acts second.

(c) Since $\pi P = \pi Q = \pi$, averaging the two updated laws gives $\pi R = \pi$. Composition also preserves the law, because $\pi(PQ) = (\pi P)Q = \pi Q = \pi$.

Solution to Exercise 3.9.

(a) Since $(P\mathbf{1}_B)(x) = P(x,B)$, the identity follows from $L = P - I$.

(b) By linearity, $\pi(L\varphi) = \pi(P\varphi) - \pi(\varphi)$. Thus stationarity gives zero expected drift. Conversely, if this difference is zero for every bounded measurable φ, kernel duality gives $(\pi P)(\varphi) = \pi(\varphi)$ for every such φ. Test functions determine laws, so $\pi P = \pi$. Under a stationary distribution, the expected value of every bounded test function stays the same after one step, so its average change is zero.

(c) Part (b) gives $\pi(L\varphi) = 0$. The function $-L\varphi$ is nonnegative and has integral zero under π, so it is zero almost surely under π.

Solution to Exercise 3.10.

(a) On $\{\tau \le k\}$, both stopped indices equal τ, and the increment is zero. On $\{\tau > k\}$, the stopped indices are k and $k+1$, proving the stated identity. Since $\{\tau > k\} \in \mathcal{F}_k$ and the martingale increment has conditional mean zero, so does the stopped increment. The stopped process is adapted and integrable, since it only uses the finitely many integrable values $M_0^\varphi, \dots, M_N^\varphi$. Thus it is a martingale.

(b) Its initial value is zero, so $\mathbb{E}M_\tau^\varphi = 0$ by taking expectations at time N of the stopped process. Substituting the definition of M_τ^φ gives (3.3.22). A deterministic choice $\tau = n$ recovers (3.3.17).

Solution to Exercise 3.11.

(a) The kernel preserves nonnegativity, and stationarity gives $\pi(f_n) = \pi(f_0) = 1$. Repeated use of reversibility shows that $P_n = P^n$ is self-adjoint. Duality therefore gives

$$\begin{aligned} \mu_n(B) &= \mu_0(P_n \mathbf{1}_B) = \langle f_0, P_n \mathbf{1}_B \rangle_\pi \\ &= \langle P_n f_0, \mathbf{1}_B \rangle_\pi = \int_B f_n(x)\, \pi(dx). \end{aligned} \tag{B.76}$$

Thus f_n is the density of μ_n with respect to π.

(b) Since $\pi(f_0) = 1$, the function $f_0 - 1$ has mean zero. Apply Theorem 3.26 to $f_0 - 1$ and use $P_n(f_0 - 1) = f_n - 1$ to obtain $\|f_n - 1\|_\pi \le (1-\lambda)^n \|f_0 - 1\|_\pi$. Squaring and using (3.4.27) gives (3.4.30).

(c) Part (a) gives $\pi(f_n - 1) = 0$. Its positive and negative parts therefore each have integral $\frac{1}{2}\pi(|f_n - 1|)$. For every measurable B, Cauchy–Schwarz gives

$$|\mu_n(B) - \pi(B)| = \left| \int_B (f_n - 1)\, d\pi \right| \le \frac{1}{2} \int_E |f_n - 1|\, d\pi \le \frac{1}{2} \|f_n - 1\|_\pi. \tag{B.77}$$

Taking the supremum over B and using $\|f_n - 1\|_\pi = \sqrt{\chi^2(\mu_n \| \pi)}$ proves (3.4.31). Part (b) then gives

$$d_{\mathrm{TV}}(\mu_n, \pi) \le \frac{1}{2}(1-\lambda)^n \sqrt{\chi^2(\mu_0 \| \pi)}. \tag{B.78}$$

Solutions to Chapter 4

Solution to Exercise 4.1.

(a) Since X_0 has mean and variance zero, it equals zero almost surely. Each increment is a centered Gaussian variable, and

$$\mathrm{Var}(X_t - X_s) = t + s - 2s = t - s, \qquad s < t. \tag{B.79}$$

(b) For $s < t \le u < v$, the covariance of the two increments is

$$\mathrm{Cov}(X_t - X_s, X_v - X_u) = t - t - s + s = 0. \tag{B.80}$$

Any finite vector of such increments is jointly Gaussian, since it is a linear transformation of a finite vector of observations of X. Its distinct coordinates have zero covariance, hence are independent.

(c) Parts (a) and (b), together with the assumed path continuity, verify every condition in the definition of standard Brownian motion.

Solution to Exercise 4.2.

(a) Each path is a constant multiple of the continuous function $\sqrt{t}$. Also $X_t \sim \mathcal{N}(0, t)$, which is the law of B_t.

(b) Both $X_1 = Z$ and $X_2 - X_1 = (\sqrt{2} - 1)Z$ are nonconstant multiples of the same variable. In particular,

$$\mathrm{Cov}(X_1, X_2 - X_1) = \sqrt{2} - 1 \neq 0, \tag{B.81}$$

so the increments on $[0, 1]$ and $[1, 2]$ are not independent.

(c) One-time laws do not describe the dependence between different observations. The random-walk proof also uses independence of disjoint increment blocks to identify the joint Gaussian limit.

Solution to Exercise 4.3.

(a) On each grid interval, $\widetilde{B}_t^{(N)}$ is a convex average of the two endpoint values, so it cannot exceed their maximum. Each grid value is attained by the interpolated path, which proves (4.1.16).

(b) Continuous functions attain their maxima on $[0,1]$. For every $t \in [0,1]$, we have $f(t) \le g(t) + d(f,g) \le M(g) + d(f,g)$. Taking the maximum gives $M(f) - M(g) \le d(f,g)$. Interchanging f and g proves $|M(f) - M(g)| \le d(f,g)$, hence continuity.

(c) The function $F = \varphi \circ M$ is continuous by part (b) and bounded because φ is bounded. Donsker's theorem and part (a) give

$$\begin{aligned} \mathbb{E}\,\varphi\left(\frac{1}{\sqrt{N}} \max_{0\le k\le N} S_k\right) &= \mathbb{E}F(\widetilde{B}^{(N)}) \\ &\longrightarrow \mathbb{E}F(B) = \mathbb{E}\,\varphi\left(\max_{0\le t\le 1} B_t\right). \end{aligned} \tag{B.82}$$

This holds for every bounded continuous φ, so the real-valued maxima converge in distribution as claimed.

Solution to Exercise 4.4.

(a) We have $f_n(0) = 0$ for every n. For a fixed $t > 0$, eventually $1/n + 1/n^2 < t$, so $f_n(t) = 0$. The function takes the value one at $t = 1/n$ and lies between zero and one everywhere, proving the supremum claim.

(b) For any fixed finite set of times, every coordinate tends to zero by part (a). Thus the corresponding deterministic vectors converge to the zero vector, and their point-mass laws converge to its law.

(c) The reverse triangle inequality gives

$$\left|\sup_t |f(t)| - \sup_t |g(t)|\right| \le \sup_t |f(t) - g(t)| = d(f,g). \tag{B.83}$$

Truncation at one preserves this bound, so F is continuous and bounded. However, $F(f_n) = 1$ for every n, whereas $F(0) = 0$. The expected values of this bounded continuous path test function do not converge to its value under the zero-path law, so convergence of path laws fails.

Solution to Exercise 4.5.

(a) The process $aB + bW$ has continuous paths and is a centered Gaussian process, because the two Gaussian processes are independent. Its covariance is

$$\operatorname{Cov}(aB_s + bW_s, aB_t + bW_t) = (a^2 + b^2)\min\{s,t\} = \min\{s,t\}. \tag{B.84}$$

The preceding characterization exercise shows that it is standard Brownian motion.

(b) Part (a) shows that both U and V are standard Brownian motions. Moreover, for any $s, t \geq 0$,

$$\operatorname{Cov}(U_s, V_t) = \frac{1}{2}\left(\operatorname{Cov}(B_s, B_t) - \operatorname{Cov}(W_s, W_t)\right) = 0. \tag{B.85}$$

Any finite collection of U and V observations is jointly Gaussian, and all covariances between the two groups vanish. The two observation vectors are therefore independent. Since this holds for all finite collections, the two processes are independent.

Solution to Exercise 4.6.

(a) The mean value theorem gives $|g(t_j) - g(t_{j-1})| \leq C\Delta t$. Squaring and summing gives $[g]_T^{(N)} \leq NC^2(\Delta t)^2 = C^2T^2/N$.

(b) Cauchy–Schwarz for the two vectors of increments gives

$$\left([g, B]_T^{(N)}\right)^2 \leq [g]_T^{(N)}[B]_T^{(N)}. \tag{B.86}$$

Part (a) and $\mathbb{E}[B]_T^{(N)} = T$ give the required bound after taking expectations.

(c) Expanding each squared increment gives

$$[X]_T^{(N)} = [g]_T^{(N)} + 2\sigma[g, B]_T^{(N)} + \sigma^2[B]_T^{(N)}. \tag{B.87}$$

The first term tends to zero deterministically, hence in L^2, by part (a). The second tends to zero in L^2 by part (b), and the last tends to $\sigma^2 T$ in L^2 by Proposition 4.12. Thus $[X]_t = \sigma^2 t$: the smooth increments and their products with Brownian increments disappear in the limit.

Solution to Exercise 4.7.

(a) Expanding the square on each half-interval gives

$$2B_{t_j^*}\Delta B_{t_j} = B_{t_j}^2 - B_{t_{j-1}}^2 + (B_{t_j^*} - B_{t_{j-1}})^2 - (B_{t_j} - B_{t_j^*})^2. \tag{B.88}$$

Summing over j telescopes the first difference, giving the identity in part (a).

(b) Each squared half-interval increment has mean $\Delta t/2$ and variance $2(\Delta t/2)^2$. Each sum consists of N independent terms, so its mean is $N\Delta t/2 = T/2$ and its variance is $2N(\Delta t/2)^2 = T^2/(2N)$.

(c) Part (b) shows that both sums converge to $T/2$ in L^2. Their difference tends to zero, so part (a) gives $M_N \to B_T^2/2$ in L^2. The two sums are independent, and their difference has mean zero. Thus the exact error is

$$\mathbb{E}\left[\left(M_N - \frac{B_T^2}{2}\right)^2\right] = \frac{1}{4}\left(\frac{T^2}{2N} + \frac{T^2}{2N}\right) = \frac{T^2}{4N}. \tag{B.89}$$

(d) The difference-of-squares identity gives

$$\sum_{j=1}^{N} \frac{B_{t_{j-1}} + B_{t_j}}{2}\,\Delta B_{t_j} = \frac{1}{2}\sum_{j=1}^{N}(B_{t_j}^2 - B_{t_{j-1}}^2) = \frac{B_T^2}{2}. \tag{B.90}$$

The endpoint average generally differs from the value of the path at the time midpoint. Part (c) shows that the difference of these sums has positive second moment for every fixed N, but tends to zero in L^2.

Solution to Exercise 4.8.

(a) The process W starts at zero and has continuous paths. For any finite collection of disjoint intervals, the vector of B increments has a product of centered Gaussian laws. This joint law is unchanged by multiplying all coordinates by 1 or by -1. Independence of A and B therefore shows that the vector of W increments has the same product law. Its increments are independent centered Gaussians with the required variances, proving that W is standard Brownian motion.

(b) Both variables have mean zero, and independence gives

$$\operatorname{Cov}(B_s, W_t) = \mathbb{E}[AB_sB_t] = \mathbb{E}A\,\mathbb{E}[B_sB_t] = 0. \tag{B.91}$$

(c) The two indicated events agree because $|W_t| = |B_t|$. If their common probability is $p \in (0,1)$, their intersection has probability p, whereas independence would give p^2. Thus B_t and W_t are dependent. Each is Gaussian individually, but the pair is not jointly Gaussian; otherwise zero covariance would imply independence.

(d) The identity $\Delta W_{t_j} = A\Delta B_{t_j}$ gives $[B,W]_T^{(N)} = A[B]_T^{(N)}$. Since $A^2 = 1$,

$$\mathbb{E}\left[\left([B,W]_T^{(N)} - AT\right)^2\right] = \mathbb{E}\left[([B]_T^{(N)} - T)^2\right] = \frac{2T^2}{N} \longrightarrow 0. \tag{B.92}$$

Therefore $[B,W]_t = At$. Its sign records whether the two paths move together or in opposite directions; averaging over the random sign gives zero covariance but does not remove this dependence.

Solutions to Chapter ??

Solution to Exercise ??.

(a) If F preserves norms, then $\|F(0)\| = \|0\| = 0$, so $a = 0$. Conversely, when $a = 0$, orthogonality gives $\|Qx\|^2 = x^\top Q^\top Qx = \|x\|^2$. Preservation of distances for arbitrary a follows from **??**.

(b) For any matrix D, applying U to each column preserves its Euclidean norm, so $\|UD\|_F = \|D\|_F$. Applying V to each row likewise gives $\|DV\|_F = \|D\|_F$. Subtracting the two images cancels C, and therefore

$$\|(UAV + C) - (UA'V + C)\|_F = \|U(A - A')V\|_F = \|A - A'\|_F. \tag{B.93}$$

(c) Subtraction gives $(h + q) - (k + q) = h - k$, so the distance is unchanged. Taking $h = 0$ and $q(t) = 1$ changes the norm from zero to $\sqrt{T}$.

Solution to Exercise ??.

(a) Define the matrix D by $D_{ki} = x_i\sqrt{\mu_{k,i}}$. Then

$$\|D\|_F^2 = \sum_{k=0}^{N}\sum_{i=1}^{m} \mu_{k,i}\, x_i^2 = \mathbb{E}\left[\sum_{k=0}^{N} X_k^2\right] = \|X\|_N^2. \tag{B.94}$$

(b) At each time, $\mathbb{E}X_k^2 = (1-\varepsilon)x_1^2 + \varepsilon x_m^2$. Thus $\|X\|_N^2 = (N+1)((1-\varepsilon)x_1^2 + \varepsilon x_m^2)$. The rare state contributes its squared size x_m^2 multiplied by its probability ε; its contribution depends on both quantities.

(c) For the piecewise constant path, its squared time-integrated norm is $\sum_{k=0}^{N} \Delta t\, X_k^2$. Taking expectations gives $\Delta t\, \|X\|_N^2$. Repeating each value doubles the discrete squared norm, but each new interval has half the length, so the expected squared continuous-time path norm is unchanged.

Solution to Exercise ??.

(a) On the jth interval, the difference is the increment $B_{t_{j-1}} - B_t$. Its second moment is $t - t_{j-1}$, giving

$$\|H^{(N)} - B\|_T^2 = \sum_{j=1}^{N} \int_{t_{j-1}}^{t_j} (t - t_{j-1})\, dt = \frac{N(\Delta t)^2}{2} = \frac{T^2}{2N}. \tag{B.95}$$

(b) The integral of $H^{(N)}$ is exactly the left-endpoint sum. Apply the distance form of the isometry to $H^{(N)}$ and B, then use part (a), to obtain (**??**).

(c) On each interval, $\mathbb{E}[(B_{t_j} - B_t)^2] = t_j - t$. Integrating gives the same value $(\Delta t)^2/2$ per interval, hence $\|K^{(N)} - B\|_T^2 = T^2/(2N)$. However, the two weighted sums satisfy

$$\sum_{j=1}^{N} B_{t_j} \Delta B_{t_j} = \sum_{j=1}^{N} B_{t_{j-1}} \Delta B_{t_j} + [B]_T^{(N)} \longrightarrow \int_0^T B_t\, dB_t + T \quad \text{in probability.} \tag{B.96}$$

The left sum converges by part (b), and the squared-increment sum converges to T by Proposition 4.12. There is no contradiction: $K^{(N)}$ uses the future value B_{t_j} before time t_j, so it is not adapted. Its right-endpoint sum is not an Itô integral of a simple process, and the isometry does not apply to that sum.

Solution to Exercise ??.

(a) Linearity and the isometry give

$$\mathbb{E}[(I_T(H) + I_T(K))^2] = \mathbb{E} \int_0^T (H_t + K_t)^2\, dt. \tag{B.97}$$

Expand both squares and cancel $\mathbb{E}[I_T(H)^2] = \mathbb{E} \int_0^T H_t^2\, dt$ and the corresponding terms for K. Dividing the remaining equality by two proves (**??**). The product is integrable by Cauchy–Schwarz, and both integrals have mean zero, so this expectation is their covariance.

(b) For any $a, b \in \mathbb{R}$, linearity gives $aI_T(h) + bI_T(k) = I_T(ah + bk)$. This is Gaussian by **??**, so the pair is jointly Gaussian. A jointly Gaussian pair is independent exactly when its covariance is zero, which by part (a) is the condition $\int_0^T h(t)k(t)\, dt = 0$. For the given functions,

$$\int_0^T \left(t - \frac{T}{2}\right) dt = 0. \tag{B.98}$$

Thus B_T and $\int_0^T (t - T/2)\, dB_t$ are independent, even though both integrals use Brownian increments over the same time interval.